REFERANS KİTAP

Cisco CLI Komut Kitabı

200 Temel IOS Komutu ile Ağ Yönetimi

Kapsam: IOS / IOS-XE / NX-OS

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CISCO NETWORKING REFERENCE

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1. Temel Yapilandirma Komutlari

> Cihazin temel kimlik bilgileri, yonetim erisimi ve genel sistem ayarlarini iceren komutlar. Cihaza ilk erisimden itibaren kullanilan temel yonetim komutlaridir.

1 enable

ACİKLAMA

Kullaniciyi ayricalikli EXEC modundan (Privileged EXEC) yukseltir. Bu modda tum show ve debug komutlari calistirilabilir.

KULLANİM AMACİ

Yonetici yetkilerine gecmek, ayricalikli modda yapilandirma ve izleme islemleri yapmak.

SYNTAX

```
Router> enable [privilege-level]
```

ORNEK KULLANİM

```
Router> enable
Router#
```

SİK YAPİLAN HATA

Enable sifresi unutulursa cihaza erisim kaybolur. Sifre recovery islemi gerekir.

CİKTİ ORNEGİ

```
Router> enable
Password: *****
Router#
! Prompt '#' ayricalikli modu gosterir
```

2 configure terminal

ACİKLAMA Genel yapilandirma moduna (Global Configuration Mode) gecisi saglar. Tum sistem ayarlari bu modda yapilir.

KULLANİM AMACİ Hostname, domain name, sifreler, servis ayarlari gibi tum genel yapilandirmalari girmek.

SYNTAX

```
Router# configure terminal
```

ORNEK KULLANİM

```
Router# configure terminal
Router(config)#
```

SİK YAPİLAN HATA Kisa yazim 'conf t' calisir ancak 'configure' yazip Enter'a basinca alt modlar listelenir, dogrudan terminale gecilmez.

CİKTİ ORNEGİ

```
Router# configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
! '(config)#' global yapilandirma modunu gosterir
```

3 hostname

ACİKLAMA Cihazin ag uzerindeki adini (hostname) belirler. Prompt uzerinde gorunur ve yonetimi kolaylastirir.

KULLANİM AMACİ Cihaza tanimlayici bir isim vermek, coklu cihaz yonetiminde karmasayi onlemek.

SYNTAX

```
Router(config)# hostname name
```

ORNEK KULLANİM

```
Router(config)# hostname IST-CORE-SW01
IST-CORE-SW01(config)#
```

SİK YAPİLAN HATA Bosluk iceren isimler kullanilamaz. Tire (-) veya alt cizgi (_) kullanilmalidir.

CİKTİ ORNEGİ

```
Router(config)# hostname ANK-BORDER-RTR01
ANK-BORDER-RTR01(config)#
! Prompt otomatik olarak yeni hostname ile guncellendi
```

4 enable secret

ACİKLAMA Ayricalikli moda (enable mode) erisim icin MD5 hash ile sifrelenmis guvenli sifre belirler. 'enable password' dan daha guvenlidir.

KULLANİM AMACİ Enable moduna erisim icin guclu bir sifre tanimlamak, yetkisiz erisimi onlemek.

SYNTAX

```
Router(config)# enable secret [level level] {password | [encryption-type] encrypted-password}
```

ORNEK KULLANİM

```
Router(config)# enable secret Cisco123!
```

SİK YAPİLAN HATA Hem 'enable password' hem 'enable secret' tanimlanirsa secret gecerlidir. Ancak password clear-text olarak kalir ve guvenlik riski olusturur.

CİKTİ ORNEGİ

```
Router(config)# enable secret Cisco123!
! Sifre running-config'de hash'lenmis olarak gorunur
Router# show running-config | include enable secret
enable secret 5 $1$mERr$IL3c8lSNH2QmLGOv/4u.w/
```

5 service password-encryption

ACİKLAMA Yapilandirmadaki tum clear-text sifreleri zayif bir sifreleme algoritmasi (Type 7) ile sifreler. Tam guvenlik saglamaz ancak sifrelerin gozle gorunmesini engeller.

KULLANİM AMACİ Yapilandirma dosyasinda gorunen sifreleri sifrelemek, basit guvenlik saglamak.

SYNTAX

```
Router(config)# service password-encryption
```

ORNEK KULLANİM

```
Router(config)# service password-encryption
```

SİK YAPİLAN HATA Type 7 sifreleme kolayca kirilabilir. Gercek guvenlik icin 'secret' komutu (Type 5/Type 9) kullanilmalidir.

CİKTİ ORNEGİ

```
Router(config)# service password-encryption
! Oncesi: username admin password Cisco123
! Sonrasi:
username admin password 7 0822455D0A16
! 7 = Type 7 sifreleme ( zayif )
```

6 username

ACİKLAMA

Yerel veritabaninda kullanici adi ve sifre tanimlar. AAA yerel kimlik dogrulamasi, konsol/vty erisimi icin kullanilir.

KULLANİM AMACİ

Cihaza erisim icin yerel kullanici hesaplari olusturmak, role-based erisim tanimlamak.

SYNTAX

```
Router(config)# username name [privilege level] {password | secret} value
```

ORNEK KULLANİM

```
Router(config)# username admin privilege 15 secret Admin@123
Router(config)# username operator privilege 5 secret Op@2024
```

SİK YAPİLAN HATA

'password' yerine 'secret' kullanilmazsa sifre zayif olarak saklanir. Her zaman secret ile tanimlayin.

CİKTİ ORNEGİ

```
Router# show running-config | section username
username admin privilege 15 secret 9 $9$vFpM.lzQn...
username operator privilege 5 secret 9 $9$GnKp.qRm...
! 9 = SHA256 tabanli guclu sifreleme (Type 9)
```

7 banner motd

ACİKLAMA Message of the Day (MOTD) banner tanimlar. Kullanici cihaza eristiginde ilk gorunen mesajdir. Yasal uyarilar icin kullanilir.

KULLANİM AMACİ Yasal uyarilar, erisim politikalari, guvenlik bildirimleri gostermek.

SYNTAX

```
Router(config)# banner motd delimiter message delimiter
```

ORNEK KULLANİM

```
Router(config)# banner motd #
Authorized Access Only!
All activities are monitored.
#
```

SİK YAPİLAN HATA

Delimiter karakteri (#) mesaj icinde kullanilirsa banner erken sonlanir. Mesajda kullanilmayacak benzersiz bir delimiter secin.

CİKTİ ORNEGİ

```
Authorized Access Only!
All activities are monitored.

User Access Verification

Username:
```

8 line console 0

ACİKLAMA Konsol baglantisi (Console 0) icin yapilandirma moduna girer. Fiziksel konsol port uzerinden erisim ayarlari yapilir.

KULLANİM AMACİ Konsol erisimine sifre koymak, zaman asimi, logging ayarlari yapmak.

SYNTAX

```
Router(config)# line console 0
```

ORNEK KULLANİM

```
Router(config)# line console 0
Router(config-line)# password Con@2024
Router(config-line)# login
Router(config-line)# exec-timeout 5 0
```

SİK YAPİLAN HATA 'login' komutu verilmezse sifre istenmez ve cihaz korunmasiz kalir. Her zaman login komutunu ekleyin.

CİKTİ ORNEGİ

```
Router(config)# line console 0
Router(config-line)# do show line console 0
Tty Typ     Tx/Rx    A Modem  Roty AccO AccI  Uses  Noise Overruns
Int
   0 CTY              -    -      -    -    -     0      0    0/0
-

Line 0, Location: "", Type: ""
Length: 24 lines, Width: 80 columns
Status: Ready, Active
```

9 line vty

ACİKLAMA

Sanal terminal hatlari (VTY) uzerinden uzaktan erisim (SSH/Telnet) yapilandirmasi yapilir. Varsayilan olarak 0-4 veya 0-15 araligi kullanilir.

KULLANİM AMACİ

SSH/Telnet erisimine sifre koymak, protokol kisitlamalari yapmak, zaman asimi ayarlamak.

SYNTAX

```
Router(config)# line vty first-line [last-line]
```

ORNEK KULLANİM

```
Router(config)# line vty 0 15
Router(config-line)# transport input ssh
Router(config-line)# login local
Router(config-line)# exec-timeout 10 0
```

SİK YAPİLAN HATA

'transport input all' veya tanimlanmamissa hem SSH hem Telnet acik kalir. Guvenlik icin sadece 'transport input ssh' kullanin.

CİKTİ ORNEGİ

```
Router(config-line)# do show line vty 0
Tty Typ     Tx/Rx    A Modem  Roty AccO AccI  Uses  Noise Overruns
Int
*  2 VTY              -    -      -    -    -     5      0    0/0
-

Line 2, Location: "", Type: ""
Session limit is not set.
Allowed input transports are ssh.
Preferred transport is ssh.
```

10 copy running-config startup-config

ACİKLAMA

Gecerli yapilandirmayi (running-config) kalici bellege (startup-config / NVRAM) kaydeder. Cihaz yeniden baslatildiginda yapilandirma korunur.

KULLANİM AMACİ

Yapilan tum degisiklikleri kalici olarak saklamak, cihaz yeniden baslatildiginda kaybolmasini onlemek.

SYNTAX

```
Router# copy running-config startup-config
```

ORNEK KULLANİM

```
Router# copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
```

SİK YAPİLAN HATA

'wr' kisaltmasi kullanilabilir ancak unutulan kaydetme islemi cihaz yeniden baslatilinca tum yapilandirmayi siler. Her degisiklikten sonra kaydedin.

CİKTİ ORNEGİ

```
Router# copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
Compressed configuration from 4521 bytes to 2187 bytes[OK]
! Veya kisa versiyon:
Router# write memory
Building configuration...
[OK]
```

2. Arayuz Yapilandirma Komutlari

Fiziksel ve sanal ag arayuzlerinin yapilandirilmasi, IP adresi ataması, hiz/duplex ayarlari ve port yonetimi komutlarini icerir.

11 interface

ACİKLAMA

Belirtilen arayuzun yapilandirma moduna (Interface Configuration Mode) girer. Ethernet, Serial, Loopback, VLAN arayuzleri yapilandirilir.

KULLANİM AMACİ

Herhangi bir ag arayuzunun IP adresi, hiz, duplex, acma/kapama gibi ayarlarini yapmak.

SYNTAX

```
Router(config)# interface type slot/port
Router(config)# interface type slot/subslot/port
```

ORNEK KULLANİM

```
Router(config)# interface GigabitEthernet0/0
Router(config-if)#
```

SİK YAPİLAN HATA

Arayuz adi yanlis yazilirsa yeni bir arayuz olusturulur (ornegin 'GigabitEthernet0/0/0' yerine 'GigabitEthernet0/0'). Modul yapisini dogrulayin.

CİKTİ ORNEGİ

```
Router(config)# interface GigabitEthernet0/0/1
Router(config-if)#
! (config-if) = Interface Configuration Mode
```

12 ip address

ACİKLAMA Bir arayuze IPv4 adresi ve subnet maskesi atar. Temel L3 yapilandirma komutudur.

KULLANİM AMACİ Arayuz uzerinden IP iletisimini saglamak, ag baglantisi kurmak.

SYNTAX

```
Router(config-if)# ip address ip-address subnet-mask [secondary]
```

ORNEK KULLANİM

```
Router(config-if)# ip address 192.168.1.1 255.255.255.0
```

SİK YAPİLAN HATA Ayni agdaki IP farkli cihazlarda tekrarlanirsa IP conflict olusur. Ayrica arayuz 'shutdown' ise IP atanmis olsa bile iletisim olmaz.

CİKTİ ORNEGİ

```
Router(config-if)# ip address 10.1.1.1 255.255.255.0
Router# show ip interface GigabitEthernet0/0/1
GigabitEthernet0/0/1 is up, line protocol is up
  Internet address is 10.1.1.1/24
  Broadcast address is 255.255.255.255
  Address determined by setup command
```

13 no shutdown

ACİKLAMA Kapali (administratively down) durumdaki arayuzu aktif hale getirir. Varsayilan olarak tum arayuzler kapalidir.

KULLANİM AMACİ Yapilandirilan arayuzu calisir duruma getirmek, iletisimi baslatmak.

SYNTAX

```
Router(config-if)# no shutdown
```

ORNEK KULLANİM

```
Router(config-if)# no shutdown
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN:      Line      protocol      on      Interface
GigabitEthernet0/0, changed state to up
```

SİK YAPİLAN HATA

IP adresi atandiktan sonra 'no shutdown' unutulursa arayuz 'administratively down' kalir. Bu en yaygin unutulan komuttur.

CİKTİ ORNEGİ

```
Router(config-if)# no shutdown
%LINK-5-CHANGED: Interface GigabitEthernet0/0/1, changed state to
up
%LINEPROTO-5-UPDOWN:      Line      protocol      on      Interface
GigabitEthernet0/0/1, changed state to up

Router# show ip interface brief
Interface                  IP-Address        OK? Method Status
Protocol
GigabitEthernet0/0/1       10.1.1.1            YES  manual  up
```

14 description

ACİKLAMA Arayuze aciklayici bir metin ekler. Yapilandirmada gorunur ancak islevsel bir etkisi yoktur. Dokumantasyon amaclidir.

KULLANİM AMACİ Arayuzun amacini, bagli cihazlari, lokasyonu belgelemek, sorun gidermede kolaylik saglamak.

SYNTAX

```
Router(config-if)# description text
```

ORNEK KULLANİM

```
Router(config-if)# description *** Uplink to Core-SW01 Gi1/0/1 ***
```

SİK YAPİLAN HATA Bosluklu tanimlamalar icin tirnak isareti kullanilmaz ancak sorun cikmaz. Aciklama guncellendiginde eski tamamen silinir.

CİKTİ ORNEGİ

```
Router# show interfaces GigabitEthernet0/0/1 description
Interface                      Status         Protocol Description
Gi0/0/1                         up             up       *** Uplink
to Core-SW01 ***
```

15 speed

ACİKLAMA Ethernet arayuzunun calisacagi hizi (10, 100, 1000, 10000 Mbps) manuel olarak ayarlar. Varsayilan 'auto'dur.

KULLANİM AMACİ Otomatik mutabakat sorunlarinda sabit hiz ayarlamak, performans optimizasyonu yapmak.

SYNTAX

```
Router(config-if)# speed {10 | 100 | 1000 | 10000 | auto}
```

ORNEK KULLANİM

```
Router(config-if)# speed 1000
```

SİK YAPİLAN HATA Bir tarafta 'speed 1000' diger tarafta 'auto' veya 'speed 100' ise mismatch olusur. Her iki taraf da eslesmelidir.

CİKTİ ORNEGİ

```
Router(config-if)# speed 1000
Router# show interfaces GigabitEthernet0/0/1
GigabitEthernet0/0/1 is up, line protocol is up
  Hardware is Gigabit Ethernet, address is aabb.ccdd.ee01
  MTU 1500 bytes, BW 1000000 Kbit/sec, DLY 10 usec,
     reliability 255/255, txload 1/255, rxload 1/255
  Full-duplex, 1000Mb/s, media type is RJ45
```

16 duplex

ACİKLAMA

Ethernet arayuzunun calisma modunu (half veya full duplex) manuel olarak ayarlar. Full duplex hem gonderim hem alim ayni anda yapilir.

KULLANİM AMACİ

Otomatik mutabakat sorunlarinda duplex modunu sabitlemek, performans sorunlarini cozmek.

SYNTAX

```
Router(config-if)# duplex {auto | full | half}
```

ORNEK KULLANİM

```
Router(config-if)# duplex full
```

SİK YAPİLAN HATA

Bir tarafta 'duplex full' diger tarafta 'duplex auto' olursa duplex mismatch olusur ve performans duser (late collisions artar).

CİKTİ ORNEGİ

```
Router(config-if)# duplex full
Router# show interfaces GigabitEthernet0/0/1 | include duplex
  Full-duplex, 1000Mb/s, media type is RJ45
  input flow-control is off, output flow-control is unsupported
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored
    0 watchdog, 0 multicast, 0 pause input
```

17 interface range

ACİKLAMA

Birden fazla arayuzu ayni anda secerek toplu yapilandirma imkani verir. Zamandan tasarruf saglar.

KULLANİM AMACİ

Toplu port yapilandirmasi, VLAN atamasi, port kapatma/acma gibi toplu islemler yapmak.

SYNTAX

```
Router(config)# interface range type slot/port - port
Router(config)# interface range type slot/port , type slot/port
```

ORNEK KULLANİM

```
Switch(config)# interface range GigabitEthernet0/1 - 24
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 10
```

SİK YAPİLAN HATA

Aralik (range) icinde olmayan bir port yazilirsa hata verir. Virgul ile ayrik portlarda her portun tam adi yazilmalidir.

CİKTİ ORNEGİ

```
Switch(config)# interface range gi0/1 - 4
Switch(config-if-range)# description Access Ports
! Tum 4 port ayni anda yapilandirildi
Switch# show running-config interface range gi0/1 - 4
interface GigabitEthernet0/1
 description Access Ports
!
interface GigabitEthernet0/2
 description Access Ports
!
```

18 interface loopback

ACİKLAMA

Sanal (mantiksal) bir loopback arayuzu olusturur. Fiziksel baglanti gerektirmez ve her zaman 'up' durumundadir.

KULLANİM AMACİ

Router ID, yonetim IP'si, yonlendirme protokolu stabilitesi saglamak icin kullanilir.

SYNTAX

```
Router(config)# interface loopback number
```

ORNEK KULLANİM

```
Router(config)# interface loopback 0
Router(config-if)# ip address 1.1.1.1 255.255.255.255
```

SİK YAPİLAN HATA

Loopback IP'si yonlendirme protokolunde Router ID olarak kullanilirsa, IP degisikligi protokolun yeniden baslamasina neden olur.

CİKTİ ORNEGİ

```
Router(config)# interface loopback 0
Router(config-if)# ip address 10.255.1.1 255.255.255.255
Router# show ip interface brief | include Loopback
Loopback0                      10.255.1.1          YES  manual  up
up
! Loopback her zaman up/up durumdadir
```

19 show ip interface brief

ACİKLAMA Tum arayuzlerin IP adresi, durum (Status/Protocol) ve yapilandirma bilgilerini ozet tablo halinde gosterir.

KULLANİM AMACİ Hizli bir sekilde tum arayuzlerin durumunu kontrol etmek, IP atamalarini dogrulamak.

SYNTAX

```
Router# show ip interface brief [interface-type]
```

ORNEK KULLANİM

```
Router# show ip interface brief
```

SİK YAPİLAN HATA

'Status up' ama 'Protocol down' olabilir (ornegin karsi taraf kapaliysa). Her iki degerin de 'up' olmasi gerekir.

CİKTİ ORNEGİ

```
Router# show ip interface brief
Interface                  IP-Address      OK? Method Status
Protocol
GigabitEthernet0/0/0   unassigned      YES unset  administratively
down down
GigabitEthernet0/0/1       10.1.1.1                YES  manual  up
up
Serial0/1/0                 172.16.1.1          YES  manual  up
down
Loopback0                   1.1.1.1             YES  manual  up
```

20 show interfaces

ACİKLAMA

Belirtilen arayuzun veya tum arayuzlerin detayli istatistiklerini, hiz/duplex ayarlarini, hata sayilarini gosterir.

KULLANİM AMACİ

Arayuz performansini analiz etmek, hata sayilarini incelemek, fiziksel katman sorunlarini tespit etmek.

SYNTAX

```
Router# show interfaces [interface-type number] [stats]
```

ORNEK KULLANİM

```
Router# show interfaces GigabitEthernet0/0/1
```

SİK YAPİLAN HATA

Cikti cok uzundur. 'include', 'exclude', 'begin' filtreleri ile arama yaparak istenen bilgiye ulasin.

CİKTİ ORNEGİ

```
Router# show interfaces gi0/0/1 | include rate|duplex|error
  Full-duplex, 1000Mb/s, media type is RJ45
  5 minute input rate 24000 bits/sec, 45 packets/sec
  5 minute output rate 18000 bits/sec, 32 packets/sec
     0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored
     0 output errors, 0 collisions, 0 interface resets
```

3. IP Yonlendirme Komutlari

> Statik ve dinamik yonlendirme yapilandirmalari, route tablosu yonetimi ve yonlendirme protokolleri temel komutlarini icerir.

21 ip route

ACİKLAMA

Statik bir yonlendirme kaydi ekler. Hedef ag, sonraki atlamanin (next-hop) IP adresi veya cikis arayuzu belirtilir.

KULLANİM AMACİ

Dinamik yonlendirme kullanilmayan aglarda, varsayilan yol (default route) veya ozel yollar tanimlamak.

SYNTAX

```
Router(config)# ip route prefix mask {next-hop-ip | exit-interface} [admin-distance]
```

ORNEK KULLANİM

```
Router(config)# ip route 0.0.0.0 0.0.0.0 192.168.1.254
Router(config)# ip route 10.2.0.0 255.255.0.0 Serial0/0/0 150
```

SİK YAPİLAN HATA

Next-hop IP'si erisilebilir degilse route 'incomplete' olarak gorunur. Cikis arayuzu ile verilen route'lar sadece point-to-point baglantilarda calisir.

CİKTİ ORNEGİ

```
Router(config)# ip route 192.168.2.0 255.255.255.0 10.1.1.2
Router# show ip route static
S*    0.0.0.0/0 [1/0] via 192.168.1.254
S     192.168.2.0/24 [1/0] via 10.1.1.2
! S = Static, [1/0] = AD/Metric
```

22 show ip route

ACİKLAMA

IP yonlendirme tablosunu goruntuler. Her bir route'un kaynak protokolunu, yonetim mesafesini, metrik degerini ve sonraki atlamayi gosterir.

KULLANİM AMACİ

Yonlendirme tablosunu incelemek, belirli bir hedef icin yol varligini kontrol etmek, yonlendirme sorunlarini gidermek.

SYNTAX

```
Router# show ip route [connected | static | ospf | eigrp | bgp | summary]
```

ORNEK KULLANİM

```
Router# show ip route
Router# show ip route 192.168.1.0 255.255.255.0
```

SİK YAPİLAN HATA

'show ip route' ile gorunen tablo RIB'dir (Routing Information Base). FIB (Forwarding Information Base) icin 'show ip cef' kullanilmalidir.

CİKTİ ORNEGİ

```
Router# show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

Gateway of last resort is 192.168.1.254 to network 0.0.0.0

S*    0.0.0.0/0 [1/0] via 192.168.1.254
C     10.1.1.0/24 is directly connected, GigabitEthernet0/0/1
```

23 ip default-gateway

ACİKLAMA

Layer 2 switch'lerde IP iletisim icin varsayilan gecis (default gateway) adresini tanimlar. Router'larda kullanilmaz.

KULLANİM AMACİ

Switch'in kendi VLAN IP'sinden farkli aglara erismesi gerektiginde gateway tanimlamak.

SYNTAX

```
Switch(config)# ip default-gateway ip-address
```

ORNEK KULLANİM

```
Switch(config)# ip default-gateway 192.168.1.254
```

SİK YAPİLAN HATA

Router uzerinde 'ip default-gateway' calismaz. Router'larda 'ip route 0.0.0.0 0.0.0.0' kullanilir. Ayrica yanlis subnet'teki gateway yazilirsa erisim olmaz.

CİKTİ ORNEGİ

```
Switch(config)# ip default-gateway 192.168.10.1
Switch# show running-config | include default-gateway
ip default-gateway 192.168.10.1
Switch# ping 8.8.8.8 source vlan 10
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 8.8.8.8, timeout is 2 seconds:
Packet sent with a source address of 192.168.10.2
!!!!!
Success rate is 100 percent (5/5)
```

24 ip forwarding

ACİKLAMA

Bir cihazda IP yonlendirmeyi (routing) etkinlestirir. Varsayilan olarak Layer 3 cihazlarda aciktir. Layer 2 switch'lerde kapatilabilir.

KULLANİM AMACİ

Switch'te Layer 3 ozelliklerini (SVI routing, routed port) aktif hale getirmek.

SYNTAX

```
Router(config)# ip routing
```

ORNEK KULLANİM

```
Switch(config)# ip routing
```

SİK YAPİLAN HATA

Multilayer switch'te 'ip routing' kapaliysa VLAN'lar arasi routing calismaz. Bu komut unutulursa SVI'ler up/up olsa bile iletisim olmaz.

CİKTİ ORNEGİ

```
Switch(config)# ip routing
Switch# show ip route
Codes: L - local, C - connected, S - static

Gateway of last resort is not set

C     192.168.10.0/24 is directly connected, Vlan10
L     192.168.10.1/32 is directly connected, Vlan10
C     192.168.20.0/24 is directly connected, Vlan20
L     192.168.20.1/32 is directly connected, Vlan20
```

25 ip cef

ACİKLAMA

Cisco Express Forwarding (CEF) ozelligini etkinlestirir. Paket anahtarlamayi hizlandiran, RIB tabanli FIB olusturan bir mekanizmadir.

KULLANİM AMACİ

Yuksek performansli paket iletimi saglamak, process switching yerine optimized switching kullanmak.

SYNTAX

```
Router(config)# ip cef [distributed]
```

ORNEK KULLANİM

```
Router(config)# ip cef
```

SİK YAPİLAN HATA

Bazi platformlarda CEF varsayilan olarak aciktir. Kapatildiginda 'no ip cef' ile tekrar acmak gerekebilir. MPLS ve IPsec icin CEF zorunludur.

CİKTİ ORNEGİ

```
Router(config)# ip cef
Router# show ip cef
Prefix              Next Hop            Interface
0.0.0.0/0           192.168.1.254       GigabitEthernet0/0/0
10.1.1.0/24         attached            GigabitEthernet0/0/1
10.1.1.1/32         receive             GigabitEthernet0/0/1
192.168.1.0/24      attached            GigabitEthernet0/0/0
192.168.1.1/32      receive             GigabitEthernet0/0/0
```

26 show ip cef

ACİKLAMA

CEF Forwarding Information Base (FIB) tablosunu goruntuler. Gercek zamanli paket iletim kararlarinin alindigi tablodur.

KULLANİM AMACİ

Forwarding tablosunu kontrol etmek, belirli bir hedef icin cikis arayuzunu dogrulamak, forwarding sorunlarini gidermek.

SYNTAX

```
Router# show ip cef [prefix] [detail]
```

ORNEK KULLANİM

```
Router# show ip cef 192.168.1.0
```

SİK YAPİLAN HATA

'show ip route' ile gorunen yol FIB'de gorunmuyorsa CEF'te sorun vardir. CEF devre disi birakilmis olabilir.

CİKTİ ORNEGİ

```
Router# show ip cef 10.2.2.0 detail
10.2.2.0/24, version 27, epoch 0
0 packets, 0 bytes
  via 10.1.1.2, GigabitEthernet0/0/1, 0 dependencies
    next hop 10.1.1.2, GigabitEthernet0/0/1
    valid adjacency
```

27 ip prefix-list

ACİKLAMA

IP prefix'leri filtrelemek icin kullanilan listeler olusturur. ACL'lerden daha gucludur, subnet maskesi kontrolu yapabilir.

KULLANİM AMACİ

BGP route filtering, route redistribution filtering, policy-based routing icin prefix filtreleri olusturmak.

SYNTAX

```
Router(config)# ip prefix-list list-name [seq seq-value] {permit | deny} prefix/len [ge ge-value] [le le-value]
```

ORNEK KULLANİM

```
Router(config)# ip prefix-list BLOCK-RFC1918 seq 5 deny 10.0.0.0/8 le 32
Router(config)# ip prefix-list BLOCK-RFC1918 seq 10 deny 172.16.0.0/12 le 32
Router(config)# ip prefix-list BLOCK-RFC1918 seq 15 permit 0.0.0.0/0 le 32
```

SİK YAPİLAN HATA

Prefix-list'te implicit deny vardir. Tum trafige izin vermek icin son satirda 'permit 0.0.0.0/0 le 32' eklenmelidir.

CİKTİ ORNEGİ

```
Router# show ip prefix-list
ip prefix-list BLOCK-RFC1918: 3 entries
   seq 5 deny 10.0.0.0/8 le 32
   seq 10 deny 172.16.0.0/12 le 32
   seq 15 permit 0.0.0.0/0 le 32
```

28 maximum-paths

ACİKLAMA Bir yonlendirme protokolunun ayni hedefe gitmek icin kullanabilecegi maksimum es maliyetli yol (ECMP) sayisini belirler.

KULLANİM AMACİ Load balancing yapilacak paralel yol sayisini ayarlamak, bant genisligini optimize etmek.

SYNTAX

```
Router(config-router)# maximum-paths number
```

ORNEK KULLANİM

```
Router(config-router)# maximum-paths 4
```

SİK YAPİLAN HATA OSPF ve EIGRP'de varsayilan 4'tur. BGP'de varsayilan 1'dir (load balancing kapali). BGP'de multi-path icin ayrica 'maximum-paths eibgp' gerekebilir.

CİKTİ ORNEGİ

```
Router(config)# router ospf 1
Router(config-router)# maximum-paths 8
Router# show ip route 10.1.1.0
Routing entry for 10.1.1.0/24
  Known via "ospf 1", distance 110, metric 20, type intra area
  Last update from 10.2.2.2 on GigabitEthernet0/0/1, 00:01:23 ago
  Routing Descriptor Blocks:
    10.2.2.2, from 1.1.1.1, 00:01:23 ago, via GigabitEthernet0/0/1
    10.3.3.3, from 1.1.1.1, 00:01:23 ago, via GigabitEthernet0/0/2
  ! 2 es maliyetli yol yuk dengeliyor
```

29 passive-interface

ACİKLAMA

Belirtilen arayuz uzerinden yonlendirme guncellemelerinin gonderilmesini engeller ancak gelen guncellemeleri kabul eder.

KULLANİM AMACİ

Yonlendirme guncellemelerinin gereksiz yere LAN arayuzlerine gonderilmesini onlemek, guvenligi artirmak.

SYNTAX

```
Router(config-router)# passive-interface interface-type number
Router(config-router)# passive-interface default
```

ORNEK KULLANİM

```
Router(config)# router ospf 1
Router(config-router)# passive-interface GigabitEthernet0/0/0
```

SİK YAPİLAN HATA

'passive-interface default' verdikten sonra komsuluk kurulmasi gereken arayuzler icin 'no passive-interface' ile acilmalidir.

CİKTİ ORNEGİ

```
Router# show ip protocols
Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Passive Interface(s):
    GigabitEthernet0/0/0
    GigabitEthernet0/0/1
  Routing for Networks:
    10.1.1.0 0.0.0.255 area 0
```

30 redistribute

ACİKLAMA

Bir yonlendirme protokolunden ogrenilen yollari baska bir yonlendirme protokolune aktarir.

KULLANİM AMACİ

Farkli yonlendirme protokolleri arasinda iletisimi saglamak, statik/bagli yollarin dinamik protokole aktarimi.

SYNTAX

```
Router(config-router)# redistribute {static | connected | bgp | ospf | eigrp} [process-id] [metric] [route-map]
```

ORNEK KULLANİM

```
Router(config)# router ospf 1
Router(config-router)# redistribute connected subnets
Router(config-router)# redistribute eigrp 100 metric 100 subnets
```

SİK YAPİLAN HATA

Redistribution'da routing loop olusabilir. Route-map ve tag kullanarak kontrollu redistribution yapilmalidir.

CİKTİ ORNEGİ

```
Router# show ip route ospf
O   E2      172.16.0.0/16   [110/20]   via   10.1.1.2,   00:05:12,
GigabitEthernet0/0/1
! O E2 = OSPF External Type 2 ( redistributed route )
```

4. VLAN ve Trunking

Sanal LAN (VLAN) olusturma, trunk port yapilandirmasi, VTP ve inter-VLAN routing komutlarini icerir. Layer 2 ag segmentasyonu icin temel komutlardir.

31 vlan

ACİKLAMA Yeni bir VLAN olusturur ve VLAN yapilandirma moduna girer. VLAN ID (1-4094) ve istege bagli isim tanimlanabilir.

KULLANİM AMACİ Ag segmentasyonu icin VLAN'lar olusturmak, broadcast domain'leri ayirmak.

SYNTAX

```
Switch(config)# vlan vlan-id
Switch(config-vlan)# name vlan-name
```

ORNEK KULLANİM

```
Switch(config)# vlan 10
Switch(config-vlan)# name SALES
Switch(config-vlan)# exit
Switch(config)# vlan 20
Switch(config-vlan)# name ENGINEERING
```

SİK YAPİLAN HATA VLAN 1 varsayilandir ve yonetim icin kullanilmamalidir. VLAN 1002-1005 FDDI/Token Ring icin ayrilmistir, kullanilmamalidir.

CİKTİ ORNEGİ

```
Switch(config)# vlan 10
Switch(config-vlan)# name SALES
Switch(config-vlan)# do show vlan brief
VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------
-------------
1    default                               active    Gi0/0, Gi0/1,
Gi0/2
10   SALES                            active
20   ENGINEERING                      active
```

32 switchport mode access

ACİKLAMA

Bir switch portunu access (erisim) moduna ayarlar. Bu modda port sadece tek bir VLAN'a ait olur.

KULLANİM AMACİ

Endpoint cihazlarin (PC, printer, IP telefon) baglandigi portlari access moduna almak.

SYNTAX

```
Switch(config-if)# switchport mode access
```

ORNEK KULLANİM

```
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 10
```

SİK YAPİLAN HATA

DTP (Dynamic Trunking Protocol) nedeniyle port trunk olarak kalabilir. 'switchport nonegotiate' veya 'switchport mode access' ile DTP kapatilmalidir.

CİKTİ ORNEGİ

```
Switch(config-if)# switchport mode access
Switch# show interfaces gi0/1 switchport
Name: Gi0/1
Switchport: Enabled
Administrative Mode: static access
Operational Mode: static access
Administrative Trunking Encapsulation: negotiate
Access Mode VLAN: 10 (SALES)
```

33 switchport mode trunk

ACİKLAMA

Bir switch portunu trunk moduna ayarlar. Trunk portu uzerinden birden fazla VLAN'in trafigi etiketlenerek (tag) tasinir.

KULLANİM AMACİ

Switch-to-switch veya switch-to-router baglantilarinda birden fazla VLAN trafigini tek port uzerinden tasimak.

SYNTAX

```
Switch(config-if)# switchport mode trunk
Switch(config-if)# switchport trunk encapsulation {dot1q | isl}
```

ORNEK KULLANİM

```
Switch(config-if)# switchport trunk encapsulation dot1q
Switch(config-if)# switchport mode trunk
Switch(config-if)# switchport trunk allowed vlan 10,20,30
```

SİK YAPİLAN HATA

'allowed vlan' listesinde native VLAN (varsayilan 1) unutulursa trunk calismaz. Ayrica her iki switch'te de trunk encapsulation eslesmelidir.

CİKTİ ORNEGİ

```
Switch(config-if)# switchport mode trunk
Switch# show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Gi0/24      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Gi0/24      10,20,30,99

Port        Vlans allowed and active in management domain
Gi0/24      10,20,30,99
```

34 switchport trunk native vlan

ACİKLAMA Trunk uzerinde etiketlenmemis (untagged) trafigin ait oldugu VLAN'i (Native VLAN) belirler. Varsayilan VLAN 1'dir.

KULLANİM AMACİ Guvenlik icin Native VLAN'i varsayilandan degistirmek, VLAN hopping saldirilarini onlemek.

SYNTAX

```
Switch(config-if)# switchport trunk native vlan vlan-id
```

ORNEK KULLANİM

```
Switch(config-if)# switchport trunk native vlan 99
```

SİK YAPİLAN HATA Trunk'un iki ucundaki Native VLAN farkliysa Native VLAN mismatch olusur ve CDP hata mesaji verir. Her iki taraf da eslesmelidir.

CİKTİ ORNEGİ

```
Switch(config-if)# switchport trunk native vlan 99
Switch# show interfaces gi0/24 switchport | include Native
Trunking Native Mode VLAN: 99 (NATIVE)
! Guvenlik icin Native VLAN kullanilmayan bir VLAN olmalidir
```

35 switchport access vlan

ACİKLAMA Access modundaki bir portu belirtilen VLAN'a atar. Port uzerinden gelen trafig bu VLAN'a ait olur.

KULLANİM AMACİ Kullanici portlarini ilgili VLAN'lara atamak, ag segmentasyonunu fiziksel portlara uygulamak.

SYNTAX

```
Switch(config-if)# switchport access vlan vlan-id
```

ORNEK KULLANİM

```
Switch(config-if)# switchport access vlan 20
```

SİK YAPİLAN HATA Atanan VLAN daha once olusturulmamis bile olsa switch otomatik olarak olusturur. Ancak bu istenmeyen bir davranistir, once VLAN'i olusturun.

CİKTİ ORNEGİ

```
Switch(config-if)# switchport access vlan 20
Switch# show vlan id 20
VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------
-------------
20   ENGINEERING                      active    Gi0/1, Gi0/2,
Gi0/3
```

36 interface vlan

ACİKLAMA

Switched Virtual Interface (SVI) olusturur. Layer 3 switch'lerde VLAN'lar arasi routing icin kullanilan sanal arayuzdur.

KULLANİM AMACİ

VLAN'lar arasi yonlendirme (inter-VLAN routing) yapmak, switch uzerinden VLAN'lara gateway IP'si saglamak.

SYNTAX

```
Switch(config)# interface vlan vlan-id
Switch(config-if)# ip address ip-address subnet-mask
```

ORNEK KULLANİM

```
Switch(config)# interface vlan 10
Switch(config-if)# ip address 192.168.10.1 255.255.255.0
Switch(config-if)# no shutdown
```

SİK YAPİLAN HATA

SVI 'up/up' olmasi icin o VLAN'da en az bir fiziksel portun 'up' olmasi veya 'no shutdown' olmasi gerekir. Aksi halde 'up/down' kalir.

CİKTİ ORNEGİ

```
Switch(config)# interface vlan 10
Switch(config-if)# ip address 192.168.10.1 255.255.255.0
Switch(config-if)# no shutdown
Switch# show ip interface brief | include Vlan
Vlan1                  unassigned      YES unset  administratively
down down
Vlan10                 192.168.10.1    YES manual up
up
Vlan20                 192.168.20.1    YES manual up
up
```

37 show vlan brief

ACİKLAMA

Tum VLAN'larin kisa ozet listesini gosterir. VLAN ID, isim, durum ve port uyeliklerini icerir.

KULLANİM AMACİ

VLAN yapilandirmasini hizlica gozden gecirmek, port-VLAN eslesmelerini dogrulamak.

SYNTAX

```
Switch# show vlan brief
```

ORNEK KULLANİM

```
Switch# show vlan brief
```

SİK YAPİLAN HATA

'show vlan' komutu uzundur. Kisa ozet icin 'show vlan brief' tercih edilmelidir. Trunk portlar VLAN listesinde gorunmez.

CİKTİ ORNEGİ

```
Switch# show vlan brief
VLAN Name                              Status    Ports
---- --------------------------------- --------- -------------------
-------------
1    default                           active    Gi0/0
10   SALES                             active    Gi0/1, Gi0/2
20    ENGINEERING                           active     Gi0/3, Gi0/4,
Gi0/5
99   NATIVE                            active
1002 fddi-default                      act/unsup
```

38 show interfaces trunk

ACİKLAMA

Tum trunk portlarinin durumunu, encapsulation turunu, native VLAN ve izin verilen VLAN listesini gosterir.

KULLANİM AMACİ

Trunk baglantilarini dogrulamak, VLAN trafiginin hangi portlardan gectigini kontrol etmek.

SYNTAX

```
Switch# show interfaces trunk
```

ORNEK KULLANİM

```
Switch# show interfaces trunk
```

SİK YAPİLAN HATA

Trunk portlarda 'not-trunking' gorunuyorsa DTP sorunu olabilir. 'switchport mode trunk' ile zorlamak gerekir.

CİKTİ ORNEGİ

```
Switch# show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Gi0/24      on           802.1q         trunking      99
Gi0/23      on           802.1q         trunking      99

Port        Vlans allowed on trunk
Gi0/24      10,20,30,99
Gi0/23      10,20,30,99

Port        Vlans in spanning tree forwarding state
```

39 switchport port-security

ACİKLAMA

Access portu uzerinde port guvenligini etkinlestirir. MAC adresi tabanli erisim kontrolu saglar.

KULLANİM AMACİ

Yetkisiz cihazlarin aga baglanmasini onlemek, MAC flooding saldirilarina karsi koruma saglamak.

SYNTAX

```
Switch(config-if)# switchport port-security
Switch(config-if)# switchport port-security maximum value
Switch(config-if)# switchport port-security mac-address {sticky |
mac-addr}
Switch(config-if)# switchport port-security violation {protect |
restrict | shutdown}
```

ORNEK KULLANİM

```
Switch(config-if)# switchport port-security
Switch(config-if)# switchport port-security maximum 2
Switch(config-if)# switchport port-security mac-address sticky
Switch(config-if)# switchport port-security violation shutdown
```

SİK YAPİLAN HATA

Port-security sadece access modunda calisir. Trunk veya dynamic modda calismaz. 'err-disabled' durumuna duserse manuel acilmasi gerekir.

CİKTİ ORNEGİ

```
Switch# show port-security interface gi0/1
Port Security              : Enabled
Port Status                : Secure-up
Violation Mode             : Shutdown
Maximum MAC Addresses      : 2
Total MAC Addresses        : 2
Configured MAC Addresses   : 0
Sticky MAC Addresses       : 2
Last Source Address:Vlan   : aabb.ccdd.ee01:10
Security Violation Count   : 0
```

40 vtp mode

ACİKLAMA

VLAN Trunking Protocol (VTP) calisma modunu ayarlar. Server, Client, Transparent veya Off modlari mevcuttur.

KULLANİM AMACİ

VLAN bilgilerinin switch'ler arasinda otomatik senkronizasyonunu saglamak veya devre disi birakmak.

SYNTAX

```
Switch(config)# vtp mode {server | client | transparent | off}
```

ORNEK KULLANİM

```
Switch(config)# vtp mode transparent
Switch(config)# vtp domain COMPANY
Switch(config)# vtp version 2
```

SİK YAPİLAN HATA

VTP server'a yeni switch (higher revision number ile) baglanirsa tum VLAN'lar silinebilir. 'VTP pruning' kullaniminda dikkatli olunmalidir.

CİKTİ ORNEGİ

```
Switch# show vtp status
VTP Version                     : 2
Configuration Revision          : 5
Maximum VLANs supported locally : 255
Number of existing VLANs        : 12
VTP Operating Mode              : Server
VTP Domain Name                 : COMPANY
VTP Pruning Mode                : Disabled
VTP V2 Mode                     : Enabled
```

5. Spanning Tree Protocol (STP)

> Spanning Tree Protocol ailesi (STP/RSTP/MST) ile loop onleme, root bridge secimi, port rolleri ve STP guvenlik ozelliklerinin yapilandirilmasi komutlaridir.

41 spanning-tree mode

ACİKLAMA

Kullanilacak Spanning Tree protokolunu secer. pvst+, rapid-pvst, mst secenekleri mevcuttur.

KULLANİM AMACİ

Mevcut ag mimarisine uygun STP modunu secmek, hizli konverjans (RSTP) saglamak.

SYNTAX

```
Switch(config)# spanning-tree mode {pvst | rapid-pvst | mst}
```

ORNEK KULLANİM

```
Switch(config)# spanning-tree mode rapid-pvst
```

SİK YAPİLAN HATA

Farkli switch'lerde farkli STP modu kullanilirsa uyumsuzluk olusabilir. Tum switch'lerde ayni mod kullanilmalidir.

CİKTİ ORNEGİ

```
Switch(config)# spanning-tree mode rapid-pvst
Switch# show spanning-tree summary
Switch is in rapid-pvst mode
Root bridge for: VLAN0001, VLAN0010
Extended system ID           is enabled
Portfast Default             is disabled
PortFast BPDU Guard Default  is enabled
Portfast BPDU Filter Default is disabled
Loopguard Default            is disabled
```

42 spanning-tree vlan root

ACİKLAMA

Belirtilen VLAN icin switch'i root bridge olarak ayarlar. Bridge priority degerini otomatik olarak dusurur.

KULLANİM AMACİ

Root bridge secimini kontrol etmek, ag mimarisine uygun sekilde root'u merkez switch'e yerlestirmek.

SYNTAX

```
Switch(config)# spanning-tree vlan vlan-id root {primary | secondary} [diameter diameter]
```

ORNEK KULLANİM

```
Switch(config)# spanning-tree vlan 1,10,20 root primary
Switch(config)# spanning-tree vlan 1,10,20 root secondary
```

SİK YAPİLAN HATA

'root primary' bridge priority'yi 24576 yapar. Daha dusuk priority'ye sahip baska bir switch varsa root olamaz.

CİKTİ ORNEGİ

```
Switch(config)# spanning-tree vlan 10 root primary
Switch# show spanning-tree vlan 10
VLAN0010
  Spanning tree enabled protocol rstp
  Root ID    Priority    24586
             Address     aabb.cc00.0100
             This bridge is the root
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
  Bridge ID  Priority    24586   (priority 24576 sys-id-ext 10)
```

43 spanning-tree portfast

ACİKLAMA

Access portunda STP'nin Listening/Learning asamalarini atlayarak portu hemen Forwarding durumuna gecirir.

KULLANİM AMACİ

Endpoint cihazlarin baglandigi portlarda STP gecikmesini ortadan kaldirmak (DHCP, PXE boot gibi).

SYNTAX

```
Switch(config-if)# spanning-tree portfast
Switch(config)# spanning-tree portfast default
```

ORNEK KULLANİM

```
Switch(config-if)# spanning-tree portfast
%Warning: portfast should only be enabled on ports connected to a
single host.
```

SİK YAPİLAN HATA

Trunk veya uplink portlarda portfast acilirsa loop olusabilir. Mutlaka sadece access portlarda ve BPDU Guard ile birlikte kullanin.

CİKTİ ORNEGİ

```
Switch(config-if)# spanning-tree portfast
Warning: portfast should only be enabled on ports connected to
a single host. Connecting hubs, concentrators, switches, bridges,
etc...
to this interface when portfast is enabled, can cause temporary
bridging loops. Use with CAUTION.

%Portfast has been configured on GigabitEthernet0/1 but will only
have effect when the interface is in a non-trunking mode.
```

44 spanning-tree bpduguard enable

ACİKLAMA

PortFast etkin portlarda BPDU (Bridge Protocol Data Unit) alinmasi durumunda portu hemen kapatir (err-disabled).

KULLANİM AMACİ

Yetkisiz switch'lerin baglanmasini onlemek, STP tabanli saldirilari (rogue switch) engellemek.

SYNTAX

```
Switch(config-if)# spanning-tree bpduguard enable
Switch(config)# spanning-tree portfast bpduguard default
```

ORNEK KULLANİM

```
Switch(config-if)# spanning-tree portfast
Switch(config-if)# spanning-tree bpduguard enable
```

SİK YAPİLAN HATA

BPDU Guard ile kapatilan port 'err-disabled' durumuna gecer. 'shutdown' + 'no shutdown' ile acilmasi gerekir.

CİKTİ ORNEGİ

```
Switch# show spanning-tree summary
Root bridge for: VLAN0001
PortFast BPDU Guard Default is enabled

! Port BPDU alinca kapanir:
%SPANTREE-2-BLOCK_BPDUGUARD:     Received     BPDU     on     port
GigabitEthernet0/1
 with BPDU Guard enabled. Disabling port.
%PM-4-ERR_DISABLE:  bpduguard  error  detected  on  Gi0/1,  putting
Gi0/1 in err-disable state
```

45 spanning-tree guard root

ACİKLAMA

Root Guard, port uzerinden daha dusuk priority'li (superior) BPDU alinmasini engeller. Boylece root bridge secimi kontrol altinda tutulur.

KULLANİM AMACİ

Root bridge pozisyonunu korumak, yetkisiz switch'lerin root olmasini engellemek.

SYNTAX

```
Switch(config-if)# spanning-tree guard root
```

ORNEK KULLANİM

```
Switch(config-if)# spanning-tree guard root
```

SİK YAPİLAN HATA

Root Guard uplink portlarda kullanilmalidir. Root bridge'un kendisinde kullanilmasi gereksizdir. Root Guard ve Loop Guard ayni portta kullanilamaz.

CİKTİ ORNEGİ

```
Switch# show spanning-tree interface gi0/24 detail
 Port 24 (GigabitEthernet0/24) of VLAN0001 is root forwarding
   Port path cost: 4, Port priority: 128, Port Identifier: 128.24.
   Designated root has priority 24576, address aabb.cc00.1000
   Root guard is enabled on this port
   Number of transitions to forwarding state: 1
   Link type is point-to-point by default
```

46 show spanning-tree

ACİKLAMA

Spanning Tree protokolunun durumunu, root bridge bilgilerini, port rollerini ve maliyetlerini detayli olarak gosterir.

KULLANİM AMACİ

STP topolojisini analiz etmek, root bridge, designated port, blocked port'lari tespit etmek.

SYNTAX

```
Switch# show spanning-tree [vlan vlan-id] [summary | detail | root]
```

ORNEK KULLANİM

```
Switch# show spanning-tree vlan 10
Switch# show spanning-tree summary
```

SİK YAPİLAN HATA

Cikti her VLAN icin ayridir. Cok VLAN'li ortamlarda ozet gorunum icin 'summary' parametresi kullanilmalidir.

CİKTİ ORNEGİ

```
Switch# show spanning-tree vlan 10
VLAN0010
  Spanning tree enabled protocol rstp
  Root ID    Priority    24586
             Address     aabb.cc00.1000
             Cost        4
             Port        24 (GigabitEthernet0/24)
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
  Bridge ID  Priority    32778  (priority 32768 sys-id-ext 10)
```

47 spanning-tree cost

ACİKLAMA Bir portun STP maliyet (cost) degerini manuel olarak ayarlar. Dusuk maliyet = tercih edilen yol.

KULLANİM AMACİ Trafigin hangi yolu kullanacagini etkilemek, yedek yollarin tercih sirasini belirlemek.

SYNTAX

```
Switch(config-if)# spanning-tree [vlan vlan-id] cost value
```

ORNEK KULLANİM

```
Switch(config-if)# spanning-tree vlan 10 cost 10
```

SİK YAPİLAN HATA Cost degeri sadece o portun outbound trafigini etkiler. Root bridge'e giden toplam yol maliyetidir (cumulative).

CİKTİ ORNEGİ

```
Switch(config-if)# spanning-tree cost 10
Switch# show spanning-tree interface gi0/2

Port 2 (GigabitEthernet0/2) of VLAN0001 is designated forwarding
   Port path cost: 10, Port priority: 128, Port Identifier: 128.2.
```

48 spanning-tree vlan priority

ACİKLAMA Belirli bir VLAN icin switch'in bridge priority degerini manuel olarak ayarlar. Deger 4096'in katlari olmalidir.

KULLANİM AMACİ Root bridge secimini dogrudan kontrol etmek, belirli switch'leri root veya backup root olarak tanimlamak.

SYNTAX

```
Switch(config)# spanning-tree vlan vlan-id priority value
```

ORNEK KULLANİM

```
Switch(config)# spanning-tree vlan 10 priority 4096
Switch(config)# spanning-tree vlan 20 priority 8192
```

SİK YAPİLAN HATA Priority degeri 0, 4096, 8192, 12288, 16384, 20480, 24576, 28672, 32768, 36864, 40960, 45056, 49152, 53248, 57344, 61440 olabilir.

CİKTİ ORNEGİ

```
Switch(config)# spanning-tree vlan 10 priority 4096
Switch# show spanning-tree vlan 10 | include Priority
Bridge ID  Priority    4106   (priority 4096 sys-id-ext 10)
! sys-id-ext = VLAN ID (10) eklenir: 4096 + 10 = 4106
```

49 spanning-tree link-type

ACİKLAMA

RSTP'de portun baglanti tipini (point-to-point veya shared) manuel olarak ayarlar. Point-to-point dogrudan baglanti demektir.

KULLANİM AMACİ

RSTP konverjansini hizlandirmak, port tipini otomatik algilamaya birakmamak.

SYNTAX

```
Switch(config-if)# spanning-tree link-type {point-to-point | shared}
```

ORNEK KULLANİM

```
Switch(config-if)# spanning-tree link-type point-to-point
```

SİK YAPİLAN HATA

Full-duplex portlar otomatik olarak point-to-point olarak algilanir. Half-duplex'te shared varsayilir. Manuel mudahale genellikle gerekmez.

CİKTİ ORNEGİ

```
Switch(config-if)# spanning-tree link-type point-to-point
Switch# show spanning-tree interface gi0/24 detail
 Port 24 (GigabitEthernet0/24) of VLAN0001 is root forwarding
   Link type is point-to-point by default
   BPDU: sent 2341, received 452
```

50 errdisable recovery

ACİKLAMA

Err-disabled (hata nedeniyle devre disi kalmis) portlarin belirli bir sure sonra otomatik olarak kurtarilmasini saglar.

KULLANİM AMACİ

BPDU Guard, Port Security gibi nedenlerle kapanan portlarin otomatik acilmasini saglamak.

SYNTAX

```
Switch(config)# errdisable recovery cause {all | bpduguard | psecure-violation | ...}
Switch(config)# errdisable recovery interval seconds
```

ORNEK KULLANİM

```
Switch(config)# errdisable recovery cause bpduguard
Switch(config)# errdisable recovery cause psecure-violation
Switch(config)# errdisable recovery interval 300
```

SİK YAPİLAN HATA

'cause all' tum nedenleri kapsar ancak guvenlik icin spesifik nedenler belirtilmelidir. Cok dusuk interval saldirganin tekrar denemesine izin verebilir.

CİKTİ ORNEGİ

```
Switch# show errdisable recovery
ErrDisable Reason            Timer Status
-----------------            -------------
arp-inspection               Disabled
bpduguard                    Enabled
dhcp-rate-limit              Disabled
dtp-flap                     Disabled
psecure-violation            Enabled

Timer interval: 300 seconds
```

6. DHCP ve DNS

> DHCP sunucu/relay istemci yapilandirmasi, IP adres havuzu yonetimi ve DNS ozelliklerinin yapilandirilmasi komutlaridir.

51 ip dhcp pool

ACİKLAMA

Bir DHCP IP adres havuzu (pool) olusturur ve DHCP yapilandirma moduna girer. Istemcilere IP ataması yapilir.

KULLANİM AMACİ

Router veya switch uzerinde DHCP sunucusu olarak calismak, otomatik IP dagitimi yapmak.

SYNTAX

```
Router(config)# ip dhcp pool pool-name
Router(dhcp-config)# network network-address subnet-mask
Router(dhcp-config)# default-router ip-address
Router(dhcp-config)# dns-server ip-address
```

ORNEK KULLANİM

```
Router(config)# ip dhcp pool VLAN10-POOL
Router(dhcp-config)# network 192.168.10.0 255.255.255.0
Router(dhcp-config)# default-router 192.168.10.1
Router(dhcp-config)# dns-server 8.8.8.8 8.8.4.4
Router(dhcp-config)# domain-name company.local
```

SİK YAPİLAN HATA

DHCP havuzu olusturulurken 'ip dhcp excluded-address' ile gateway ve statik IP'ler haric tutulmazsa cakisma olusabilir.

CİKTİ ORNEGİ

```
Router# show ip dhcp pool
Pool VLAN10-POOL :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 254
 Leased addresses               : 12
 Excluded addresses             : 3
 Pending event                  : none

 1 subnet is currently in the pool :
```

52 ip dhcp excluded-address

ACİKLAMA

DHCP havuzundan haric tutulacak (rezerve) IP adreslerini veya araligini tanimlar.

KULLANİM AMACİ

Gateway, sunucular, yazicilar gibi statik IP kullanan cihazlar icin DHCP dagitim araligindan cikarmak.

SYNTAX

```
Router(config)# ip dhcp excluded-address low-address [high-address]
```

ORNEK KULLANİM

```
Router(config)# ip dhcp excluded-address 192.168.10.1 192.168.10.10
Router(config)# ip dhcp excluded-address 192.168.10.254
```

SİK YAPİLAN HATA

Excluded-address global olarak tanimlanir, havuz icinde degil. Birden fazla havuz varsa tumu icin gecerlidir.

CİKTİ ORNEGİ

```
Router# show ip dhcp excluded-address
IP address      Client ID/
                Hardware addr/
                User class
192.168.10.1
192.168.10.2
...
192.168.10.10
192.168.10.254
```

53 ip helper-address

ACİKLAMA

DHCP broadcast isteklerini (UDP 67/68) belirtilen IP adresine unicast olarak yonlendirir. DHCP relay islevi gorur.

KULLANİM AMACİ

Farkli subnet'teki istemcilerin uzaktaki DHCP sunucusuna erismesini saglamak.

SYNTAX

```
Router(config-if)# ip helper-address dhcp-server-ip
```

ORNEK KULLANİM

```
Router(config-if)# ip helper-address 10.1.1.10
```

SİK YAPİLAN HATA

ip helper-address istemcilerin bulundugu arayuze (SVI veya L3 interface) yazilmalidir, DHCP sunucusunun arayuzune degil.

CİKTİ ORNEGİ

```
Router# show ip interface vlan 10 | include Helper
  Helper address is 10.1.1.10
Router# show ip dhcp relay
Information about the relay agent
  Helper addresses are configured on the following interfaces:
    Vlan10: 10.1.1.10
```

54 show ip dhcp binding

ACİKLAMA DHCP sunucusu tarafindan kiralana (lease) tum IP adreslerini, MAC adreslerini ve lease suresini listeler.

KULLANİM AMACİ Hangi IP'nin kime atandigini kontrol etmek, IP cakismalarini tespit etmek.

SYNTAX

```
Router# show ip dhcp binding [ip-address]
```

ORNEK KULLANİM

```
Router# show ip dhcp binding
```

SİK YAPİLAN HATA Binding tablosu sadece DHCP sunucusu uzerinde gorunur. Relay uzerinde degil. Lease suresi dolan IP'ler otomatik silinir.

CİKTİ ORNEGİ

```
Router# show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address          Client-ID/                    Lease expiration
Type
                    Hardware address/
                    User name
192.168.10.11     0060.5c5e.7a8b              Mar 15 2025 08:30 AM
Automatic
192.168.10.12     00e0.a3b1.c2d3              Mar 15 2025 08:32 AM
Automatic
```

55 ip name-server

ACİKLAMA Cihazin kendisinin kullanacagi DNS sunucularini tanimlar. Cihaz uzerinden isim cozumlemesi yapilir.

KULLANİM AMACİ Cihazin NTP sunucusu, syslog sunucusu gibi hedefleri isimle erismesini saglamak.

SYNTAX

```
Router(config)# ip name-server ip-address [ip-address2 ...]
```

ORNEK KULLANİM

```
Router(config)# ip name-server 8.8.8.8 8.8.4.4
```

SİK YAPİLAN HATA ip name-server cihazin kendi DNS'i icindir, DHCP istemcilerine dagitilan DNS degildir. Onun icin 'dns-server' komutu kullanilir.

CİKTİ ORNEGİ

```
Router# show running-config | include name-server
ip name-server 8.8.8.8 8.8.4.4
Router# ping ntp.pool.org
Translating "ntp.pool.org"...domain server (8.8.8.8) [OK]
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 162.159.200.1, timeout is 2
seconds:
!!!!!
```

56 ip domain-name

ACİKLAMA Cihazin varsayilan domain adini tanimlar. SSH sertifikasi olustururken ve DNS tamamlama icin kullanilir.

KULLANİM AMACİ SSH RSA key olustururken domain adi eklemek, DNS cozumlemesinde kullanmak.

SYNTAX

```
Router(config)# ip domain-name domain-name
```

ORNEK KULLANİM

```
Router(config)# ip domain-name company.local
```

SİK YAPİLAN HATA ip domain-name tanimlanmadan 'crypto key generate rsa' calistirilirsa SSH calismaz. Once domain-name tanimlanmalidir.

CİKTİ ORNEGİ

```
Router# show running-config | include domain
ip domain-name company.local
ip name-server 8.8.8.8
Router(config)# crypto key generate rsa
The name for the keys will be: Router.company.local
```

57 ip dns server

ACİKLAMA

Router'i DNS sunucusu olarak calistirir. Yerel DNS kayitlari tanimlanabilir veya istekleri baska DNS'e yonlendirir.

KULLANİM AMACİ

Kucuk ofislerde basit DNS sunucusu olarak calistirmak, yerel isim cozumlemesi saglamak.

SYNTAX

```
Router(config)# ip dns server
Router(config)# ip host name ip-address
```

ORNEK KULLANİM

```
Router(config)# ip dns server
Router(config)# ip host server1 192.168.1.10
Router(config)# ip host printer-hq 192.168.1.20
```

SİK YAPİLAN HATA

ip dns server calismasi icin 'ip domain-lookup' aktif olmalidir (varsayilan olarak aktiftir). 'no ip domain-lookup' verildiyse calismaz.

CİKTİ ORNEGİ

```
Router# show ip dns server
DNS server is enabled
Router# show ip hosts
Default domain is company.local
Name/address lookup uses domain service
Name servers are 8.8.8.8

Host              Address
server1           192.168.1.10
printer-hq        192.168.1.20
```

58 ip dhcp relay information option

ACİKLAMA

DHCP relay agent'inin Option 82 (Agent Information Option) eklemesini saglar. DHCP istemcisinin hangi switch/port'tan geldigini belirtir.

KULLANİM AMACİ

DHCP sunucusuna istemcinin fiziksel konum bilgisini iletmek, guvenli IP dagitimi saglamak.

SYNTAX

```
Router(config)# ip dhcp relay information option
```

ORNEK KULLANİM

```
Router(config)# ip dhcp relay information option
Router(config)# ip dhcp relay information option insert
```

SİK YAPİLAN HATA

Bazi DHCP sunuculari Option 82 olmayan istekleri reddeder. Ayrica switch'lerde 'ip dhcp snooping information option allow-untrusted' gerekebilir.

CİKTİ ORNEGİ

```
Router# show ip dhcp relay information trusted-sources
Trusted sources are not configured.
! Option 82 aktif:
Router# show running-config | include relay
ip dhcp relay information option
```

59 clear ip dhcp binding

ACİKLAMA

DHCP binding tablosundaki IP-MAC eslesmelerini temizler. Belirli IP veya tum binding'leri siler.

KULLANİM AMACİ

DHCP IP cakismalarini cozmek, gecersiz lease kayitlarini temizlemek, IP degisikligi sonrasi tabloyu yenilemek.

SYNTAX

```
Router# clear ip dhcp binding {ip-address | *}
```

ORNEK KULLANİM

```
Router# clear ip dhcp binding *
Router# clear ip dhcp binding 192.168.10.11
```

SİK YAPİLAN HATA

'*' ile tum binding'ler silinir, istemciler yeni IP almak zorunda kalir. Uretim ortaminda dikkatli kullanilmalidir.

CİKTİ ORNEGİ

```
Router# clear ip dhcp binding *
Router# show ip dhcp binding
! Tumu silindi, bos liste
Router# clear ip dhcp binding 192.168.10.11
Router# show ip dhcp binding
! Sadece 192.168.10.11 silindi, digerleri duruyor
```

60 service dhcp

ACİKLAMA

Cihaz uzerindeki DHCP sunucu ve relay hizmetlerini etkinlestirir. Varsayilan olarak aktiftir.

KULLANİM AMACİ

DHCP sunucusu veya relay olarak calismak veya bu hizmeti devre disi birakmak.

SYNTAX

```
Router(config)# service dhcp
Router(config)# no service dhcp
```

ORNEK KULLANİM

```
Router(config)# no service dhcp
```

SİK YAPİLAN HATA

DHCP calismiyorsa once 'show running-config | include service dhcp' ile devre disi olup olmadigini kontrol edin.

CİKTİ ORNEGİ

```
Router(config)# no service dhcp
% DHCP server and relay services are disabled.
Router(config)# service dhcp
% DHCP server and relay services are enabled.
```

7. ACL ve NAT

> Erisim kontrol listeleri (ACL) ile trafik filtreleme, Network Address Translation (NAT) ile IP adres donusumu ve PAT yapilandirmasi komutlaridir.

61 access-list (standard)

ACİKLAMA

Standart ACL (1-99, 1300-1999) olusturur. Sadece kaynak IP adresine gore filtreleme yapar.

KULLANİM AMACİ

Basit IP bazli erisim kontrolu yapmak, belirli IP'lerin trafigine izin vermek veya engellemek.

SYNTAX

```
Router(config)# access-list number {permit | deny} source [source-wildcard] [log]
```

ORNEK KULLANİM

```
Router(config)# access-list 10 permit 192.168.1.0 0.0.0.255
Router(config)# access-list 10 deny any log
```

SİK YAPİLAN HATA

Standart ACL hedefe yakin uygulanir. Kaynak IP'ye gore filtreler ancak hedef ve port bilgisine erisemez.

CİKTİ ORNEGİ

```
Router# show access-lists 10
Standard IP access list 10
    10 permit 192.168.1.0, wildcard bits 0.0.0.255 (25 matches)
    20 deny   any log (5 matches)
```

62 access-list (extended)

ACİKLAMA

Extended ACL (100-199, 2000-2699) olusturur. Kaynak/hedef IP, protokol, port numarasi gibi detayli filtreleme yapar.

KULLANİM AMACİ

Detayli trafik filtreleme yapmak, belirli uygulamalara erisimi kontrol etmek.

SYNTAX

```
Router(config)# access-list number {permit | deny} protocol source source-wildcard [operator port] destination destination-wildcard [operator port] [established] [log]
```

ORNEK KULLANİM

```
Router(config)# access-list 101 permit tcp 192.168.1.0 0.0.0.255 any eq 80
Router(config)# access-list 101 permit tcp 192.168.1.0 0.0.0.255 any eq 443
Router(config)# access-list 101 deny ip any any log
```

SİK YAPİLAN HATA

ACL'ler sirali islenir, ilk eslesme uygulanir. 'permit' kurallarindan sonra 'deny any' eklenmezse tum trafige izin verilir (implicit deny).

CİKTİ ORNEGİ

```
Router# show access-lists 101
Extended IP access list 101
    10 permit tcp 192.168.1.0 0.0.0.255 any eq www (120 matches)
    20 permit tcp 192.168.1.0 0.0.0.255 any eq 443 (85 matches)
    30 deny ip any any log (12 matches)
```

63 ip access-group

ACİKLAMA Olusturulan ACL'yi bir arayuze gelen (in) veya giden (out) trafik icin uygular.

KULLANİM AMACİ ACL'yi aktif hale getirmek, belirli arayuz uzerinde trafik filtrelemeyi uygulamak.

SYNTAX

```
Router(config-if)# ip access-group {number | name} {in | out}
```

ORNEK KULLANİM

```
Router(config-if)# ip access-group 101 in
```

SİK YAPİLAN HATA

ACL 'in' yerine 'out' olarak yanlis yonde uygulanirsa beklenmeyen sonuclar olusur. Trafigin yonunu dogru analiz edin.

CİKTİ ORNEGİ

```
Router# show ip interface GigabitEthernet0/0/0 | include access
  Inbound  access list is 101
  Outbound access list is not set
Router# show access-lists 101
Extended IP access list 101
    10 permit tcp 192.168.1.0 0.0.0.255 any eq www (245 matches)
    20 deny ip any any log (18 matches)
```

64 ip nat inside / outside

ACİKLAMA

NAT icin bir arayuzun inside (ic) veya outside (dis) oldugunu belirtir. NAT islemi bu tanimlamalara gore yapilir.

KULLANİM AMACİ

NAT isleminde hangi arayuzun dahili aga, hangisinin dis aga baktigini tanimlamak.

SYNTAX

```
Router(config-if)# ip nat inside
Router(config-if)# ip nat outside
```

ORNEK KULLANİM

```
Router(config)# interface GigabitEthernet0/0/0
Router(config-if)# ip nat inside
Router(config)# interface GigabitEthernet0/0/1
Router(config-if)# ip nat outside
```

SİK YAPİLAN HATA

'ip nat inside' ve 'ip nat outside' tanimlanmadan 'ip nat source' komutlari calismaz. Her iki yon de tanimlanmalidir.

CİKTİ ORNEGİ

```
Router# show ip interface brief
Interface                  IP-Address      OK? Method  Status
Protocol
GigabitEthernet0/0/0       192.168.1.1     YES manual  up
up
GigabitEthernet0/0/1       203.0.113.1     YES manual  up
up
Router# show running-config interface g0/0/0 | include nat
 ip nat inside
Router# show running-config interface g0/0/1 | include nat
```

65 ip nat inside source

ACİKLAMA

Iç (inside) IP adreslerini dis (outside) IP adresine NAT donusumunu tanimlar. Static NAT, Dynamic NAT ve PAT (overload) destekler.

KULLANİM AMACİ

Ic agdan dis aga cikista IP adreslerini donusturmek, birden fazla ic IP'yi tek dis IP'ye (PAT) eslemek.

SYNTAX

```
Router(config)# ip nat inside source list access-list-number
interface type overload
Router(config)# ip nat inside source static local-ip global-ip
```

ORNEK KULLANİM

```
Router(config)# access-list 1 permit 192.168.1.0 0.0.0.255
Router(config)# ip nat inside source list 1 interface
GigabitEthernet0/0/1 overload
```

SİK YAPİLAN HATA

'overload' eklenmezse sadece Dynamic NAT yapilir ve her ic IP icin ayri dis IP gerekir. Internet erisimi icin overload (PAT) zorunludur.

CİKTİ ORNEGİ

```
Router# show ip nat translations
Pro  Inside  global          Inside  local           Outside  local
Outside global
tcp     203.0.113.1:51234         192.168.1.10:44321     8.8.8.8:443
8.8.8.8:443
tcp     203.0.113.1:51235         192.168.1.11:51234     1.1.1.1:443
1.1.1.1:443
udp   203.0.113.1:51236       192.168.1.10:53            8.8.8.8:53
8.8.8.8:53
! Inside global = PAT IP:port, Inside local = gercek IP:port
```

66 show ip nat translations

ACİKLAMA

Aktif NAT cevirme tablosunu gosterir. Ic IP, dis IP, protokol ve port numaralarini listeler.

KULLANİM AMACİ

NAT isleminin calistigini dogrulamak, aktif cevirmeleri izlemek, sorun gidermek.

SYNTAX

```
Router# show ip nat translations [verbose]
```

ORNEK KULLANİM

```
Router# show ip nat translations
```

SİK YAPİLAN HATA

NAT tablosu bos gorunuyorsa, ACL yanlis tanimlanmis, inside/outside arayuzler atanmamis veya trafigin yonu yanlis olabilir.

CİKTİ ORNEGİ

```
Router# show ip nat translations
Pro  Inside  global          Inside  local           Outside  local
Outside global
--- 203.0.113.1        192.168.1.10       ---                ---
tcp   203.0.113.1:1024       192.168.1.10:49152   93.184.216.34:443
93.184.216.34:443
udp    203.0.113.1:1025           192.168.1.11:57521    8.8.8.8:53
8.8.8.8:53
```

67 ip access-list extended (named)

ACİKLAMA

Isimlendirilmis (named) extended ACL olusturur. Numarali ACL'lere gore daha esnek duzenleme imkani sunar.

KULLANİM AMACİ

Kolay hatirlanan isimlerle ACL yonetimi yapmak, kurallari sira numaralariyla duzenlemek.

SYNTAX

```
Router(config)# ip access-list extended name
Router(config-ext-nacl)#  [sequence-number]  {permit  |  deny}
protocol source destination [log]
```

ORNEK KULLANİM

```
Router(config)# ip access-list extended BLOCK-TELNET
Router(config-ext-nacl)# deny tcp any any eq 23 log
Router(config-ext-nacl)# permit ip any any
```

SİK YAPİLAN HATA

Named ACL'de kural eklerken sequence number belirtilmezse en sona eklenir. Araya kural eklemek icin 'sequence-number' kullanilmalidir.

CİKTİ ORNEGİ

```
Router# show ip access-list BLOCK-TELNET
Extended IP access list BLOCK-TELNET
    10 deny tcp any any eq telnet log (15 matches)
    20 permit ip any any (452 matches)
```

68 clear ip nat translation

ACİKLAMA

NAT cevirme tablosundaki kayitlari temizler. Belirli bir cevirme veya tumu silinebilir.

KULLANİM AMACİ

NAT sorunlarini cozmek, IP degisikliklerinden sonra tabloyu temizlemek, NAT bellek sizintisini onlemek.

SYNTAX

```
Router# clear ip nat translation {* | inside global-ip local-ip | outside local-ip global-ip}
```

ORNEK KULLANİM

```
Router# clear ip nat translation *
Router# clear ip nat translation inside 203.0.113.1 192.168.1.10
```

SİK YAPİLAN HATA

'*' ile tum translation'lar silinir ve aktif baglantilar kesilir. Uretim ortaminda dikkatli kullanilmalidir.

CİKTİ ORNEGİ

```
Router# clear ip nat translation *
Router# show ip nat translations
! Tumu silindi, bos liste
```

69 ip nat pool

ACİKLAMA

Dynamic NAT icin kullanilacak dis (global) IP adres havuzunu tanimlar.

KULLANİM AMACİ

PAT yerine gercek Dynamic NAT yapmak, ic IP'leri havuzdaki dis IP'lere birebir eslemek.

SYNTAX

```
Router(config)# ip nat pool pool-name start-ip end-ip {netmask subnet-mask | prefix-length length}
```

ORNEK KULLANİM

```
Router(config)# ip nat pool PUBLIC-POOL 203.0.113.10 203.0.113.20 netmask 255.255.255.0
Router(config)# access-list 1 permit 192.168.1.0 0.0.0.255
Router(config)# ip nat inside source list 1 pool PUBLIC-POOL
```

SİK YAPİLAN HATA

Havuzdaki IP sayisi ic agdaki cihaz sayisindan azsa, fazla cihaz NAT yapamaz. PAT icin 'overload' eklenmelidir.

CİKTİ ORNEGİ

```
Router# show ip nat pool
Name               Total      Available  Low        High
PUBLIC-POOL            11             11             203.0.113.10
203.0.113.20
```

70 ip access-list resequence

ACİKLAMA

Named ACL'deki kural sira numaralarini yeniden duzenler. Araliga yeni kurallar eklemek icin kullanilir.

KULLANİM AMACİ

ACL kurallarinin sira numaralarini yeniden duzenlemek, yeni kurallar icin bosluk yaratmak.

SYNTAX

```
Router(config)# ip access-list resequence access-list-name
starting-sequence-number increment
```

ORNEK KULLANİM

```
Router(config)# ip access-list resequence BLOCK-TELNET 10 10
```

SİK YAPİLAN HATA

Resequence sonrasi ACL uygulanan arayuzlerde kisa bir kesinti olabilir. Bakim penceresinde yapilmalidir.

CİKTİ ORNEGİ

```
Router# show ip access-list BLOCK-TELNET
Extended IP access list BLOCK-TELNET
    10 deny tcp any any eq telnet log
    20 permit ip any any
Router(config)# ip access-list resequence BLOCK-TELNET 10 10
! Kural sirasi 10, 20, 30... seklinde duzenlendi
```

8. OSPF Yonlendirme Protokolu

> Open Shortest Path First (OSPF) link-state yonlendirme protokolunun temel ve ileri seviye yapilandirma komutlaridir.

71 router ospf

ACİKLAMA

OSPF yonlendirme surecini baslatir ve router yapilandirma moduna girer. Process ID sadece lokal anlamlidir.

KULLANİM AMACİ

OSPF yonlendirmesini etkinlestirmek, OSPF parametrelerini yapilandirmak.

SYNTAX

```
Router(config)# router ospf process-id
```

ORNEK KULLANİM

```
Router(config)# router ospf 1
Router(config-router)#
```

SİK YAPİLAN HATA

Farkli router'larda farkli process ID kullanilabilir. Process ID'lerin eslesmesi gerekmez. Area ID'lerin eslesmesi sarttir.

CİKTİ ORNEGİ

```
Router(config)# router ospf 1
Router(config-router)#
! (config-router) = Router Configuration Mode
```

72 network (OSPF)

ACİKLAMA

Belirtilen ag araligindaki arayuzleri OSPF'e dahil eder. Wildcard maskesi kullanilir ve area atanir.

KULLANİM AMACİ

Hangi arayuzlerin OSPF komsulugu kuracagini ve hangi area'ya ait olacagini belirlemek.

SYNTAX

```
Router(config-router)# network ip-address wildcard-mask area area-id
```

ORNEK KULLANİM

```
Router(config-router)# network 10.1.1.0 0.0.0.255 area 0
Router(config-router)# network 192.168.1.0 0.0.0.255 area 10
```

SİK YAPİLAN HATA

Wildcard mask subnet mask degildir. Subnet mask'in tersidir. Ornegin /24 subnet mask 255.255.255.0 ise wildcard 0.0.0.255'tir.

CİKTİ ORNEGİ

```
Router# show ip ospf interface brief
Interface    PID   Area            IP Address/Mask    Cost  State
Nbrs F/C
Gi0/0/1      1     0                10.1.1.1/24         1     DR
1/1
Gi0/0/2      1     10               192.168.1.1/24      1     BDR
1/1
Lo0          1     0                1.1.1.1/32          1     P2P
0/0
```

73 router-id

ACİKLAMA

OSPF router ID'sini manuel olarak belirler. En yuksek loopback IP'si varsayilan olarak secilir. Manuel router-id onceliklidir.

KULLANİM AMACİ

OSPF router ID'sini sabitlemek, degismesini onlemek, kolay tanimlanabilir ID kullanmak.

SYNTAX

```
Router(config-router)# router-id ip-address
```

ORNEK KULLANİM

```
Router(config-router)# router-id 1.1.1.1
```

SİK YAPİLAN HATA

Router ID degistirildiginde OSPF sureci yeniden baslar ve komsuluklar kesilir. clear ip ospf process komutu gerekir.

CİKTİ ORNEGİ

```
Router(config-router)# router-id 1.1.1.1
Reload or use "clear ip ospf process" command, for this to take
effect
Router# clear ip ospf process
Reset ALL OSPF processes? [no]: yes
Router# show ip ospf | include Router ID
 Router ID: 1.1.1.1
```

74 show ip ospf neighbor

ACİKLAMA

OSPF komsularinin (neighbor) durumunu, router ID, priority, durum (FULL, 2WAY, INIT) ve dead timer bilgilerini gosterir.

KULLANİM AMACİ

OSPF komsuluklarinin calistigini dogrulamak, komsuluk sorunlarini gidermek.

SYNTAX

```
Router# show ip ospf neighbor [detail]
```

ORNEK KULLANİM

```
Router# show ip ospf neighbor
```

SİK YAPİLAN HATA

'FULL' disinda bir durum (INIT, 2WAY, EXSTART) uzun sure kaliyorsa MTU uyusmazligi, area mismatch veya authentication sorunu olabilir.

CİKTİ ORNEGİ

```
Router# show ip ospf neighbor
Neighbor ID       Pri    State                Dead Time    Address
Interface
2.2.2.2             1    FULL/BDR           00:00:35     10.1.1.2
GigabitEthernet0/0/1
3.3.3.3             1    FULL/DR            00:00:38     192.168.1.2
GigabitEthernet0/0/2
! FULL = Komsuluk tamamlandi, DR/BDR rolleri belirlendi
```

75 show ip ospf database

ACİKLAMA

OSPF link-state database (LSDB) icerigini goruntuler. Router LSA, Network LSA, Summary LSA gibi kayitlari icerir.

KULLANİM AMACİ

OSPF database'ini incelemek, LSA tiplerini analiz etmek, yonlendirme sorunlarini gidermek.

SYNTAX

```
Router# show ip ospf database [router | network | summary | external | nssa-external]
```

ORNEK KULLANİM

```
Router# show ip ospf database
Router# show ip ospf database router 1.1.1.1
```

SİK YAPİLAN HATA

Cok buyuk network'lerde database cok buyuk olabilir. Spesifik LSA tiplerini filtreleyerek incelemek daha verimlidir.

CİKTİ ORNEGİ

```
Router# show ip ospf database
            OSPF Router with ID (1.1.1.1) (Process ID 1)
                Router Link States (Area 0)
Link ID          ADV Router       Age          Seq#        Checksum
Link count
1.1.1.1         1.1.1.1         482         0x80000003 0x00A1B2 2
2.2.2.2         2.2.2.2         356         0x80000002 0x00C3D4 2

                Net Link States (Area 0)
Link ID         ADV Router      Age         Seq#       Checksum
```

76 ip ospf cost

ACİKLAMA

Bir arayuzun OSPF maliyet (cost) degerini manuel olarak ayarlar. Dusuk maliyet = tercih edilen yol.

KULLANİM AMACİ

Trafigin hangi yolu kullanacagini etkilemek, bant genisligine gore yol tercihini ayarlamak.

SYNTAX

```
Router(config-if)# ip ospf cost value
```

ORNEK KULLANİM

```
Router(config-if)# ip ospf cost 100
```

SİK YAPİLAN HATA

Varsayilan cost = Reference Bandwidth / Interface Bandwidth. 10Gbps+ arayuzlerde varsayilan cost 1 olabilir. 'auto-cost reference-bandwidth' guncellenmelidir.

CİKTİ ORNEGİ

```
Router(config-if)# ip ospf cost 50
Router# show ip ospf interface GigabitEthernet0/0/1
GigabitEthernet0/0/1 is up, line protocol is up
  Internet Address 10.1.1.1/24, Area 0
   Process ID 1, Router ID 1.1.1.1, Network Type BROADCAST, Cost:
50
```

77 ip ospf priority

ACİKLAMA OSPF DR/BDR seciminde kullanilan priority degerini ayarlar. En yuksek priority = DR secilir (0 = aday olmaz).

KULLANİM AMACİ DR/BDR rollerini kontrol etmek, istenen switch/router'i DR yapmak.

SYNTAX

```
Router(config-if)# ip ospf priority number
```

ORNEK KULLANİM

```
Router(config-if)# ip ospf priority 100
```

SİK YAPİLAN HATA

Priority degisikligi aninda etki etmez. Mevcut DR/BDR degismez. Clear ip ospf process veya interface reset gerekir.

CİKTİ ORNEGİ

```
Router# show ip ospf interface GigabitEthernet0/0/1
GigabitEthernet0/0/1 is up, line protocol is up
   Internet Address 10.1.1.1/24, Area 0, Attached via Network
Statement
  Process ID 1, Router ID 1.1.1.1, Network Type BROADCAST, Cost: 1
  Topology-MTID    Cost    Disabled    Shutdown      Topology Name
        0           1       no          no            Base
  Transmit Delay is 1 sec, State DR, Priority 100
  Designated Router (ID) 1.1.1.1, Interface address 10.1.1.1
```

78 area authentication

ACİKLAMA

OSPF area uzerinde authentication (kimlik dogrulama) etkinlestirir. Null, plain-text veya MD5 secenekleri vardir.

KULLANİM AMACİ

OSPF komsuluklarini yetkisiz router'lardan korumak, routing security saglamak.

SYNTAX

```
Router(config-router)# area area-id authentication [message-digest]
```

ORNEK KULLANİM

```
Router(config-router)# area 0 authentication message-digest
Router(config-if)# ip ospf message-digest-key 1 md5 Cisco123
```

SİK YAPİLAN HATA

Tum router'larda ayni area'da ayni authentication tipi ve key kullanilmalidir. Uyumsuzluk komsulugun kopmasina neden olur.

CİKTİ ORNEGİ

```
Router# show ip ospf interface GigabitEthernet0/0/1 | include auth
  Message digest authentication enabled
    Youngest key id is 1
Router# show ip ospf | include Area
    Area BACKBONE(0)
        Number of interfaces in this area is 3
        Area has message digest authentication
```

79 default-information originate

ACİKLAMA

Router'daki varsayilan yolun (default route) OSPF ile diger router'lara dagitilmasini saglar.

KULLANİM AMACİ

Internet cikisi olan router'in diger router'lara default route ogretmesini saglamak.

SYNTAX

```
Router(config-router)# default-information originate [always] [metric value] [metric-type type]
```

ORNEK KULLANİM

```
Router(config-router)# default-information originate always metric 10
```

SİK YAPİLAN HATA

'always' kullanilmazsa router'da aktif bir default route olmasi gerekir. Yoksa OSPF default route yayinlamaz.

CİKTİ ORNEGİ

```
Router# show ip route ospf | include gateway
Gateway of last resort is 203.0.113.1 to network 0.0.0.0

O*E2        0.0.0.0/0    [110/10]    via    10.1.1.1,    00:15:23,
GigabitEthernet0/0/1
! O*E2 = OSPF External Type 2 default route
```

80 area stub / totally stubby / nssa

ACİKLAMA

OSPF stub area cesitlerini tanimlar. Stub area'lar dis LSA'yi (Type 5) almaz, NSSA ise sinirli dis route kabul eder.

KULLANİM AMACİ

Kucuk ofis/uzak sube router'larinda LSDB boyutunu kucultmek, bellek ve CPU kullanimini azaltmak.

SYNTAX

```
Router(config-router)# area area-id stub
Router(config-router)# area area-id stub no-summary
Router(config-router)# area area-id nssa
```

ORNEK KULLANİM

```
Router(config-router)# area 10 stub
Router(config-router)# area 10 stub no-summary
```

SİK YAPİLAN HATA

Stub area'da tum router'lar ayni stub tipinde olmalidir. Backbone (Area 0) stub olamaz. ASBR iceren area NSSA olmalidir.

CİKTİ ORNEGİ

```
Router# show ip ospf | include Area 10
    Area 10
        Number of interfaces in this area is 1
        It is a stub area
        generates stub default route with cost 1
        Area has no authentication
```

9. EIGRP Yonlendirme Protokolu

> Enhanced Interior Gateway Routing Protocol (EIGRP) gelismis mesafe vektoru yonlendirme protokolunun yapilandirma komutlaridir.

81 router eigrp

ACİKLAMA

Named EIGRP yapilandirma moduna girer. Named mode, autonomous system (AS) numarasi ile birlikte isimlendirilmis konfigurasyon saglar.

KULLANİM AMACİ

EIGRP yonlendirmesini etkinlestirmek, EIGRP parametrelerini yapilandirmak.

SYNTAX

```
Router(config)# router eigrp virtual-instance-name
Router(config-router)# address-family {ipv4 | ipv6} autonomous-system as-number
```

ORNEK KULLANİM

```
Router(config)# router eigrp COMPANY-NET
Router(config-router)# address-family ipv4 autonomous-system 100
Router(config-router-af)# af-interface GigabitEthernet0/0/1
Router(config-router-af-interface)#    summary-address    10.1.0.0 255.255.0.0
```

SİK YAPİLAN HATA

Eski 'router eigrp [AS]' (classic mode) ile named mode arasinda konfigurasyon farklidir. Named mode tercih edilmelidir.

CİKTİ ORNEGİ

```
Router# show ip protocols
Routing Protocol is "eigrp COMPANY-NET"
  Outgoing update filter list for all interfaces is not set
  Default networks flagged in outgoing updates
  EIGRP-IPv4 Protocol for AS(100)
```

82 network (EIGRP)

ACİKLAMA Belirtilen ag araligindaki arayuzleri EIGRP surecine dahil eder. Wildcard maskesi kullanilabilir.

KULLANİM AMACİ Hangi arayuzlerin EIGRP komsulugu kuracagini belirlemek, ag araliklarini tanimlamak.

SYNTAX

```
Router(config-router-af)# network ip-address [wildcard-mask]
```

ORNEK KULLANİM

```
Router(config-router-af)# network 10.1.1.0 0.0.0.255
Router(config-router-af)# network 192.168.1.0
```

SİK YAPİLAN HATA Wildcard mask belirtilmezse classful network varsayilir. Ornegin 'network 10.0.0.0' tum 10.x.x.x arayuzleri dahil eder.

CİKTİ ORNEGİ

```
Router# show ip eigrp interfaces
EIGRP-IPv4 Interfaces for AS(100)
                                    Xmit Queue    PeerQ         Mean
Pacing Time   Multicast    Pending
Interface                    Peers   Un/Reliable   Un/Reliable   SRTT
Un/Reliable   Flow Timer   Routes
Gi0/0/1                       1         0/0         0/0           10
0/0           50           0
Gi0/0/2                       1         0/0         0/0           15
0/0           75           0
```

83 eigrp router-id

ACİKLAMA EIGRP router ID'sini manuel olarak belirler. OSPF gibi en yuksek loopback IP'si varsayilan olarak secilir.

KULLANİM AMACİ EIGRP router ID'sini sabitlemek, EIGRP topolojisinde karisikligi onlemek.

SYNTAX

```
Router(config-router-af)# eigrp router-id ip-address
```

ORNEK KULLANİM

```
Router(config-router-af)# eigrp router-id 1.1.1.1
```

SİK YAPİLAN HATA Ayni router ID'ye sahip iki router varsa EIGRP komsuluk kuramaz. Her router benzersiz ID kullanmalidir.

CİKTİ ORNEGİ

```
Router# show ip eigrp topology | include router
EIGRP-IPv4 Topology Entry for AS(100)/ID(1.1.1.1)
```

84 show ip eigrp neighbors

ACİKLAMA EIGRP komsularini, durumlarini (Peer Status), smooth round-trip time (SRTT) ve Retransmission Timeout (RTO) gosterir.

KULLANİM AMACİ EIGRP komsuluklarinin calistigini dogrulamak, komsu iliskilerini izlemek.

SYNTAX

```
Router# show ip eigrp neighbors [detail]
```

ORNEK KULLANİM

```
Router# show ip eigrp neighbors
```

SİK YAPİLAN HATA Komsu 'H' (Hold) suresi dolarsa komsuluk kopar. AS numarasi, authentication veya subnet uyumsuzlugu buna neden olabilir.

CİKTİ ORNEGİ

```
Router# show ip eigrp neighbors
EIGRP-IPv4 Neighbors for AS(100)
H    Address                 Interface               Hold Uptime
SRTT   RTO  Q  Seq
                                                        (sec)
(ms)       Cnt Num
0   10.1.1.2                Gi0/0/1                 14 00:23:45
10   100  0  15
1   192.168.1.2             Gi0/0/2                 12 00:20:12
15   150  0  22
```

85 show ip eigrp topology

ACİKLAMA

EIGRP topoloji tablosunu goruntuler. Feasible Distance (FD), Reported Distance (RD), successor ve feasible successor bilgilerini icerir.

KULLANİM AMACİ

EIGRP topolojisini incelemek, yedek yollari (feasible successor) tespit etmek, DUAL algoritmasi davranisini analiz etmek.

SYNTAX

```
Router# show ip eigrp topology [all-links]
```

ORNEK KULLANİM

```
Router# show ip eigrp topology
Router# show ip eigrp topology all-links
```

SİK YAPİLAN HATA

'show ip route' ile gorunen successor yoldur. Yedek yollar (feasible successor) sadece 'show ip eigrp topology' ile gorunur.

CİKTİ ORNEGİ

```
Router# show ip eigrp topology
EIGRP-IPv4 Topology Entry for AS(100)/ID(1.1.1.1) for 10.2.2.0/24
  State is Passive, Query origin flag is 1, 1 Successor(s), FD is
307200
  Routing Descriptor Blocks:
  10.1.1.2 (GigabitEthernet0/0/1), from 10.1.1.2, Send flag is 0x0
      Composite metric is (307200/281600), route is Internal
      Vector metric:
        Minimum bandwidth is 100000 Kbit
        Total delay is 2000 microseconds
```

86 variance

ACİKLAMA EIGRP'de es maliyetli olmayan yollarin (unequal cost load balancing) yuk dengelenmesini saglar. Varsayilan deger 1'dir.

KULLANİM AMACİ Farkli maliyetli yollar arasinda trafigi paylastirmak, bant genisligini daha verimli kullanmak.

SYNTAX

```
Router(config-router-af)# variance multiplier
```

ORNEK KULLANİM

```
Router(config-router-af)# variance 2
```

SİK YAPİLAN HATA Variance degeri FD'nin kac kati olacagini belirler. Feasibility condition (AD < FD) saglanmali, sadece variance yetmez.

CİKTİ ORNEGİ

```
Router(config-router-af)# variance 2
Router# show ip route eigrp
D            10.2.2.0/24   [90/307200]   via   10.1.1.2,   00:15:32,
GigabitEthernet0/0/1
                           [90/412160]  via  192.168.1.2,  00:15:32,
GigabitEthernet0/0/2
! Iki yol da routing tablosunda (unequal cost load balancing)
```

87 ip bandwidth-percent eigrp

ACİKLAMA EIGRP'nin bir arayuz uzerinde kullanabilecegi maksimum bant genisligi yuzdesini ayarlar. Varsayilan %50'dir.

KULLANİM AMACİ Dusuk bant genislikli baglantilarda (Frame Relay, MPLS) EIGRP trafigini sinirlandirmak veya artirmak.

SYNTAX

```
Router(config-if)# ip bandwidth-percent eigrp as-number percent
```

ORNEK KULLANİM

```
Router(config-if)# ip bandwidth-percent eigrp 100 75
```

SİK YAPİLAN HATA 'bandwidth' komutu yalnizca EIGRP/OSPF metrik hesaplamasi icindir, gercek hizi sinirlamaz. Ancak EIGRP buna gore oran hesaplar.

CİKTİ ORNEGİ

```
Router#  show  running-config  interface  serial0/0/0  |  include
bandwidth
 ip bandwidth-percent eigrp 100 75
 bandwidth 1544
! EIGRP max kullanimi: 1544 * 0.75 = 1158 Kbps
```

88 eigrp stub

ACİKLAMA EIGRP stub router olarak yapilandirir. Stub router sadece belirli route'leri yayinlar, query'lere yanit vermez.

KULLANİM AMACİ Uzak sube router'larinda query kapsamini sinirlandirmak, EIGRP konverjansini hizlandirmak.

SYNTAX

```
Router(config-router-af)# eigrp stub [receive-only | connected | static | summary | redistributed]
```

ORNEK KULLANİM

```
Router(config-router-af)# eigrp stub connected summary
```

SİK YAPİLAN HATA Stub router'dan uzak aglara ulasmak gerekiyorsa 'connected summary' yeterlidir. Stub yanlis tanimlanirsa uzak aglar erisilemez.

CİKTİ ORNEGİ

```
Router# show ip eigrp neighbors detail | include Stub
  Stub Peer Advertising ( CONNECTED SUMMARY ) Routes
```

89 summary-address (EIGRP)

ACİKLAMA EIGRP arayuzu uzerinden ozet (summary) route yayinlanmasini saglar. Route summarization yapilir.

KULLANİM AMACİ EIGRP topoloji tablosunu kucultmek, routing guncellemelerini azaltmak, ag performansini artirmak.

SYNTAX

```
Router(config-router-af-interface)# summary-address ip-address subnet-mask
```

ORNEK KULLANİM

```
Router(config-router-af-interface)# summary-address 10.1.0.0 255.255.0.0
```

SİK YAPİLAN HATA Summary route null0 interface'e yonelir. Eger alt aglardan biri down olursa blackhole olusabilir. Dikkatli planlama gerekir.

CİKTİ ORNEGİ

```
Router# show ip route eigrp
D     10.1.0.0/16 is a summary, 00:10:25, Null0
```

90 timers active-time

ACİKLAMA

EIGRP'de active state'de kalan route'un ne kadar sure query gondereceğini belirler. Varsayilan 3 dakikadir.

KULLANİM AMACİ

EIGRP query kapsamini ve suresini sinirlandirmak, SIA (Stuck in Active) sorunlarini onlemek.

SYNTAX

```
Router(config-router-af)# timers active-time [time-limit | disabled]
```

ORNEK KULLANİM

```
Router(config-router-af)# timers active-time 1
```

SİK YAPİLAN HATA

Cok dusuk deger SIA durumlarina neden olabilir. Stub router kullanmak SIA sorunlarini cozmenin en iyi yoludur.

CİKTİ ORNEGİ

```
Router# show ip eigrp topology | include Active
! Aktif route yok (iyi durum)
Router# show running-config | include timers
 timers active-time 1
```

10. BGP Yonlendirme Protokolu

> Border Gateway Protocol (BGP) dis yonlendirme protokolunun temel ve ileri seviye yapilandirma komutlaridir.

91 router bgp

ACİKLAMA

BGP yonlendirme surecini baslatir. AS numarasi (autonomous system) ile birlikte tanimlanir.

KULLANİM AMACİ

Internet servis saglayicilari ve buyuk kurumlar arasi yonlendirme icin BGP'yi etkinlestirmek.

SYNTAX

```
Router(config)# router bgp autonomous-system
```

ORNEK KULLANİM

```
Router(config)# router bgp 65001
Router(config-router)#
```

SİK YAPİLAN HATA

iBGP (ic BGP) komsulari AS numarasi ayni, eBGP (dis BGP) komsulari AS numarasi farkli olmalidir. Komsu AS numarasi yanlis yazilirsa komsuluk kurulmaz.

CİKTİ ORNEGİ

```
Router# show ip protocols | include BGP
Routing Protocol is "bgp 65001"
  Outgoing update filter list for all interfaces is not set
  IGP synchronization is disabled
  Automatic route summarization is disabled
```

92 neighbor

ACİKLAMA

BGP komsusu (peer) tanimlar. Komsunun IP adresi ve AS numarasi belirtilir. BGP'de komsuluk manuel olarak tanimlanir.

KULLANİM AMACİ

BGP komsularini tanimlamak, komsu iliskilerini kurmak.

SYNTAX

```
Router(config-router)# neighbor ip-address remote-as autonomous-system
```

ORNEK KULLANİM

```
Router(config-router)# neighbor 203.0.113.2 remote-as 65002
Router(config-router)# neighbor 192.168.1.2 remote-as 65001
```

SİK YAPİLAN HATA

BGP komsulugu kurulmasi icin TCP 179 portu acik olmalidir. Varsayilan olarak eBGP'de TTL=1'dir, multihop gerekiyorsa 'ebgp-multihop' kullanilmalidir.

CİKTİ ORNEGİ

```
Router# show ip bgp summary
BGP router identifier 1.1.1.1, local AS number 65001
BGP table version is 25, main routing table version 25
5 network entries using 700 bytes of memory
5 path entries using 400 bytes of memory

Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ
Up/Down  State/PfxRcd
203.0.113.2     4        65002    1523    1521       25    0    0
1d02h    5
```

93 network (BGP)

ACİKLAMA BGP tarafindan yayinlanacak (advertise) agi tanimlar. Agin routing tablosunda tam olarak eslesmesi gerekir.

KULLANİM AMACİ Sahip olunan IP bloklarini BGP ile dis dunyaya duyurmak.

SYNTAX

```
Router(config-router)# network network-address [mask subnet-mask]
```

ORNEK KULLANİM

```
Router(config-router)# network 203.0.113.0 mask 255.255.255.0
```

SİK YAPİLAN HATA Network komutu routing tablosunda tam eslesen route arar. Mask belirtilmezse classful network varsayilir. Yanlis mask ile yayin yapilmaz.

CİKTİ ORNEGİ

```
Router# show ip bgp
BGP table version is 25, local router ID is 1.1.1.1
Status codes: s suppressed, d damped, h history, * valid, > best,
i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path,
f RT-Filter,
                    x best-external, a additional-path, c RIB-
compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
```

94 show ip bgp

ACİKLAMA

BGP yonlendirme tablosunu (RIB - Routing Information Base) goruntuler. Tum yol bilgilerini, attributelari ve durumlari icerir.

KULLANİM AMACİ

BGP route'lerini incelemek, path selection surecini analiz etmek, BGP sorunlarini gidermek.

SYNTAX

```
Router# show ip bgp [prefix] [longer-prefixes]
```

ORNEK KULLANİM

```
Router# show ip bgp
Router# show ip bgp 8.8.8.0/24
```

SİK YAPİLAN HATA

'show ip bgp' BGP RIB'ini gosterir, ana routing tablosu degildir. '>' isareti best path'i gosterir. h = history, d = dampened.

CİKTİ ORNEGİ

```
Router# show ip bgp
   Network          Next Hop            Metric LocPrf Weight Path
* i192.168.1.0/24   10.1.1.2                  0    100      0 i
*>                   0.0.0.0                  0         32768 i
*  8.8.8.0/24       203.0.113.2              0             0 65002
i
! * = valid, > = best path, i = learned via iBGP
```

95 neighbor shutdown

ACİKLAMA

Belirli bir BGP komsusunu gracefully kapatir. Komsuyu tamamen kaldirmadan gecici olarak devre disi birakir.

KULLANİM AMACİ

Bakim islemleri sirasinda BGP komsusunu gecici olarak kapatmak, sorun giderme amaciyla izole etmek.

SYNTAX

```
Router(config-router)# neighbor ip-address shutdown
```

ORNEK KULLANİM

```
Router(config-router)# neighbor 203.0.113.2 shutdown
```

SİK YAPİLAN HATA

'no neighbor' ile kaldirmak yapilandirmayi siler. 'shutdown' ile sadece gecici olarak kapatilir, konfigurasyon korunur.

CİKTİ ORNEGİ

```
Router(config-router)# neighbor 203.0.113.2 shutdown
! Log mesaji:
%BGP-5-ADJCHANGE: neighbor 203.0.113.2 Down Admin. shutdown

Router# show ip bgp summary | include 203.0.113.2
203.0.113.2     4        65002       0       0        0    0    0
never    Idle (Admin)
```

96 route-map

ACİKLAMA

Eslesme (match) ve eylem (set) kurallari tanimlayarak trafigi/routelari manipule eden guclu bir aractir. BGP'de siklikla kullanilir.

KULLANİM AMACİ

BGP path attribute'larini (local-preference, MED, AS-path) manipule etmek, routing policy uygulamak.

SYNTAX

```
Router(config)# route-map name {permit | deny} [sequence]
Router(config-route-map)# match {ip address | as-path | community
| ...}
Router(config-route-map)# set {local-preference | metric | weight
| ...}
```

ORNEK KULLANİM

```
Router(config)# route-map SET_LP permit 10
Router(config-route-map)# match ip address prefix-list FROM-ISP2
Router(config-route-map)# set local-preference 200
```

SİK YAPİLAN HATA

Route-map'in sonunda implicit deny vardir. Eslesmeyen trafige izin vermek icin son satirda 'permit' bos route-map eklenmelidir.

CİKTİ ORNEGİ

```
Router# show route-map SET_LP
route-map SET_LP, permit, sequence 10
  Match clauses:
    ip address prefix-lists: FROM-ISP2
  Set clauses:
    local-preference 200
  Policy routing matches: 0 packets, 0 bytes
```

97 neighbor route-map

ACİKLAMA Bir route-map'i BGP komsusu uzerinde gelen (in) veya giden (out) yonlendirme guncellemelerine uygular.

KULLANİM AMACİ BGP komsularina ozel routing policy uygulamak, inbound/outbound route filtering yapmak.

SYNTAX

```
Router(config-router)# neighbor ip-address route-map name {in | out}
```

ORNEK KULLANİM

```
Router(config-router)# neighbor 203.0.113.2 route-map SET_LP in
```

SİK YAPİLAN HATA Route-map degisikligi sonrasi BGP session'u soft clear yapilmalidir: 'clear ip bgp neighbor-ip soft [in|out]'. Hard clear kesinti yapar.

CİKTİ ORNEGİ

```
Router# show ip bgp neighbors 203.0.113.2 | include route-map
  route-map SET_LP in
Router# clear ip bgp 203.0.113.2 soft in
! Soft inbound reset - kesinti olmadan route-map uygulandi
```

98 aggregate-address

ACİKLAMA Birden fazla BGP yolunu tek bir ozet (aggregate) route haline getirir. Routing tablosunu kucultur.

KULLANİM AMACİ Sahip olunan IP bloklarini ozetleyerek daha az prefix duyurmak, Internet routing tablosuna katkida bulunmak.

SYNTAX

```
Router(config-router)# aggregate-address ip-address subnet-mask [summary-only] [as-set]
```

ORNEK KULLANİM

```
Router(config-router)# aggregate-address 203.0.113.0 255.255.255.0 summary-only
```

SİK YAPİLAN HATA 'summary-only' eklenmezse hem ozet route hem de alt prefix'ler yayinlanir. AS-path bilgisini korumak icin 'as-set' kullanilmalidir.

CİKTİ ORNEGİ

```
Router# show ip bgp
   Network          Next Hop            Metric LocPrf Weight Path
*> 203.0.113.0/24   0.0.0.0                  0         32768 i
s> 203.0.113.0/25   0.0.0.0                  0         32768 i
s> 203.0.113.128/25 0.0.0.0                  0         32768 i
! s = suppressed (summary-only ile gizlendi)
```

99 bgp bestpath

ACİKLAMA

BGP best path selection algoritmasinin davranisini degistirir. MED, AS-path gibi attribute'larin karsilastirma seklini etkiler.

KULLANİM AMACİ

BGP path selection politikasini ozellestirmek, coklu ISP baglantilarinda trafigi yonlendirmek.

SYNTAX

```
Router(config-router)# bgp bestpath {med missing-as-worst | compare-routerid | as-path ignore}
```

ORNEK KULLANİM

```
Router(config-router)# bgp bestpath med missing-as-worst
Router(config-router)# bgp bestpath compare-routerid
```

SİK YAPİLAN HATA

'bgp always-compare-med' olmadan sadece ayni AS'den gelen MED'ler karsilastirilir. Farkli AS'lerden gelen MED'leri karsilastirmak icin bu komut gerekir.

CİKTİ ORNEGİ

```
Router# show ip bgp 8.8.8.0
Paths: (2 available, best #2)
  Advertised to update-groups:
     1
  65002 65003
    203.0.113.2 from 203.0.113.2 (2.2.2.2)
      Origin IGP, metric 0, localpref 100, valid, external
      bestpath-med: missing-as-worst applied
  65002 65004
    203.0.113.6 from 203.0.113.6 (6.6.6.6)
```

100 clear ip bgp

ACİKLAMA

BGP komsuluklarini yeniden baslatir. Hard reset (kesintili) veya soft reset (kesintisiz) yapilabilir.

KULLANİM AMACİ

BGP route-map degisikliklerini uygulamak, BGP tablosunu yenilemek, sorun giderme yapmak.

SYNTAX

```
Router# clear ip bgp {* | ip-address} [soft {in | out}] [hard]
```

ORNEK KULLANİM

```
Router# clear ip bgp * soft
Router# clear ip bgp 203.0.113.2 soft in
```

SİK YAPİLAN HATA

'clear ip bgp *' (hard reset) tum BGP session'larini keser. Uretim ortaminda kesinti olusturur. Soft reset tercih edilmelidir.

CİKTİ ORNEGİ

```
Router# clear ip bgp * soft
! Soft reset - tum komsularla kesintisiz yenileme

Router# clear ip bgp 203.0.113.2 soft in
! Sadece 203.0.113.2 komsusundan gelen routelar yenilendi
```

11. HSRP, VRRP ve GLBP

> First Hop Redundancy Protocols (FHRP) ile gateway yedekliligi saglayan HSRP, VRRP ve GLBP protokollerinin yapilandirma komutlaridir.

101 standby version / standby ip (HSRP)

ACİKLAMA

HSRP (Hot Standby Router Protocol) sanal gateway IP'si ve versiyonu tanimlar. HSRPv1 (IPv4) ve HSRPv2 destekler.

KULLANİM AMACİ

Gateway yedekliligi saglamak, bir router down oldugunda digerinin gateway gorevini devralmasini saglamak.

SYNTAX

```
Router(config-if)# standby version {1 | 2}
Router(config-if)# standby group-number ip virtual-ip
```

ORNEK KULLANİM

```
Router(config-if)# standby version 2
Router(config-if)# standby 10 ip 192.168.10.254
```

SİK YAPİLAN HATA

HSRP versiyonu her iki router'da da ayni olmalidir. v1 ile v2 birlikte calismaz. Virtual IP gercek bir interface IP'siyle ayni olamaz.

CİKTİ ORNEGİ

```
Router# show standby brief
                     P indicates configured to preempt.
                     |
Interface     Grp   Pri  P  State      Active               Standby
Virtual IP
Gi0/0/1       10    100  P  Active     local              192.168.10.2
192.168.10.254
```

102 standby priority / preempt

ACİKLAMA

HSRP priority degeri belirler (daha yuksek = aktif olma olasiligi yuksek). Preempt, yuksek priority'li router'in kontrolu geri almasini saglar.

KULLANİM AMACİ

Aktif/standby rollerini kontrol etmek, belirli router'i tercihli aktif gateway yapmak.

SYNTAX

```
Router(config-if)# standby group-number priority value
Router(config-if)# standby group-number preempt [delay {minimum | sync} seconds]
```

ORNEK KULLANİM

```
Router(config-if)# standby 10 priority 110
Router(config-if)# standby 10 preempt
```

SİK YAPİLAN HATA

Preempt varsayilan olarak kapalidir. Priority yuksek olsa bile preempt tanimlanmazsa aktif router kontrolu geri alamaz.

CİKTİ ORNEGİ

```
Router# show standby
GigabitEthernet0/0/1 - Group 10 (version 2)
  State is Active
    8 state changes, last state change 00:12:34
  Virtual IP address is 192.168.10.254
  Active virtual MAC address is 0000.0c9f.f00a
  Local virtual MAC address is 0000.0c9f.f00a (v2 default)
  Hello time 3 sec, hold time 10 sec
  Next hello sent in 1.234 secs
  Preemption enabled
```

103 vrrp ip

ACİKLAMA

VRRP (Virtual Router Redundancy Protocol) sanal gateway IP'si tanimlar. HSRP'ye benzer ancak acik standarttir (RFC 3768).

KULLANİM AMACİ

Coklu uretici ortamlarinda gateway yedekliligi saglamak. Cisco disi cihazlarla uyumlu FHRP.

SYNTAX

```
Router(config-if)# vrrp group-number ip virtual-ip [secondary]
```

ORNEK KULLANİM

```
Router(config-if)# vrrp 10 ip 192.168.10.254
Router(config-if)# vrrp 10 priority 110
Router(config-if)# vrrp 10 preempt
```

SİK YAPİLAN HATA

VRRP'de sanal IP, interface'in gercek IP'si olabilir. Bu durumda priority otomatik 255 olur (master). Diger router'lar priority < 255 olmalidir.

CİKTİ ORNEGİ

```
Router# show vrrp brief
Interface            Grp   Pri Time   Own Pre State    Master addr
Group addr
Gi0/0/1              10    110 3609   Y    Y    Master   192.168.10.1
192.168.10.254
```

104 glbp ip

ACİKLAMA

GLBP (Gateway Load Balancing Protocol) sanal gateway IP'si tanimlar. HSRP/VRRP'den farkli olarak otomatik load balancing yapar.

KULLANİM AMACİ

Gateway yedekliligi ile birlikte trafigi otomatik olarak load balancing yapmak.

SYNTAX

```
Router(config-if)# glbp group-number ip virtual-ip
```

ORNEK KULLANİM

```
Router(config-if)# glbp 10 ip 192.168.10.254
Router(config-if)# glbp 10 priority 110
Router(config-if)# glbp 10 preempt
```

SİK YAPİLAN HATA

GLBP'de AVF (Active Virtual Forwarder) sayisi kac router varsa o kadardir. Load balancing modu (round-robin, weighted, host-dependent) ayarlanmalidir.

CİKTİ ORNEGİ

```
Router# show glbp brief
Interface   Grp  Fwd Pri State    Address         Active router
Standby router
Gi0/0/1       10    -   110 Active     192.168.10.254  local
192.168.10.2
Gi0/0/1       10    1   7   Active     0007.b400.0a01  local
-
Gi0/0/1       10    2   7   Listen     0007.b400.0a02  192.168.10.2
-
```

105 show standby / show vrrp / show glbp

ACİKLAMA FHRP protokollerinin durumunu goruntuleyen show komutlaridir. Aktif/standby rolleri, sanal IP ve priority bilgilerini gosterir.

KULLANİM AMACİ Gateway yedekliliginin calistigini dogrulamak, aktif gateway'i tespit etmek.

SYNTAX

```
Router# show standby [brief]
Router# show vrrp [brief]
Router# show glbp [brief]
```

ORNEK KULLANİM

```
Router# show standby brief
Router# show vrrp brief
Router# show glbp brief
```

SİK YAPİLAN HATA

'show standby' ile 'show vrrp' farkli protokolleri gosterir. Hangi protokol yapilandirildiysa ona gore show komutu kullanilmalidir.

CİKTİ ORNEGİ

```
Router# show standby brief
Interface     Grp  Pri  P  State     Active                  Standby
Virtual IP
Gi0/0/1       10   110  P  Active   local                192.168.10.2
192.168.10.254
Router# show vrrp brief
Interface    Grp Pri Time   Own Pre State    Master addr      Group
addr
Gi0/0/1        10   110  3609    Y     Y     Master    192.168.10.1
192.168.10.254
```

106 standby track

ACİKLAMA

HSRP'nin bir arayuzu izlemesini (track) ve o arayuz down oldugunda priority dusurmesini saglar.

KULLANİM AMACİ

WAN baglantisi kopunca gateway'in otomatik olarak diger router'a gecmesini saglamak.

SYNTAX

```
Router(config-if)# standby group-number track interface-type number [decrement value]
```

ORNEK KULLANİM

```
Router(config-if)# standby 10 track GigabitEthernet0/0/0 30
```

SİK YAPİLAN HATA

Decrement degeri priority'den buyuk olursa priority negatif olabilir. Varsayilan decrement 10'dur. Preempt ile birlikte calismalidir.

CİKTİ ORNEGİ

```
Router# show standby
GigabitEthernet0/0/1 - Group 10
  Priority 110 (configured 110)
    Track interface GigabitEthernet0/0/0 state Up decrement 30
  ! WAN interface UP - Priority 110

! WAN interface DOWN oldugunda:
Router# show standby
  Priority 80 (configured 110)
    Track interface GigabitEthernet0/0/0 state Down decrement 30
```

107 vrrp authentication

ACİKLAMA

VRRP grubu icin authentication (text veya md5) tanimlar. Yetkisiz cihazlarin VRRP grubuna katilmasini engeller.

KULLANİM AMACİ

VRRP komsulugunu guvenlik altina almak, yetkisiz VRRP router'larini engellemek.

SYNTAX

```
Router(config-if)# vrrp group-number authentication text password
Router(config-if)# vrrp group-number authentication md5 key-string
password
```

ORNEK KULLANİM

```
Router(config-if)# vrrp 10 authentication md5 key-string VRRP@2024
```

SİK YAPİLAN HATA

Authentication sifresi tum grup uyelerinde ayni olmalidir. MD5 yerine text kullanilirsa sifre clear-text gorunur.

CİKTİ ORNEGİ

```
Router# show vrrp
GigabitEthernet0/0/1 - Group 10
  State is Master
  Virtual IP address is 192.168.10.254
  Virtual MAC address is 0000.5e00.010a
  Advertisement interval is 1.000 sec
  Preemption enabled
  Authentication text "VRRPKEY"
  Priority is 110
```

108 glbp load-balancing

ACİKLAMA GLBP load balancing algoritmasini secer. Round-robin, weighted, host-dependent secenekleri vardir.

KULLANİM AMACİ Trafigin farkli router'lar arasinda nasil dagilacagini belirlemek.

SYNTAX

```
Router(config-if)# glbp group-number load-balancing {round-robin | weighted | host-dependent}
```

ORNEK KULLANİM

```
Router(config-if)# glbp 10 load-balancing round-robin
```

SİK YAPİLAN HATA Host-dependent modda ayni MAC her zaman ayni router'a yonlenir. NAT ortamlarinda kullanislidir. Round-robin NAT session'lari bozabilir.

CİKTİ ORNEGİ

```
Router# show glbp
GigabitEthernet0/0/1 - Group 10
  State is Active
    2 state changes, last state change 00:45:12
  Virtual IP address is 192.168.10.254
  Hello time 3 sec, hold time 10 sec
  Next hello sent in 1.234 secs
  Redirect time 600 sec, forwarder timeout 14400 sec
  Preemption enabled, min delay 0 sec
  Active is local
```

109 standby timers

ACİKLAMA

HSRP hello ve hold timer degerlerini ayarlar. Daha dusuk degerler daha hizli failover ancak daha fazla CPU yuku demektir.

KULLANİM AMACİ

HSRP failover hizini ayarlamak, ag gereksinimlerine gore timer'lari optimize etmek.

SYNTAX

```
Router(config-if)# standby group-number timers [msec] hello-time [msec] hold-time
```

ORNEK KULLANİM

```
Router(config-if)# standby 10 timers 1 3
Router(config-if)# standby 10 timers msec 200 msec 750
```

SİK YAPİLAN HATA

Timer degerleri tum grup uyelerinde ayni olmalidir. Hold time = en az 3 x hello time olmalidir. Milisaniye (msec) kullaniminda tutarlilik sarttir.

CİKTİ ORNEGİ

```
Router# show standby
GigabitEthernet0/0/1 - Group 10
  Hello time 1 sec, hold time 3 sec
  Next hello sent in 0.234 secs
  Preemption enabled
  Active router is local
  Priority 110 (configured 110)
```

110 track object

ACİKLAMA

IP SLA veya arayuz durumunu izleyen bir tracking nesnesi olusturur. HSRP/VRRP/GLBP ile birlikte kullanilir.

KULLANİM AMACİ

Daha gelismis tracking mantigi olusturmak, sadece arayuz degil IP erisilebilirligini izlemek.

SYNTAX

```
Router(config)# track object-number interface type number {line-protocol | ip routing}
Router(config)# track object-number ip sla sla-operation-number {reachability | state}
```

ORNEK KULLANİM

```
Router(config)# track 10 interface GigabitEthernet0/0/0 line-protocol
Router(config-if)# standby 10 track 10 decrement 30
```

SİK YAPİLAN HATA

Track nesnesi olusturulmadan HSRP'de kullanilirsa hata vermez ancak calismaz. Once track nesnesi tanimlanmalidir.

CİKTİ ORNEGİ

```
Router# show track
Track 10
  Interface GigabitEthernet0/0/0 line-protocol
  Line protocol is Up (hw state up, ip state up)
    1 change, last change 00:23:45
  Tracked by:
    HSRP GigabitEthernet0/0/1 10
```

12. SNMP, Syslog ve NTP

> Ag cihazlarinin izlenmesi, loglanmasi ve zaman senkronizasyonu icin kullanilan SNMP, Syslog ve NTP yapilandirma komutlaridir.

111 snmp-server community

ACİKLAMA

SNMP community string (sifre) tanimlar. RO (Read-Only) veya RW (Read-Write) erisim saglar.

KULLANİM AMACİ

SNMP ile cihaz bilgilerinin okunmasini veya yazilmasini saglamak, NMS (Network Management Station) erisimi vermek.

SYNTAX

```
Router(config)# snmp-server community string {ro | rw} [acl-number]
```

ORNEK KULLANİM

```
Router(config)# snmp-server community CompanySNMP RO 10
Router(config)# snmp-server community AdminSNMP RW 11
```

SİK YAPİLAN HATA

Varsayilan community string'ler (public, private) kesinlikle kullanilmamalidir. Guclu, rastgele string'ler kullanilmalidir.

CİKTİ ORNEGİ

```
Router# show snmp community
Community name: CompanySNMP
Community Index: CompanySNMP
Community SecurityName: CompanySNMP
storage-type: nonvolatile        active

Community name: AdminSNMP
Community Index: AdminSNMP
Community SecurityName: AdminSNMP
storage-type: nonvolatile        active
```

112 snmp-server host

ACİKLAMA

SNMP trap'lerin veya inform'larin gonderilecegi hedef sunucuyu (NMS) tanimlar.

KULLANİM AMACİ

Cihaz olaylarinin (link up/down, config change) SNMP trap olarak NMS'e gonderilmesini saglamak.

SYNTAX

```
Router(config)# snmp-server host host-address [version {1 | 2c | 3}] community-string [trap-type]
```

ORNEK KULLANİM

```
Router(config)# snmp-server host 192.168.1.10 version 2c CompanySNMP
```

SİK YAPİLAN HATA

Trap gondermek icin ayrica 'snmp-server enable traps' ile trap turleri etkinlestirilmelidir. Sadece host tanimlamak yeterli degildir.

CİKTİ ORNEGİ

```
Router# show snmp host
Notification host: 192.168.1.10    udp-port: 162   type: trap
  user: CompanySNMP   security model: v2c
```

113 snmp-server enable traps

ACİKLAMA

Belirli SNMP trap turlerinin gonderilmesini etkinlestirir. Varsayilan olarak trap'ler kapalidir.

KULLANİM AMACİ

Istenen olay turlerinin (link, envmon, config, bgp, ospf) NMS'e trap olarak gonderilmesini saglamak.

SYNTAX

```
Router(config)# snmp-server enable traps [trap-type [sub-type]]
```

ORNEK KULLANİM

```
Router(config)# snmp-server enable traps snmp linkdown linkup
Router(config)# snmp-server enable traps envmon
Router(config)# snmp-server enable traps config
```

SİK YAPİLAN HATA

'snmp-server enable traps' tum trap'leri acar. Bu gereksiz yuku artirir. Sadece ihtiyac duyulan trap'leri acin.

CİKTİ ORNEGİ

```
Router# show snmp
Chassis: FOC1234A1B2
0 SNMP packets input
  0 Bad SNMP version errors
  0 Unknown community name
  0 Illegal operation for community name supplied
SNMP Logging: enabled
  Logging to 192.168.1.10.162, 0/10 received
```

114 logging host

ACİKLAMA Syslog mesajlarinin gonderilecegi hedef sunucuyu tanimlar. UDP 514 portunu kullanir.

KULLANİM AMACİ Cihaz loglarinin merkezi bir syslog sunucusunda toplanmasini saglamak.

SYNTAX

```
Router(config)# logging host ip-address [transport udp port | transport tcp port]
```

ORNEK KULLANİM

```
Router(config)# logging host 192.168.1.10
Router(config)# logging trap informational
```

SİK YAPİLAN HATA

'logging trap' seviyesi cok yuksek (error) ayarlanirsa bilgilendirici loglar gitmez. Dusuk seviye (informational) daha detayli log saglar.

CİKTİ ORNEGİ

```
Router# show logging
Syslog logging: enabled (0 messages dropped, 2 messages rate-limited, 0 flushes, 0 overruns, xml disabled, filtering disabled)

No Active Message Discriminator.

No Inactive Message Discriminator.

      Console logging: level debugging, 245 messages logged, xml disabled,
```

115 ntp server

ACİKLAMA

NTP (Network Time Protocol) sunucusu tanimlar. Cihazin zamanini senkronize eder.

KULLANİM AMACİ

Cihazin sistem saatini dogru tutmak, log zaman damgalarini esitlemek, sertifika validasyonu icin zaman senkronizasyonu saglamak.

SYNTAX

```
Router(config)# ntp server ip-address [prefer] [source interface]
```

ORNEK KULLANİM

```
Router(config)# ntp server 192.168.1.10 prefer
Router(config)# ntp server 10.1.1.5
```

SİK YAPİLAN HATA

NTP icin firewall'da UDP 123 portu acik olmalidir. 'prefer' anahtar kelimesi birincil NTP sunucusu olarak isaretler.

CİKTİ ORNEGİ

```
Router# show ntp associations
  address         ref clock       st   when   poll reach  delay
offset   disp
*~192.168.1.10    192.168.1.1      2     56     64   377  0.521
1.234  0.876
+~10.1.1.5        192.168.1.1      2     45     64   377  1.234
2.567  0.654
 * sys.peer, # selected, + candidate, - outlyer, x falseticker, ~
configured
```

116 ntp authenticate

ACİKLAMA

NTP authentication'u etkinlestirir. NTP sunucusuyla guvenli zaman senkronizasyonu saglar.

KULLANİM AMACİ

NTP spoofing saldirilarini onlemek, sadece yetkili NTP sunucularindan zaman almak.

SYNTAX

```
Router(config)# ntp authenticate
Router(config)# ntp authentication-key number md5 password
Router(config)# ntp trusted-key key-number
Router(config)# ntp server ip-address key key-number
```

ORNEK KULLANİM

```
Router(config)# ntp authenticate
Router(config)# ntp authentication-key 1 md5 NTP@Key2024
Router(config)# ntp trusted-key 1
Router(config)# ntp server 192.168.1.10 key 1
```

SİK YAPİLAN HATA

'ntp authenticate' tek basina yetmez. authentication-key, trusted-key ve server'da key tanimlanmalidir. Key numeralari eslesmelidir.

CİKTİ ORNEGİ

```
Router# show ntp status
Clock is synchronized, stratum 3, reference is 192.168.1.10
nominal freq is 250.0000 Hz, actual freq is 249.9987 Hz, precision
is 2**18
ntp uptime is 1234567 (1/100 of seconds), resolution is 4000
reference time is E2A3B4C5.12345678 (10:30:45.071 UTC Fri Jul 10
2025)
clock offset is 1.234 msec, root delay is 2.345 msec
root dispersion is 3.456 msec, peer dispersion is 0.876 msec
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is
```

117 clock timezone

ACİKLAMA

Cihazin sistem saati icin zaman dilimini tanimlar. UTC'den farki saat ve dakika olarak belirtilir.

KULLANİM AMACİ

Cihazin yerel zaman dilimine gore log ve zaman damgalari gostermesini saglamak.

SYNTAX

```
Router(config)# clock timezone zone hours-offset [minutes-offset]
```

ORNEK KULLANİM

```
Router(config)# clock timezone IST 3
Router(config)# clock timezone CET 1
Router(config)# clock timezone EST -5
```

SİK YAPİLAN HATA

NTP zaman UTC olarak alinir, timezone yerel gosterim icindir. Timezone NTP senkronizasyonunu etkilemez, sadece gorunumu degistirir.

CİKTİ ORNEGİ

```
Router# show clock
13:30:45.071 IST Fri Jul 10 2025
Router# show clock detail
13:30:45.071 IST Fri Jul 10 2025
Time source is NTP
Summer time starts 02:00:00 IST Sun Mar 30 2025
Summer time ends 03:00:00 IST Sun Oct 26 2025
```

118 clock summer-time

ACİKLAMA

Yaz saati uygulamasini (Daylight Saving Time) tanimlar. Belirli tarihlerde otomatik olarak saat 1 saat ileri/geri alinir.

KULLANİM AMACİ

Yaz-kis saati gecislerini otomatik yonetmek, log zaman damgalarinin tutarli olmasini saglamak.

SYNTAX

```
Router(config)# clock summer-time zone recurring [week day month hh:mm week day month hh:mm [offset]]
Router(config)# clock summer-time zone date date month year hh:mm date month year hh:mm [offset]
```

ORNEK KULLANİM

```
Router(config)# clock summer-time EEST recurring last Sun Mar 2:00 last Sun Oct 3:00
```

SİK YAPİLAN HATA

'recurring' kullanilmazsa her yil manuel guncelleme gerekir. Bazi bolgelerde DST kaldirilmistir, kullanilmamali.

CİKTİ ORNEGİ

```
Router# show clock detail
13:30:45.071 IST Fri Jul 10 2025
Time source is NTP
Summer time starts 02:00:00 IST Sun Mar 30 2025
Summer time ends 03:00:00 IST Sun Oct 26 2025
```

119 logging trap

ACİKLAMA

Syslog mesajlarinin gonderilecegi minimum onem seviyesini (severity level) belirler.

KULLANİM AMACİ

Gereksiz log trafigini azaltmak, sadece onemli loglarin syslog sunucusuna gitmesini saglamak.

SYNTAX

```
Router(config)# logging trap {level | severity-level-name}
```

ORNEK KULLANİM

```
Router(config)# logging trap warnings
Router(config)# logging trap 4
```

SİK YAPİLAN HATA

'logging trap debugging' tum loglari gonderir ve ag trafigini artirir. Uretimde 'informational' veya 'notifications' yeterlidir.

CİKTİ ORNEGİ

```
Router# show logging
Trap logging: level warnings, 45 message lines logged
    Logging to 192.168.1.10 (udp port 514)

Severity level guide:
0=Emergencies  1=Alerts  2=Critical  3=Errors  4=Warnings
5=Notifications  6=Informational  7=Debugging
```

120 service timestamps

ACİKLAMA

Syslog mesajlari ve debug ciktilarina zaman damgasi ekler. Varsayilan olarak kapalidir.

KULLANİM AMACİ

Log mesajlarinin ne zaman olustugunu belirlemek, sorun gidermede zaman cizelgesi olusturmak.

SYNTAX

```
Router(config)# service timestamps debug {datetime [msec] [localtime] [show-timezone] | uptime}
Router(config)# service timestamps log {datetime [msec] [localtime] [show-timezone] | uptime}
```

ORNEK KULLANİM

```
Router(config)# service timestamps log datetime msec localtime show-timezone
Router(config)# service timestamps debug datetime msec
```

SİK YAPİLAN HATA

'uptime' seceneginde zaman damgasi cihazin calisma suresi olarak gorunur. Kullanisli degildir, 'datetime' tercih edilmelidir.

CİKTİ ORNEGİ

```
Router# show logging
%LINK-5-CHANGED: Interface GigabitEthernet0/0/1, changed state to administratively down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/1, changed state to down

! Timestamps aktif oldugunda:
Jul 10 2025 13:45:32.123 IST: %LINK-5-CHANGED: Interface GigabitEthernet0/0/1, changed state to up
Jul 10 2025 13:45:33.456 IST: %LINEPROTO-5-UPDOWN: Line protocol
```

13. SSH, Telnet ve Konsol Yonetimi

> Cihaza uzaktan guvenli erisim (SSH), Telnet ve konsol yonetimi ile ilgili yapilandirma ve guvenlik komutlaridir.

121 crypto key generate rsa

ACİKLAMA SSH baglantisi icin RSA sifreleme anahtari olusturur. SSH calismasi icin zorunludur.

KULLANİM AMACİ SSH sunucusu olarak calismak icin gerekli RSA anahtar ciftini olusturmak.

SYNTAX

```
Router(config)# crypto key generate rsa [general-keys | usage-keys] [modulus modulus-size]
```

ORNEK KULLANİM

```
Router(config)# crypto key generate rsa modulus 2048
```

SİK YAPİLAN HATA

'ip domain-name' tanimlanmadan bu komut calistirilamaz. Oncelikle domain-name tanimlanmalidir.

CİKTİ ORNEGİ

```
Router(config)# crypto key generate rsa modulus 2048
The name for the keys will be: Router.company.local

% The key modulus size is 2048 bits
% Generating 2048 bit RSA keys, keys will be non-exportable...
[OK] (elapsed time was 3 seconds)

Router(config)#
%SSH-5-ENABLED: SSH 2.0 has been enabled
```

122 ip ssh version

ACİKLAMA Kullanilacak SSH protokol versiyonunu belirler. SSHv2 guvenlidir, SSHv1 zafiyetlidir.

KULLANİM AMACİ Sadece SSHv2 baglantilarina izin vermek, guvenlik aciklarini onlemek.

SYNTAX

```
Router(config)# ip ssh version 2
```

ORNEK KULLANİM

```
Router(config)# ip ssh version 2
```

SİK YAPİLAN HATA

'ip ssh version 2' calistirilinca once RSA key olusturulmus olmalidir. Ayrica IOS surumu SSHv2'yi desteklemelidir.

CİKTİ ORNEGİ

```
Router# show ip ssh
SSH Enabled - version 2.0
Authentication timeout: 120 secs; Authentication retries: 3
Minimum expected Diffie Hellman key size : 2048 bits
IOS Keys in SECSH format(ssh-rsa, base64 encoded):
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAABAQC...
```

123 transport input ssh

ACİKLAMA

VTY hatlari uzerinden sadece SSH erisimine izin verir. Telnet erisimini engeller.

KULLANİM AMACİ

Uzaktan yonetimde sadece guvenli SSH protokolunu kullanmak, Telnet'i tamamen devre disi birakmak.

SYNTAX

```
Router(config-line)# transport input ssh
```

ORNEK KULLANİM

```
Router(config)# line vty 0 15
Router(config-line)# transport input ssh
```

SİK YAPİLAN HATA

'transport input ssh' yapilmadan once SSH etkinlestirilmis olmalidir. Yoksa VTY'ye hicbir sekilde erisilemez.

CİKTİ ORNEGİ

```
Router# show line vty 0
Tty Typ     Tx/Rx    A Modem  Roty AccO AccI  Uses  Noise Overruns
Int
*  2 VTY              -    -      -    -    -     5      0    0/0
-

Line 2, Location: "", Type: ""
Session limit is not set.
Allowed input transports are ssh.
Preferred transport is ssh.
```

124 exec-timeout

ACİKLAMA Konsol veya VTY oturumunun inaktif kalma suresini dakika ve saniye olarak belirler. Sure doldugunda oturum otomatik kapanir.

KULLANİM AMACİ Yonetim oturumlarinin acik unutulmasini onlemek, guvenligi artirmak.

SYNTAX

```
Router(config-line)# exec-timeout minutes [seconds]
```

ORNEK KULLANİM

```
Router(config-line)# exec-timeout 5 0
Router(config-line)# exec-timeout 0 0
```

SİK YAPİLAN HATA 'exec-timeout 0 0' oturumun asla kapanmamasini saglar. Guvenlik riski olusturur, sadece laboratuvar ortaminda kullanilmalidir.

CİKTİ ORNEGİ

```
Router# show line console 0
Tty Typ     Tx/Rx    A Modem  Roty AccO AccI  Uses  Noise Overruns
Int
   0 CTY              -    -      -    -    -     0      0    0/0
-

Line 0, Location: "", Type: ""
Length: 24 lines, Width: 80 columns
Status: Ready, Active
  Timeout: 5:00 (minutes:seconds)
```

125 logging synchronous

ACİKLAMA Konsol uzerinde log mesajlarinin komut girisini bozmasini engeller. Log mesaji geldiginde mevcut komut korunur.

KULLANİM AMACİ Konsolda calisirken log mesajlarinin komut yazmayi engellemesini onlemek.

SYNTAX

```
Router(config-line)# logging synchronous [level severity-level | all]
```

ORNEK KULLANİM

```
Router(config)# line console 0
Router(config-line)# logging synchronous
```

SİK YAPİLAN HATA VTY hatlarinda da kullanilabilir ancak sadece kullanici deneyimi icindir. Guvenlik veya islevsellik acisindan kritik degildir.

CİKTİ ORNEGİ

```
Router(config)# interface g
%LINK-5-CHANGED: Interface GigabitEthernet0/0/1, changed state to up
! logging synchronous ile: komut bozulmaz, cursor alt satira gecer
```

126 ip ssh time-out

ACİKLAMA SSH baglanti kurma (authentication) timeout suresini saniye olarak ayarlar.

KULLANİM AMACİ SSH brute-force saldirilarini zorlastirmak, bosta kalan SSH baglantilarini otomatik kapatmak.

SYNTAX

```
Router(config)# ip ssh time-out seconds
```

ORNEK KULLANİM

```
Router(config)# ip ssh time-out 60
```

SİK YAPİLAN HATA Cok dusuk timeout (5-10 sn) yavas baglantilarda kullanicilarin baglanamamasina neden olabilir. 30-120 sn arasi uygundur.

CİKTİ ORNEGİ

```
Router# show ip ssh
SSH Enabled - version 2.0
Authentication timeout: 60 secs; Authentication retries: 3
```

127 ip ssh authentication-retries

ACİKLAMA SSH baglantisinda basarisiz authentication deneme sayisini sinirlar.

KULLANİM AMACİ SSH brute-force saldirilarini onlemek, deneme sayisini sinirlandirmak.

SYNTAX

```
Router(config)# ip ssh authentication-retries number
```

ORNEK KULLANİM

```
Router(config)# ip ssh authentication-retries 2
```

SİK YAPİLAN HATA

Cok dusuk deger (1) kullanici hatalarinda baglanti kopmasina neden olabilir. 2-3 uygun degerdir.

CİKTİ ORNEGİ

```
Router# show ip ssh
SSH Enabled - version 2.0
Authentication timeout: 60 secs; Authentication retries: 2
```

128 privilege level

ACİKLAMA Konsol veya VTY hatlari icin varsayilan ayricalik seviyesini (privilege level) belirler. 0-15 arasidir.

KULLANİM AMACİ Farkli kullanici seviyeleri icin yetki ataması yapmak, kisitli kullanicilar olusturmak.

SYNTAX

```
Router(config-line)# privilege level level
```

ORNEK KULLANİM

```
Router(config)# line vty 0 4
Router(config-line)# privilege level 5
```

SİK YAPİLAN HATA

Seviye 15 = full admin yetkisi. Seviye 1 = sadece show komutlari. Ozel komut seviyeleri 'privilege exec' ile ayarlanabilir.

CİKTİ ORNEGİ

```
Router# show privilege
Current privilege level is 15
Router# show user
    Line       User       Host(s)              Idle       Location
*  0 con 0                 idle                 00:00:00
  Interface    User               Mode         Idle     Peer
Address
! VTY 0-4 privilege level 5 ile baglanan kullanici:
Router> show privilege
Current privilege level is 5
```

129 ip ssh client algorithm

ACİKLAMA

SSH istemcisi olarak calisirken kullanilacak sifreleme algoritmalarini ayarlar.

KULLANİM AMACİ

Guvensiz SSH algoritmalarini (DES, RC4, MD5) devre disi birakmak, sadece guvenli algoritmalari kullanmak.

SYNTAX

```
Router(config)# ip ssh client algorithm encryption {aes128-ctr | aes192-ctr | aes256-ctr}
Router(config)# ip ssh server algorithm encryption {aes128-ctr | aes192-ctr | aes256-ctr}
```

ORNEK KULLANİM

```
Router(config)# ip ssh server algorithm encryption aes256-ctr aes192-ctr aes128-ctr
```

SİK YAPİLAN HATA

Cok katı algoritma kisitlamasi eski SSH istemcilerinin baglanmasini engelleyebilir. Uyumlu algoritmalar secilmelidir.

CİKTİ ORNEGİ

```
Router# show ip ssh
SSH Enabled - version 2.0
Authentication timeout: 60 secs; Authentication retries: 3
Minimum expected Diffie Hellman key size : 2048 bits
IOS Keys in SECSH format(ssh-rsa, base64 encoded):
ssh-rsa AAAAB3NzaC1yc2E...

Encryption Algorithms:
aes256-ctr aes192-ctr aes128-ctr
```

130 ip ssh pubkey-chain

ACİKLAMA

SSH public key authentication (anahtar tabanli kimlik dogrulama) icin kullanici public key'lerini tanimlar.

KULLANİM AMACİ

Sifre yerine SSH anahtar ciftiyle kimlik dogrulama yapmak, guvenligi artirmak.

SYNTAX

```
Router(config)# ip ssh pubkey-chain
Router(config-ssh-pubkey-user)# username username
Router(config-ssh-pubkey-data)# key-string
Router(config-ssh-pubkey-data)#  key-hash  {ssh-rsa  |  ssh-dss}
fingerprint
```

ORNEK KULLANİM

```
Router(config)# ip ssh pubkey-chain
Router(config-ssh-pubkey-user)# username admin
Router(config-ssh-pubkey-data)# key-string
AAAAB3NzaC1yc2EAAAADAQABAAABAQC...
Router(config-ssh-pubkey-data)# exit
Router(config-ssh-pubkey-user)# exit
```

SİK YAPİLAN HATA

Public key dogru formatta girilmelidir. Key-string modunda her satir en fazla 72 karakter olabilir.

CİKTİ ORNEGİ

```
Router# show ip ssh pubkey-chain
Username: admin
Key name: admin
Key type: ssh-rsa
Key fingerprint: E582:C8E3:FF12:7A45:9B22:33D1:4F56:7890:ABCD:EF12
Key is valid
```

14. Port Security ve Guvenlik

> Port bazli guvenlik, DHCP Snooping, Dynamic ARP Inspection, IP Source Guard ve storm control gibi Layer 2 guvenlik komutlaridir.

131 switchport port-security maximum

ACİKLAMA Bir port uzerinde ogrenilebilecek maksimum MAC adresi sayisini belirler. Varsayilan 1'dir.

KULLANİM AMACİ Bir porta baglanabilecek cihaz sayisini sinirlandirmak.

SYNTAX

```
Switch(config-if)# switchport port-security maximum value [vlan {vlan-id | access | voice}]
```

ORNEK KULLANİM

```
Switch(config-if)# switchport port-security maximum 3
Switch(config-if)# switchport port-security maximum 2 vlan access
```

SİK YAPİLAN HATA IP Phone + PC bagliysa en az 2 olmalidir. Voice VLAN icin ayri maximum degeri tanimlanabilir.

CİKTİ ORNEGİ

```
Switch# show port-security interface gi0/1
Port Security              : Enabled
Port Status                : Secure-up
Violation Mode             : Shutdown
Maximum MAC Addresses      : 3
Total MAC Addresses        : 2
Configured MAC Addresses   : 0
Sticky MAC Addresses       : 2
```

132 switchport port-security mac-address sticky

ACİKLAMA

Port'a baglanan ilk cihazin MAC adresini otomatik olarak ogrenir ve yapilandirmaya (sticky) ekler.

KULLANİM AMACİ

MAC adreslerini manuel tanimlamak yerine otomatik ogrenmek, yapilandirmaya kaydetmek.

SYNTAX

```
Switch(config-if)# switchport port-security mac-address sticky
```

ORNEK KULLANİM

```
Switch(config-if)# switchport port-security mac-address sticky
```

SİK YAPİLAN HATA

Sticky MAC'ler running-config'e kaydedilir ancak startup-config'e 'write memory' ile kaydetmek gerekir. Yoksa restart sonrasi kaybolur.

CİKTİ ORNEGİ

```
Switch# show port-security interface gi0/1
Sticky MAC Addresses       : 2
Last Source Address:Vlan   : aabb.ccdd.ee02:10
Security Violation Count   : 0

Switch# show running-config interface gi0/1 | include sticky
 switchport port-security mac-address sticky aabb.ccdd.ee01. vlan
10
 switchport port-security mac-address sticky aabb.ccdd.ee02. vlan
10
```

133 switchport port-security violation

ACİKLAMA Port security ihlali (violation) oldugunda alinacak aksiyonu belirler. protect, restrict veya shutdown.

KULLANİM AMACİ Guvenlik ihlaline karsi uygun tepkiyi vermek. En sert mod shutdown'dur.

SYNTAX

```
Switch(config-if)# switchport port-security violation {protect | restrict | shutdown}
```

ORNEK KULLANİM

```
Switch(config-if)# switchport port-security violation shutdown
```

SİK YAPİLAN HATA 'shutdown' modunda port err-disabled olur ve manuel acilmasi gerekir. 'restrict' SNMP trap ve log atar. 'protect' sessizce dusurur.

CİKTİ ORNEGİ

```
! Violation (shutdown) sonrasi log:
%PM-4-ERR_DISABLE: psecure-violation error detected on Gi0/1, putting Gi0/1 in err-disable state

Switch# show interfaces status err-disabled
Port      Name               Status       Reason
Gi0/1                        err-disabled psecure-violation
```

134 ip dhcp snooping

ACİKLAMA DHCP Snooping ozelligini etkinlestirir. Guvensiz portlardan gelen DHCP sunucu mesajlarini engeller.

KULLANİM AMACİ Rogue DHCP sunucu saldirilarini onlemek, DHCP mesajlarini izlemek.

SYNTAX

```
Switch(config)# ip dhcp snooping
Switch(config)# ip dhcp snooping vlan vlan-id [,vlan-id]
Switch(config-if)# ip dhcp snooping trust
Switch(config-if)# ip dhcp snooping limit rate rate
```

ORNEK KULLANİM

```
Switch(config)# ip dhcp snooping
Switch(config)# ip dhcp snooping vlan 10,20
Switch(config)# interface gi0/24
Switch(config-if)# ip dhcp snooping trust
```

SİK YAPİLAN HATA DHCP sunucusuna bakan trunk port 'trust' olarak isaretlenmelidir. Varsayilan olarak tum portlar untrusted'dir.

CİKTİ ORNEGİ

```
Switch# show ip dhcp snooping
Switch DHCP snooping is enabled
DHCP snooping is configured on following VLANs:
  10,20

Interface        Trusted    Rate limit (pps)
---------        -------    ----------------
GigabitEthernet0/24  yes       unlimited
GigabitEthernet0/1   no        10 (default)
```

135 ip arp inspection vlan

ACİKLAMA

Dynamic ARP Inspection (DAI) ozelligini etkinlestirir. ARP paketlerini DHCP Snooping binding tablosuna gore dogrular.

KULLANİM AMACİ

ARP spoofing/poisoning saldirilarini onlemek, ARP paketlerinin gecerliligini kontrol etmek.

SYNTAX

```
Switch(config)# ip arp inspection vlan vlan-id [,vlan-id]
Switch(config-if)# ip arp inspection trust
Switch(config-if)#  ip  arp  inspection  limit  rate  rate  [burst
interval seconds]
```

ORNEK KULLANİM

```
Switch(config)# ip arp inspection vlan 10,20
Switch(config)# interface gi0/24
Switch(config-if)# ip arp inspection trust
```

SİK YAPİLAN HATA

DHCP Snooping olmadan DAI calismaz. DAI, DHCP Snooping binding tablosuna ihtiyac duyar. Statik IP'ler icin ARP access-list gerekir.

CİKTİ ORNEGİ

```
Switch# show ip arp inspection vlan 10
Source Mac Validation      : Disabled
Destination Mac Validation : Disabled
IP Address Validation      : Disabled

 Vlan      Configuration    Operation   ACL Match          Static
ACL
 ----      -------------    ---------   ---------          --------
--
   10      Enabled          Active
```

136 ip verify source

ACİKLAMA IP Source Guard ozelligini etkinlestirir. IP paketlerinin kaynak IP/MAC adresini DHCP Snooping binding tablosuna gore dogrular.

KULLANİM AMACİ IP spoofing saldirilarini onlemek, port bazli IP/MAC eslestirmesi saglamak.

SYNTAX

```
Switch(config-if)# ip verify source [port-security]
```

ORNEK KULLANİM

```
Switch(config-if)# ip verify source port-security
```

SİK YAPİLAN HATA

'port-security' secenegi MAC adresi dogrulamasi da ekler. Static IP kullanan cihazlar icin 'ip source binding' ile statik kayit gerekir.

CİKTİ ORNEGİ

```
Switch# show ip verify source
Interface    Filter-type    Filter-mode    IP-address          Mac-
address       Vlan  Log
---------    -----------    -----------    ----------        --------
---        ----   ---
Gi0/1           ip-mac             active             192.168.10.11
aabb.ccdd.ee01    10    no
Gi0/2           ip-mac             active             192.168.10.12
aabb.ccdd.ee02    10    no
Gi0/3        ip-mac         inactive       permit-all         permit-
```

137 storm-control

ACİKLAMA

Broadcast, multicast ve unknown unicast storm'larini onlemek icin trafigi sinirlar.

KULLANİM AMACİ

Broadcast storm olusturabilecek olaylari (loop, virüs) onlemek, ag performansini korumak.

SYNTAX

```
Switch(config-if)# storm-control {broadcast | multicast | unicast}
level {level [level-low] | bps bps [bps-low] | pps pps [pps-low]}
Switch(config-if)# storm-control action {shutdown | trap}
```

ORNEK KULLANİM

```
Switch(config-if)# storm-control broadcast level 20
Switch(config-if)# storm-control action shutdown
```

SİK YAPİLAN HATA

Seviye (level) yuzde olarak girilir. 20 = %20. Cok dusuk deger normal trafigi engelleyebilir.

CİKTİ ORNEGİ

```
Switch# show storm-control broadcast
Interface  Filter State   Upper        Lower        Current
---------  -------------  -----------  -----------  ----------
Gi0/1      Forwarding     20.00%       20.00%       0.00%
```

138 spanning-tree bpdufilter

ACİKLAMA

PortFast portlarinda BPDU paketlerinin gonderilmesini ve alinmasini engeller.

KULLANİM AMACİ

PortFast portlarinda BPDU trafigini tamamen engellemek. Switch baglanmayacak portlarda kullanilir.

SYNTAX

```
Switch(config-if)# spanning-tree bpdufilter enable
Switch(config)# spanning-tree portfast bpdufilter default
```

ORNEK KULLANİM

```
Switch(config-if)# spanning-tree portfast
Switch(config-if)# spanning-tree bpdufilter enable
```

SİK YAPİLAN HATA

BPDU Filter BPDU Guard'dan daha az guvenlidir. BPDU alinirsa sessizce PortFast'tan cikar. Switch baglanma riski vardir.

CİKTİ ORNEGİ

```
Switch# show spanning-tree interface gi0/1 detail
 Port 1 (GigabitEthernet0/1) of VLAN0001 is designated forwarding
   Port path cost: 4, Port priority: 128, Port Identifier: 128.1.
   Portfast is enabled (enable)
   BPDU guard is Disabled
   BPDU filter is enabled
   Link type is point-to-point by default
```

139 private-vlan

ACİKLAMA

Private VLAN (PVLAN) olusturur. Izole portlar arasi iletisimi engeller ancak promiscuous port ile iletisim saglar.

KULLANİM AMACİ

Ayni VLAN'daki cihazlarin birbiriyle dogrudan iletisimini engellemek (otel, data center).

SYNTAX

```
Switch(config)# vlan vlan-id
Switch(config-vlan)# private-vlan {primary | isolated | community}
Switch(config-vlan)# private-vlan association {secondary-vlan-list
| add secondary-vlan-list}
```

ORNEK KULLANİM

```
Switch(config)# vlan 100
Switch(config-vlan)# private-vlan primary
Switch(config-vlan)# private-vlan association 101,102
Switch(config)# vlan 101
Switch(config-vlan)# private-vlan isolated
```

SİK YAPİLAN HATA

PVLAN icin switch'in VTP transparent modda olmasi veya VTPv3 kullanmasi gerekir. VTPv1/v2'de calismaz.

CİKTİ ORNEGİ

```
Switch# show vlan private-vlan
Primary   Secondary   Type              Ports
-------   ---------   ------------       -------------------------
-
100       101         isolated          Gi0/1, Gi0/2
100       102         community         Gi0/3, Gi0/4
100       100         primary           Gi0/24
```

140 errdisable detect cause

ACİKLAMA

Hangi durumlarin errdisable (hata nedeniyle devre disi kalma) sebebi olarak algilanacagini belirler.

KULLANİM AMACİ

Belirli guvenlik veya hata durumlarinda portun otomatik kapanmasini saglamak.

SYNTAX

```
Switch(config)# errdisable detect cause {all | arp-inspection |
bpduguard | psecure-violation | dhcp-rate-limit | dtp-flap | ...}
```

ORNEK KULLANİM

```
Switch(config)# errdisable detect cause bpduguard
Switch(config)# errdisable detect cause psecure-violation
```

SİK YAPİLAN HATA

'all' secenegi tum nedenleri kapsar. Ancak bazi nedenler istenmeyen kesintilere yol acabilir. Spesifik nedenler belirtilmelidir.

CİKTİ ORNEGİ

```
Switch# show errdisable detect
ErrDisable Reason            Detection status
-----------------            ----------------
arp-inspection               Enabled
bpduguard                    Enabled
psecure-violation            Enabled
storm-control                Enabled
udld                         Enabled
```

15. VPN ve IPsec

> IPsec VPN, GRE tunnel, GETVPN, DMVPN ve FlexVPN yapilandirmasi icin temel ve ileri seviye komutlaridir.

141 crypto isakmp policy

ACİKLAMA

IKEv1 (ISAKMP) Phase 1 politikasi tanimlar. Encryption, hash, authentication, DH grubu ve lifetime belirtilir.

KULLANİM AMACİ

IPsec VPN'in Phase 1 (anahtar degisimi) parametrelerini yapilandirmak.

SYNTAX

```
Router(config)# crypto isakmp policy priority
Router(config-isakmp)# encryption {des | 3des | aes | aes 192 |
aes 256}
Router(config-isakmp)# hash {sha | sha256 | md5}
Router(config-isakmp)# authentication {rsa-sig | rsa-enc | pre-
share}
Router(config-isakmp)# group {1 | 2 | 5 | 14 | 15 | 16}
Router(config-isakmp)# lifetime seconds
```

ORNEK KULLANİM

```
Router(config)# crypto isakmp policy 10
Router(config-isakmp)# encryption aes 256
Router(config-isakmp)# hash sha256
Router(config-isakmp)# authentication pre-share
Router(config-isakmp)# group 14
Router(config-isakmp)# lifetime 86400
```

SİK YAPİLAN HATA

Her iki ucun da ayni encryption, hash, authentication ve DH grubunu kullanmasi gerekir. Priority dusuk olan tercih edilir.

CİKTİ ORNEGİ

```
Router# show crypto isakmp policy
Global IKE policy
Protection suite of priority 10
        encryption algorithm:   AES-256
        hash algorithm:         SHA256
        authentication method:  Pre-Shared Key
        Diffie-Hellman group:   14 (2048 bit)
        lifetime:               86400 seconds, no volume limit
Default protection suite
        encryption algorithm:   DES
```

142 crypto ipsec transform-set

ACİKLAMA

IPsec Phase 2 transform set'i tanimlar. ESP/AH encapsulation, encryption ve hash algoritmalarini belirler.

KULLANİM AMACİ

IPsec Phase 2 (veri trafigi) sifreleme ve bütunluk parametrelerini yapilandirmak.

SYNTAX

```
Router(config)# crypto ipsec transform-set name transform1 [transform2] [transform3] [mode {tunnel | transport}]
```

ORNEK KULLANİM

```
Router(config)# crypto ipsec transform-set VPN-TRANSFORM esp-aes 256 esp-sha-hmac
Router(cfg-crypto-trans)# mode tunnel
```

SİK YAPİLAN HATA

Her iki VPN ucu da ayni transform-set'i kullanmalidir. Transport mode sadece host-to-host VPN'de kullanilir. Site-to-site'de tunnel mode zorunludur.

CİKTİ ORNEGİ

```
Router# show crypto ipsec transform-set
Transform set VPN-TRANSFORM: { esp-256-aes esp-sha-hmac }
   will negotiate = { Tunnel, },
```

143 crypto map

ACİKLAMA

Crypto map, ISAKMP/IKE politikalarini, transform-set'leri ve erisim listelerini bir araya getirerek bir arayuze uygular.

KULLANİM AMACİ

Site-to-site IPsec VPN'in tum bilesenlerini birlestirmek ve fiziksel arayuze uygulamak.

SYNTAX

```
Router(config)# crypto map map-name seq-num ipsec-isakmp
Router(config-crypto-map)# set peer ip-address
Router(config-crypto-map)# set transform-set name
Router(config-crypto-map)# match address acl-number
Router(config-crypto-map)# set pfs [group1 | group2 | group5 |
group14]
```

ORNEK KULLANİM

```
Router(config)# crypto map VPN-MAP 10 ipsec-isakmp
Router(config-crypto-map)# set peer 203.0.113.2
Router(config-crypto-map)# set transform-set VPN-TRANSFORM
Router(config-crypto-map)# match address 100
Router(config-crypto-map)# set pfs group14
```

SİK YAPİLAN HATA

Bir arayuze sadece bir crypto map uygulanabilir. Birden fazla VPN peer icin ayni crypto map'in farkli sequence numaralari kullanilir.

CİKTİ ORNEGİ

```
Router# show crypto map
Crypto Map "VPN-MAP" 10 ipsec-isakmp
        Peer = 203.0.113.2
        Extended IP access list 100
                access-list 100 permit ip 192.168.1.0 0.0.0.255
10.2.2.0 0.0.0.255
        Current peer: 203.0.113.2
            Security association lifetime: 4608000 kilobytes/3600
seconds
        PFS (Y/N): Y
```

144 interface tunnel

ACİKLAMA

Mantiksal (sanal) bir tunnel arayuzu olusturur. GRE, IPsec over GRE veya diger tunnel protokolleri icin kullanilir.

KULLANİM AMACİ

Farkli protokolleri (multicast, non-IP) IP uzerinden tasimak, overlay network olusturmak.

SYNTAX

```
Router(config)# interface tunnel number
Router(config-if)# tunnel source {ip-address | interface-type
number}
Router(config-if)# tunnel destination ip-address
Router(config-if)# ip address ip-address subnet-mask
```

ORNEK KULLANİM

```
Router(config)# interface tunnel 0
Router(config-if)# tunnel source GigabitEthernet0/0/1
Router(config-if)# tunnel destination 203.0.113.2
Router(config-if)# ip address 172.16.1.1 255.255.255.252
```

SİK YAPİLAN HATA

Tunnel source ve destination IP'lerinin routing tablosunda erisilebilir olmasi gerekir. Tunnel IP'si farkli bir subnet olmalidir.

CİKTİ ORNEGİ

```
Router# show interface tunnel 0
Tunnel0 is up, line protocol is up
  Hardware is Tunnel
  Internet address is 172.16.1.1/30
  MTU 9972 bytes, BW 100 Kbit/sec, DLY 50000 usec,
   Tunnel source 203.0.113.1 (GigabitEthernet0/0/1), destination
203.0.113.2
  Tunnel protocol/transport GRE/IP
```

145 show crypto ipsec sa

ACİKLAMA IPsec Security Association (SA) durumunu goruntuler. Tunnel'in kurulu olup olmadigini, encryption parametrelerini gosterir.

KULLANİM AMACİ IPsec VPN tunnel'inin calistigini dogrulamak, SA istatistiklerini incelemek.

SYNTAX

```
Router# show crypto ipsec sa [peer ip-address | map map-name]
```

ORNEK KULLANİM

```
Router# show crypto ipsec sa peer 203.0.113.2
```

SİK YAPİLAN HATA SA sayisi 0 ise tunnel kurulmamistir. 'show crypto isakmp sa' ile Phase 1 kontrol edilmelidir.

CİKTİ ORNEGİ

```
Router# show crypto ipsec sa peer 203.0.113.2

interface: GigabitEthernet0/0/1
    Crypto map tag: VPN-MAP, local addr 203.0.113.1

   protected vrf: (none)
                  local           ident      (addr/mask/prot/port):
(192.168.1.0/255.255.255.0/0/0)
                    remote       ident      (addr/mask/prot/port):
(10.2.2.0/255.255.255.0/0/0)
```

146 crypto isakmp key

ACİKLAMA Pre-shared key (PSK) tabanli IKE authentication icin karsilikh sifre tanimlar.

KULLANİM AMACİ Site-to-site VPN'de IKE Phase 1 kimlik dogrulamasi icin paylasilan sifre belirlemek.

SYNTAX

```
Router(config)# crypto isakmp key key-string address peer-address
[mask]
```

ORNEK KULLANİM

```
Router(config)#   crypto   isakmp   key   VPNS@cret2024   address
203.0.113.2
```

SİK YAPİLAN HATA 0.0.0.0 mask ile wildcard PSK kullanilabilir ancak guvenlik riski olusturur. Her peer icin ayri key kullanilmalidir.

CİKTİ ORNEGİ

```
Router# show crypto isakmp key
Keyring           Hostname/Address                PSK
(default)         203.0.113.2                     VPNS@cret2024
```

147 tunnel mode gre ip

ACİKLAMA Tunnel arayuzunun GRE (Generic Routing Encapsulation) modunda calismasini saglar. IP uzerinden GRE tunnel'i olusturur.

KULLANİM AMACİ IPsec over GRE tunnel olusturmak, multicast trafigini site-to-site tasimak.

SYNTAX

```
Router(config-if)# tunnel mode gre ip
Router(config-if)# tunnel mode gre multipoint
```

ORNEK KULLANİM

```
Router(config)# interface tunnel 0
Router(config-if)# tunnel mode gre ip
```

SİK YAPİLAN HATA GRE basliksizdir (not encrypted). GRE uzerinden giden trafigi korumak icin IPsec uygulanmalidir.

CİKTİ ORNEGİ

```
Router# show interface tunnel 0 | include protocol
  Tunnel protocol/transport GRE/IP
  Key disabled, sequencing disabled
  Checksumming of packets disabled
```

148 crypto isakmp nat keepalive

ACİKLAMA NAT traversal (NAT-T) icin keepalive mesajlarini etkinlestirir. UDP 4500 uzerinden NAT icinden IPsec calistirir.

KULLANİM AMACİ IPsec VPN'in NAT/PAT arkasında calismasini saglamak.

SYNTAX

```
Router(config)# crypto isakmp nat keepalive seconds
```

ORNEK KULLANİM

```
Router(config)# crypto isakmp nat keepalive 20
```

SİK YAPİLAN HATA NAT-T her iki ucun da desteklemesi gerekir. Bazı eski IOS surumlerinde varsayilan olarak kapali olabilir.

CİKTİ ORNEGİ

```
Router# show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst             src             state          conn-id status
203.0.113.2     192.168.1.100   QM_IDLE           1002 ACTIVE
! src IP NAT'tan gecmis private IP ise NAT-T aktif demektir
```

149 show crypto isakmp sa

ACİKLAMA IKE Phase 1 Security Association (SA) durumunu goruntuler. IPsec tunnel'in ilk asamasinin kurulu olup olmadigini gosterir.

KULLANİM AMACİ VPN tunnel Phase 1'in kuruldugunu dogrulamak, sorun giderme yapmak.

SYNTAX

```
Router# show crypto isakmp sa [detail]
```

ORNEK KULLANİM

```
Router# show crypto isakmp sa
```

SİK YAPİLAN HATA 'MM_NO_STATE' veya 'AG_NO_STATE' hala Phase 1 tamamlanmadi anlamina gelir. Pre-shared key, policy uyumsuzlugu veya erisim listesi sorunu olabilir.

CİKTİ ORNEGİ

```
Router# show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst             src             state          conn-id status
203.0.113.1     203.0.113.2     QM_IDLE           1001 ACTIVE

! QM_IDLE = Quick Mode Idle = Phase 1 basariyla tamamlandi
! MM_NO_STATE = Main Mode tamamlanmadi (sorun var)
```

150 crypto ikev2 proposal / policy

ACİKLAMA

IKEv2 (Internet Key Exchange version 2) proposal ve policy tanimlar. IKEv1'e gore daha guvenli ve hizlidir.

KULLANİM AMACİ

Modern IPsec VPN'ler icin IKEv2 kullanmak, daha guclu guvenlik saglamak.

SYNTAX

```
Router(config)# crypto ikev2 proposal name
Router(config-ikev2-proposal)# encryption {aes-cbc-128 | aes-cbc-
256 | ...}
Router(config-ikev2-proposal)# integrity {sha384 | sha256 | md5}
Router(config-ikev2-proposal)# group {19 | 20 | 21 | 24}

Router(config)# crypto ikev2 policy name
Router(config-ikev2-policy)# proposal proposal-name
```

ORNEK KULLANİM

```
Router(config)# crypto ikev2 proposal IKEV2-PROPOSAL
Router(config-ikev2-proposal)# encryption aes-cbc-256
Router(config-ikev2-proposal)# integrity sha384
Router(config-ikev2-proposal)# group 19
```

SİK YAPİLAN HATA

IKEv1 ve IKEv2 birbiriyle uyumlu degildir. Her iki uc da IKEv2 kullanmali. IKEv2'de 'authentication method' ayri tanimlanir.

CİKTİ ORNEGİ

```
Router# show crypto ikev2 proposal
IKEv2 proposal: IKEV2-PROPOSAL
 Encryption : AES-CBC-256
 Integrity  : SHA384
 PRF        : SHA384
 DH Group   : DH_GROUP_256_ECP/Group 19
```

16. QoS ve Trafik Yonetimi

> Quality of Service (QoS) ile ag trafiginin onceliklendirilmesi, bant genisligi yonetimi ve hizmet kalitesi garantisi saglayan komutlardir.

151 class-map

ACİKLAMA

QoS sinif haritasi (class map) olusturur. Trafigi belirli kriterlere gore (ACL, DSCP, protocol) siniflandirir.

KULLANİM AMACİ

Farkli trafik turlerini (ses, video, veri) birbirinden ayirmak, ayri QoS politikasi uygulamak.

SYNTAX

```
Router(config)# class-map [match-all | match-any] class-name
Router(config-cmap)# match {ip dscp dscp-value | ip precedence value | access-group name | protocol protocol-name}
```

ORNEK KULLANİM

```
Router(config)# class-map match-all VOICE
Router(config-cmap)# match ip dscp ef
Router(config)# class-map match-any VIDEO
Router(config-cmap)# match ip dscp af41 af42
```

SİK YAPİLAN HATA

'match-all' = tum kriterler eslesmeli. 'match-any' = en az bir kriter eslesmeli. Varsayilan 'match-all'dir.

CİKTİ ORNEGİ

```
Router# show class-map
 Class Map match-all VOICE (id 1)
   Match ip dscp ef (46)
 Class Map match-any VIDEO (id 2)
   Match ip dscp af41 (34) af42 (36)
```

152 policy-map

ACİKLAMA QoS politika haritasi (policy map) olusturur. Her sinif icin bandwith, priority, shape gibi eylemler tanimlar.

KULLANİM AMACİ Siniflandirilmis trafige farkli QoS davranislari atamak.

SYNTAX

```
Router(config)# policy-map policy-name
Router(config-pmap)# class class-name
Router(config-pmap-c)# {priority kbps [burst] | bandwidth kbps |
shape average kbps | police kbps [burst] [conform-action | exceed-
action]}
```

ORNEK KULLANİM

```
Router(config)# policy-map QOS-POLICY
Router(config-pmap)# class VOICE
Router(config-pmap-c)# priority 1000
Router(config-pmap)# class VIDEO
Router(config-pmap-c)# bandwidth 4000
Router(config-pmap)# class class-default
Router(config-pmap-c)# bandwidth 5000
```

SİK YAPİLAN HATA 'priority' (LLQ) voice icin, 'bandwidth' (CBWFQ) diger trafige kullanilir. 'class class-default' tanimlanmazsa varsayilan best-effort olur.

CİKTİ ORNEGİ

```
Router# show policy-map
  Policy Map QOS-POLICY
    Class VOICE
      Strict Priority
      Bandwidth 1000 (kbps) Burst 25000 (Bytes)
    Class VIDEO
      Bandwidth 4000 (kbps) Max Threshold 64 (packets)
    Class class-default
      Bandwidth 5000 (kbps) Max Threshold 64 (packets)
```

153 service-policy

ACİKLAMA Olusturulan policy-map'i bir arayuze (input/output) uygular. MQC (Modular QoS CLI) zincirinin son halkasidir.

KULLANİM AMACİ QoS politikasini aktif hale getirmek, belirli arayuzde trafigi yonetmek.

SYNTAX

```
Router(config-if)# service-policy {input | output} policy-name
```

ORNEK KULLANİM

```
Router(config)# interface GigabitEthernet0/0/1
Router(config-if)# service-policy output QOS-POLICY
```

SİK YAPİLAN HATA 'input' ve 'output' ayni anda uygulanabilir ancak farkli policy-map'ler olmalidir. LLQ (priority) sadece output'ta calisir.

CİKTİ ORNEGİ

```
Router# show policy-map interface GigabitEthernet0/0/1

 GigabitEthernet0/0/1

  Service-policy output: QOS-POLICY

    Class-map: VOICE (match-all)
      5234 packets, 1046800 bytes
      5 minute offered rate 0000 bps, drop rate 0000 bps
      Match: ip dscp ef (46)
```

154 auto qos

ACİKLAMA Cisco AutoQoS ozelligi ile otomatik olarak voice ve video trafigine uygun QoS yapilandirmasi uygular.

KULLANİM AMACİ Hizli ve basit bir sekilde VoIP QoS yapilandirmasi yapmak.

SYNTAX

```
Router(config-if)# auto qos voip [trust] [cisco-phone]
Switch(config-if)# auto qos voip [cisco-phone | cisco-softphone |
trust]
```

ORNEK KULLANİM

```
Router(config)# interface GigabitEthernet0/0/1
Router(config-if)# auto qos voip trust
```

SİK YAPİLAN HATA

AutoQoS uygulandiktan sonra olusturulan class-map/policy-map'ler manuel olarak degistirilebilir. Ancak sonraki AutoQoS uygulamasi bu degisiklikleri ezer.

CİKTİ ORNEGİ

```
Router(config-if)# auto qos voip trust
% Applying new configuration. Please wait...

Router# show auto qos
AutoQoS Policy id: voip
  Template: Default

  trust dscp
  policy-map AutoQos-4.0-Output-Policy
    class AutoQos-4.0-VoIP-Queue
```

155 mls qos

ACİKLAMA Switch'lerde Multi-Layer Switching (MLS) QoS ozelligini etkinlestirir. Trust/CoS/DSCP islemleri icin gereklidir.

KULLANİM AMACİ Switch uzerinde QoS islemlerini etkinlestirmek.

SYNTAX

```
Switch(config)# mls qos
```

ORNEK KULLANİM

```
Switch(config)# mls qos
```

SİK YAPİLAN HATA 'mls qos' etkinlestirilmeden CoS/DSCP trust veya queuing islemleri calismaz. Varsayilan olarak tum portlar untrusted olur.

CİKTİ ORNEGİ

```
Switch(config)# mls qos
Switch# show mls qos
QoS is enabled
QoS ip packet dscp rewrite is enabled
```

156 mls qos trust

ACİKLAMA Switch portunun gelen trafikteki CoS veya DSCP degerlerine guvenmesini (trust) saglar.

KULLANİM AMACİ IP Phone veya router gibi DSCP/CoS isaretleyen cihazlarin isaretlerine saygi gostermek.

SYNTAX

```
Switch(config-if)# mls qos trust {cos | dscp | ip-precedence |
device cisco-phone}
```

ORNEK KULLANİM

```
Switch(config-if)# mls qos trust dscp
Switch(config-if)# mls qos trust device cisco-phone
```

SİK YAPİLAN HATA Cisco Phone CDP ile gorunur. 'device cisco-phone' ile telefon yoksa port untrusted kalir. IP Phone'suz ortamda 'trust dscp' kullanilmalidir.

CİKTİ ORNEGİ

```
Switch# show mls qos interface GigabitEthernet0/1
GigabitEthernet0/1
  trust state: trust dscp
  trust mode: trust dscp
  trust enabled flag: ena
  COS override: dis
  DSCP Mutation Map: Default DSCP Mutation Map
  Trust device: cisco-phone
```

157 priority-queue out

ACİKLAMA Switch portu uzerinde strict priority queue (PQ) cikisini etkinlestirir. Voice trafigi icin kritiktir.

KULLANİM AMACİ VoIP trafiginin diger trafige gecismeden oncelikli olarak iletilmesini saglamak.

SYNTAX

```
Switch(config-if)# priority-queue out
```

ORNEK KULLANİM

```
Switch(config-if)# priority-queue out
```

SİK YAPİLAN HATA

Sadece trunk ve routed portlarda calisir. Access portlarda 'srr-queue bandwidth share' ile ayarlanir.

CİKTİ ORNEGİ

```
Switch# show mls qos interface GigabitEthernet0/1 queueing
GigabitEthernet0/1
  Egress priority queue: enabled
  Shaped queue weights (absolute): 25 0 0 0
  Shared queue weights: 25 25 25 25
  The port bandwidth limit: 100 (Operational Bandwidth:100.0)
  The port is mapped to qset: 1
```

158 srr-queue bandwidth share

ACİKLAMA Shared Round Robin (SRR) kuyruk bant genisligi paylasim oranlarini ayarlar.

KULLANİM AMACİ 4 cikis kuyrugu arasinda bant genisligi dagilimini oranlamak.

SYNTAX

```
Switch(config-if)# srr-queue bandwidth share weight1 weight2
weight3 weight4
```

ORNEK KULLANİM

```
Switch(config-if)# srr-queue bandwidth share 30 20 25 25
```

SİK YAPİLAN HATA

Oranlar yuzde degil, agirlik degerleridir. Toplami 100 olmasi gerekmez ancak kolay hesaplama icin 100 kullanilir.

CİKTİ ORNEGİ

```
Switch# show mls qos interface GigabitEthernet0/2 queueing
GigabitEthernet0/2
  Egress priority queue: disabled
  Shaped queue weights (absolute): 25 0 0 0
  Shared queue weights: 30 20 25 25
```

159 shape average

ACİKLAMA

Trafigi belirli bir ortalama hizda sinirlar (traffic shaping). Burst trafige izin verir ancak uzun vadede hedef hiza getirir.

KULLANİM AMACİ

ISP'ye giden trafigi sinirlandirmak, CIR (Committed Information Rate) ile eslestirmek.

SYNTAX

```
Router(config-pmap-c)# shape average {target-bit-rate [sustained-bits [excess-bits]] | percent percent}
```

ORNEK KULLANİM

```
Router(config-pmap-c)# shape average 10000000
```

SİK YAPİLAN HATA

Shaping degeri bit/saniye (bps) olarak girilir. 10 Mbps = 10000000. 'police' shaping'den farklidir; policing token dusurur, shaping kuyruklar.

CİKTİ ORNEGİ

```
Router# show policy-map interface GigabitEthernet0/0/1
  Service-policy output: SHAPE-10M

    Class-map: class-default (match-any)
      15234 packets, 12456789 bytes
      5 minute offered rate 9800000 bps, drop rate 0 bps
      Match: any
      Queueing
      queue limit 64 packets
      (queue depth/total drops/no-buffer drops) 0/0/0
```

160 police

ACİKLAMA

Trafigi belirli bir hizda kati olarak sinirlar (traffic policing). Limit asildiginda paketler dusurulur veya isaretlenir.

KULLANİM AMACİ

Asiri trafigi kati olarak sinirlandirmak, SLA ihlallerinde paket dusurmek veya remarke etmek.

SYNTAX

```
Router(config-pmap-c)# police {cir cir-bps | percent percent} [bc burst] [pir pir-bps] [conform-action action] [exceed-action action] [violate-action action]
```

ORNEK KULLANİM

```
Router(config-pmap-c)# police 5000000 8000 exceed-action drop
Router(config-pmap-c)# police cir 10000000 bc 40000 pir 15000000 be 60000 conform-action transmit exceed-action set-dscp-transmit af11 violate-action drop
```

SİK YAPİLAN HATA

Policing ani dusurme yapar. Voice trafigine policing uygulanmamali. BC (burst committed) degeri cok dusuk olursa gereksiz yere paket duser.

CİKTİ ORNEGİ

```
Router# show policy-map interface
    Class-map: VIDEO (match-all)
      5234 packets, 5234000 bytes
      5 minute offered rate 5000000 bps, drop rate 500000 bps
      Match: ip dscp af41 (34)
      police:
        cir 5000000 bps, conform-burst 8000 bits
        conform: 4500 packets, 4500000 bytes  action: transmit
        exceed: 734 packets, 734000 bytes    action: drop
        violate: 0 packets, 0 bytes          action: drop
```

17. IPv6 Yapilandirma

> IPv6 adresleme, yonlendirme ve dual-stack yapilandirmasi icin temel ve ileri seviye komutlardir.

161 ipv6 enable

ACİKLAMA Bir arayuzde IPv6'yi etkinlestirir. Arayuzun link-local adresi otomatik olusur.

KULLANİM AMACİ Arayuzde IPv6 islemlerini baslatmak, link-local iletisimi saglamak.

SYNTAX

```
Router(config-if)# ipv6 enable
```

ORNEK KULLANİM

```
Router(config-if)# ipv6 enable
```

SİK YAPİLAN HATA

Sadece 'ipv6 enable' ile sadece link-local adres olusur. Global unicast iletisim icin 'ipv6 address' da tanimlanmalidir.

CİKTİ ORNEGİ

```
Router(config-if)# ipv6 enable
Router# show ipv6 interface GigabitEthernet0/0/1
GigabitEthernet0/0/1 is up, line protocol is up
  IPv6 is enabled, link-local address is FE80::AABB:CCFF:FEDD:EE01
  No global unicast address is configured
  Joined group address(es):
    FF02::1
    FF02::2
    FF02::1:FFDD:EE01
```

162 ipv6 address

ACİKLAMA Bir arayuze IPv6 global unicast adresi atar. EUI-64 formati veya tam adres tanimlanabilir.

KULLANİM AMACİ Arayuze IPv6 adresi atamak, ag uzerinden IPv6 iletisimi saglamak.

SYNTAX

```
Router(config-if)# ipv6 address ipv6-address/prefix-length [eui-64]
Router(config-if)# ipv6 address ipv6-prefix/prefix-length eui-64
```

ORNEK KULLANİM

```
Router(config-if)# ipv6 address 2001:DB8:1:1::1/64
Router(config-if)# ipv6 address 2001:DB8:1:2::/64 eui-64
```

SİK YAPİLAN HATA EUI-64'te 64 bit prefix verilir, MAC adresinden 64 bit interface ID otomatik olusur. /64 disinda prefix kullanilmamalidir.

CİKTİ ORNEGİ

```
Router# show ipv6 interface brief
GigabitEthernet0/0/1     [up/up]
    FE80::AABB:CCFF:FEDD:EE01
    2001:DB8:1:1::1
GigabitEthernet0/0/2     [up/up]
    FE80::AABB:CCFF:FEDD:EE02
    2001:DB8:1:2:AABB:CCFF:FEDD:EE02
```

163 ipv6 unicast-routing

ACİKLAMA Cihazda IPv6 yonlendirmeyi (IPv6 unicast routing) etkinlestirir. Varsayilan olarak Layer 3 cihazlarda kapalidir.

KULLANİM AMACİ IPv6 yonlendirme protokollerini calistirmak, IPv6 paketlerini yonlendirmek.

SYNTAX

```
Router(config)# ipv6 unicast-routing
```

ORNEK KULLANİM

```
Router(config)# ipv6 unicast-routing
```

SİK YAPİLAN HATA

'ipv6 unicast-routing' olmadan cihaz kendi aglarini yonlendiremez. Ping yapabilir ancak router olarak calismaz.

CİKTİ ORNEGİ

```
Router(config)# ipv6 unicast-routing
Router# show ipv6 route
IPv6 Routing Table - default - 4 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static
route
       B - BGP, R - RIP, H - NHRP, I1 - ISIS L1
        O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1, OE2 -
OSPF ext 2
C   2001:DB8:1:1::/64 [0/0]
     via GigabitEthernet0/0/1, directly connected
```

164 ipv6 route

ACİKLAMA IPv6 statik yol tanimlar. Varsayilan yol (::/0) veya ozel hedef aglar icin kullanilir.

KULLANİM AMACİ IPv6 trafigini yonlendirmek, varsayilan gateway tanimlamak.

SYNTAX

```
Router(config)# ipv6 route ipv6-prefix/prefix-length {next-hop-address | interface-type interface-number} [administrative-distance]
```

ORNEK KULLANİM

```
Router(config)# ipv6 route ::/0 2001:DB8:1:1::2
Router(config)# ipv6 route 2001:DB8:2::/64 GigabitEthernet0/0/2
```

SİK YAPİLAN HATA IPv6'da next-hop link-local adresi (FE80::) kullanilirsa cikis arayuzu belirtilmelidir. Yoksa hangi arayuzden gidilecegi belirsiz olur.

CİKTİ ORNEGİ

```
Router# show ipv6 route static
S   ::/0 [1/0]
     via 2001:DB8:1:1::2
S   2001:DB8:2::/64 [1/0]
     via GigabitEthernet0/0/2, directly connected
```

165 ipv6 ospf

ACİKLAMA Bir arayuzde OSPFv3 (IPv6 OSPF) yonlendirmesini etkinlestirir. Process ID ve area belirtilir.

KULLANİM AMACİ IPv6 trafigi icin OSPF dinamik yonlendirme saglamak.

SYNTAX

```
Router(config-if)# ipv6 ospf process-id area area-id [instance instance-id]
```

ORNEK KULLANİM

```
Router(config-if)# ipv6 ospf 1 area 0
```

SİK YAPİLAN HATA OSPFv3'te network komutu yerine arayuz bazli etkinlestirme yapilir. Ayni process hem IPv4 hem IPv6'yi yonlendirebilir (AF).

CİKTİ ORNEGİ

```
Router# show ipv6 ospf interface
GigabitEthernet0/0/1 is up, line protocol is up
  Link Local Address FE80::AABB:CCFF:FEDD:EE01, Interface ID 1
  Area 0, Process ID 1, Instance ID 0, Router ID 1.1.1.1
  Network Type BROADCAST, Cost: 1
  State DR, Priority 1
```

166 show ipv6 route

ACİKLAMA IPv6 yonlendirme tablosunu goruntuler. Connected, local, static, OSPF, BGP gibi kaynaklardan gelen yollari gosterir.

KULLANİM AMACİ IPv6 yonlendirme tablosunu kontrol etmek, yol varligini dogrulamak.

SYNTAX

```
Router# show ipv6 route [connected | static | ospf | bgp | summary]
```

ORNEK KULLANİM

```
Router# show ipv6 route
```

SİK YAPİLAN HATA

'show ip route' IPv4'u gosterir. IPv6 icin mutlaka 'show ipv6 route' kullanilmalidir.

CİKTİ ORNEGİ

```
Router# show ipv6 route
IPv6 Routing Table - default - 5 entries
C   2001:DB8:1:1::/64 [0/0]
     via GigabitEthernet0/0/1, directly connected
L   2001:DB8:1:1::1/128 [0/0]
     via GigabitEthernet0/0/1, receive
O   2001:DB8:1:2::/64 [110/2]
     via FE80::AABB:CCFF:FEDD:EE02, GigabitEthernet0/0/1
S   ::/0 [1/0]
     via 2001:DB8:1:1::2
```

167 ipv6 dhcp pool

ACİKLAMA IPv6 DHCP sunucusu icin adres havuzu (Stateful DHCPv6) olusturur. Prefix ve diger parametreleri icerir.

KULLANİM AMACİ IPv6 istemcilere otomatik IP adresi, DNS ve diger parametreleri dagitmak.

SYNTAX

```
Router(config)# ipv6 dhcp pool pool-name
Router(config-dhcp)# address prefix ipv6-prefix [lifetime valid-lifetime preferred-lifetime]
Router(config-dhcp)# dns-server ipv6-address
Router(config-dhcp)# domain-name domain-name
```

ORNEK KULLANİM

```
Router(config)# ipv6 dhcp pool VLAN10-IPv6
Router(config-dhcp)# address prefix 2001:DB8:1:1::/64
Router(config-dhcp)# dns-server 2001:DB8::8:8:8:8
Router(config-dhcp)# domain-name company.local
```

SİK YAPİLAN HATA

Stateful DHCPv6 icin 'managed-config-flag' veya 'other-config-flag' arayuzde ayarlanmalidir. Stateless DHCPv6 sadece other-config-flag gerektirir.

CİKTİ ORNEGİ

```
Router# show ipv6 dhcp pool
DHCPv6 pool: VLAN10-IPv6
    Address allocation prefix: 2001:DB8:1:1::/64 valid 172800 preferred 86400 (2 in use, 0 conflicts)
  DNS server: 2001:DB8::8:8:8:8
  Domain name: company.local
  Active clients: 2
```

168 ipv6 nd other-config-flag / managed-config-flag

ACİKLAMA

Router Advertisement (RA) mesajlarinda O-bit ve M-bit'i ayarlar. Istemcilerin DHCPv6 kullanip kullanmayacagini belirler.

KULLANİM AMACİ

DHCPv6 davranisini kontrol etmek. SLAAC, Stateless DHCPv6 veya Stateful DHCPv6 secimi.

SYNTAX

```
Router(config-if)# ipv6 nd other-config-flag
Router(config-if)# ipv6 nd managed-config-flag
```

ORNEK KULLANİM

```
Router(config-if)# ipv6 nd other-config-flag
Router(config-if)# ipv6 nd managed-config-flag
```

SİK YAPİLAN HATA

M=1, O=0 anlamsizdir. M=1 ise O'nun degeri onemsizdir. M=0, O=1 = Stateless DHCPv6. M=0, O=0 = saf SLAAC.

CİKTİ ORNEGİ

```
Router# show ipv6 interface GigabitEthernet0/0/1 | include flag
  ICMP error messages limited to one every 100 milliseconds
  ND advertised default router preference is Medium
  Hosts use DHCP to obtain routable addresses.
  Hosts use DHCP to obtain other configuration.
```

169 ipv6 access-list

ACİKLAMA IPv6 erisim kontrol listesi (ACL) olusturur. Named ACL formatinda olur. Implicit permit icin 'permit icmp any any nd-na' ve 'permit icmp any any nd-ns' gerekir.

KULLANİM AMACİ IPv6 trafigini filtrelemek, guvenlik politikasi uygulamak.

SYNTAX

```
Router(config)# ipv6 access-list name
Router(config-ipv6-acl)# {permit | deny} protocol {source-ipv6-prefix/prefix-length | any | host source-ipv6-address} [operator [port-number]] {destination-ipv6-prefix/prefix-length | any | host destination-ipv6-address} [operator [port-number]] [log]
```

ORNEK KULLANİM

```
Router(config)# ipv6 access-list BLOCK-ICMP
Router(config-ipv6-acl)# deny icmp any any echo-request log
Router(config-ipv6-acl)# permit ipv6 any any
```

SİK YAPİLAN HATA

IPv6 ACL'de ND (Neighbor Discovery) paketlerine izin verilmezse IPv6 komsuluk calismaz. Implicit deny vardir.

CİKTİ ORNEGİ

```
Router# show ipv6 access-list
IPv6 access list BLOCK-ICMP
    deny icmp any any echo-request log (15 matches) sequence 10
    permit ipv6 any any (2345 matches) sequence 20
```

170 ipv6 traffic-filter

ACİKLAMA IPv6 ACL'yi bir arayuze gelen (in) veya giden (out) trafik icin uygular.

KULLANİM AMACİ IPv6 ACL'yi aktif hale getirmek, IPv6 trafigini filtrelemek.

SYNTAX

```
Router(config-if)# ipv6 traffic-filter access-list-name {in | out}
```

ORNEK KULLANİM

```
Router(config-if)# ipv6 traffic-filter BLOCK-ICMP in
```

SİK YAPİLAN HATA

'ipv6 traffic-filter' ile 'ip access-group' farkli komutlardir. IPv4 ACL 'ip access-group', IPv6 ACL 'ipv6 traffic-filter' ile uygulanir.

CİKTİ ORNEGİ

```
Router# show ipv6 interface GigabitEthernet0/0/1 | include filter
  Inbound  access list BLOCK-ICMP
```

18. MPLS ve VRF

> Multi-Protocol Label Switching (MPLS) ve Virtual Routing and Forwarding (VRF) ile VPN ve trafik yonetimi komutlaridir.

171 ip vrf

ACİKLAMA

Bir VRF (Virtual Routing and Forwarding) omegi olusturur. Ayni cihazda bagimsiz routing tablolari olusturur.

KULLANİM AMACİ

Musteri/trafik izolasyonu saglamak, MPLS L3VPN icin CE-PE baglantilarini yonetmek.

SYNTAX

```
Router(config)# ip vrf vrf-name
Router(config-vrf)# rd route-distinguisher
Router(config-vrf)# route-target {import | export | both} route-target-ext-community
```

ORNEK KULLANİM

```
Router(config)# ip vrf CUSTOMER-A
Router(config-vrf)# rd 65001:100
Router(config-vrf)# route-target both 65001:100
```

SİK YAPİLAN HATA

VRF'ye atanmamis arayuzler global (default) routing tablosuna aittir. 'ip vrf forwarding' ile arayuz VRF'ye baglanmalidir.

CİKTİ ORNEGİ

```
Router# show ip vrf
   Name                                     Default RD
Interfaces
    CUSTOMER-A                              65001:100
Gi0/0/1.10
    CUSTOMER-B                              65001:200
Gi0/0/1.20
```

172 ip vrf forwarding

ACİKLAMA Bir arayuzu belirtilen VRF'ye atar. Arayuz uzerindeki IP yapilandirmasi silinir ve yeniden tanimlanmalidir.

KULLANİM AMACİ Fiziksel veya sanal (subinterface) arayuzu bir VRF'ye baglamak.

SYNTAX

```
Router(config-if)# ip vrf forwarding vrf-name
```

ORNEK KULLANİM

```
Router(config)# interface GigabitEthernet0/0/1.10
Router(config-subif)# encapsulation dot1Q 10
Router(config-subif)# ip vrf forwarding CUSTOMER-A
% Interface GigabitEthernet0/0/1.10 IP address removed due to
enabling VRF CUSTOMER-A
Router(config-subif)# ip address 10.1.1.1 255.255.255.0
```

SİK YAPİLAN HATA VRF atamasindan sonra IP adresi silinir. IP adresini mutlaka yeniden tanimlayin. Bu kolayca unutulan bir adimdir.

CİKTİ ORNEGİ

```
Router# show ip interface GigabitEthernet0/0/1.10 | include VRF
  IP verify source vlan is the same as VRF (CUSTOMER-A)
Router# show ip vrf interfaces
Interface                              IP-Address              VRF
Protocol
GigabitEthernet0/0/1.10    10.1.1.1                    CUSTOMER-A
up
```

173 mpls ip

ACİKLAMA Bir arayuzde MPLS etiket dagitimini (label distribution) etkinlestirir. LDP (Label Distribution Protocol) kullanilir.

KULLANİM AMACİ MPLS core network'te LSR (Label Switching Router) olarak calismak, etiket dagitimini baslatmak.

SYNTAX

```
Router(config-if)# mpls ip
```

ORNEK KULLANİM

```
Router(config)# interface GigabitEthernet0/0/1
Router(config-if)# mpls ip
```

SİK YAPİLAN HATA 'mpls ip' etkinlestirilmeden once 'mpls label protocol ldp' ve CEF acik olmalidir. MPLS sadece L3 arayuzlerde calisir.

CİKTİ ORNEGİ

```
Router# show mpls interfaces
Interface                   IP               Tunnel    BGP  Static
Operational
GigabitEthernet0/0/1   Yes (ldp)     No       No  No     Yes
GigabitEthernet0/0/2   Yes (ldp)     No       No  No     Yes
```

174 mpls ldp router-id

ACİKLAMA MPLS LDP router ID'sini manuel olarak belirler. Varsayilan olarak en yuksek loopback IP'si secilir.

KULLANİM AMACİ LDP komsuluklarinin stabil olmasini saglamak, router ID'yi sabitlemek.

SYNTAX

```
Router(config)# mpls ldp router-id interface-type interface-number
[force]
```

ORNEK KULLANİM

```
Router(config)# mpls ldp router-id Loopback0 force
```

SİK YAPİLAN HATA 'force' kullanilmazsa yeni router ID bir sonraki LDP restart'ta etkin olur. Aninda etki icin 'force' kullanilmalidir.

CİKTİ ORNEGİ

```
Router# show mpls ldp discovery
 Local LDP Identifier:
    1.1.1.1:0
    Discovery Sources:
    Interfaces:
        GigabitEthernet0/0/1 (ldp): xmit/recv
            LDP Id: 2.2.2.2:0
```

175 show mpls forwarding-table

ACİKLAMA

MPLS Label Forwarding Information Base (LFIB) tablosunu goruntuler. Gelen etiketlerin nasil islenecegini gosterir.

KULLANİM AMACİ

MPLS etiket anahtarlamasini kontrol etmek, LSP (Label Switched Path) yolunu dogrulamak.

SYNTAX

```
Router# show mpls forwarding-table [vrf vrf-name] [ip-prefix]
```

ORNEK KULLANİM

```
Router# show mpls forwarding-table
```

SİK YAPİLAN HATA

'show mpls ldp bindings' ile 'show mpls forwarding-table' farklidir. Bindings LDP tarafindan dagitilan etiketleri, LFIB gercek forwarding kararlarini gosterir.

CİKTİ ORNEGİ

```
Router# show mpls forwarding-table
Local       Outgoing    Prefix             Bytes Label    Outgoing
Next Hop
Label      Label      or Tunnel Id     Switched      interface
17           Pop Label   2.2.2.2/32         0                Gi0/0/1
10.1.1.2
18           Pop Label   3.3.3.3/32         0                Gi0/0/2
10.2.2.3
19           17           10.3.3.0/24       0                Gi0/0/1
10.1.1.2
```

176 address-family ipv4 vrf

ACİKLAMA Bir yonlendirme protokolunde VRF-specific yapilandirma moduna girer. BGP'de VRF icin route'lari tanimlamak icin kullanilir.

KULLANİM AMACİ MPLS L3VPN'de PE router uzerinde musteri VPN route'larini yapilandirmak.

SYNTAX

```
Router(config-router)# address-family ipv4 vrf vrf-name
Router(config-router-af)# neighbor ip-address remote-as as-number
Router(config-router-af)# neighbor ip-address activate
```

ORNEK KULLANİM

```
Router(config)# router bgp 65001
Router(config-router)# address-family ipv4 vrf CUSTOMER-A
Router(config-router-af)# neighbor 10.1.1.2 remote-as 65002
Router(config-router-af)# neighbor 10.1.1.2 activate
```

SİK YAPİLAN HATA VRF address-family'de 'network' komutu kullanilirken prefix VRF routing tablosunda olmalidir. Aksi halde BGP'ye eklenmez.

CİKTİ ORNEGİ

```
Router# show ip bgp vpnv4 vrf CUSTOMER-A
BGP table version is 12, local router ID is 1.1.1.1
   Network          Next Hop            Metric LocPrf Weight Path
Route Distinguisher: 65001:100 (default for vrf CUSTOMER-A)
*> 10.1.1.0/24      0.0.0.0                  0         32768 i
*>i10.2.2.0/24      2.2.2.2                  0    100      0 i
```

177 show ip route vrf

ACİKLAMA Belirli bir VRF'nin IP yonlendirme tablosunu goruntuler.

KULLANİM AMACİ Musteri VPN route'larini kontrol etmek, VRF izolasyonunu dogrulamak.

SYNTAX

```
Router# show ip route vrf vrf-name [connected | static | bgp | ospf]
```

ORNEK KULLANİM

```
Router# show ip route vrf CUSTOMER-A
```

SİK YAPİLAN HATA

'show ip route' sadece global routing tablosunu gosterir. VRF route'larini gormek icin 'vrf' anahtar kelimesi zorunludur.

CİKTİ ORNEGİ

```
Router# show ip route vrf CUSTOMER-A
Routing Table: CUSTOMER-A
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter
area

Gateway of last resort is not set

C     10.1.1.0/24 is directly connected, GigabitEthernet0/0/1.10
B     10.2.2.0/24 [200/0] via 2.2.2.2, 00:15:32
```

178 mpls ip (global)

ACİKLAMA Cihaz genelinde MPLS IP yonlendirmesini etkinlestirir. CEF zorunludur.

KULLANİM AMACİ MPLS router olarak calismak, butun MPLS ozelliklerini etkinlestirmek.

SYNTAX

```
Router(config)# mpls ip
Router(config)# mpls label protocol ldp
```

ORNEK KULLANİM

```
Router(config)# mpls ip
Router(config)# mpls label protocol ldp
```

SİK YAPİLAN HATA

'mpls ip' etkinlestirilmeden once 'ip cef' acik olmalidir. MPLS CEF'e bagimlidir. 'no mpls ip' tum MPLS islemlerini durdurur.

CİKTİ ORNEGİ

```
Router# show mpls ip
MPLS IP is enabled
LDP IP Notification: enabled
    GigabitEthernet0/0/1 (ldp): xmit/recv
    GigabitEthernet0/0/2 (ldp): xmit/recv
```

179 router bgp address-family vpnv4

ACİKLAMA

BGP'de VPNv4 address-family'i etkinlestirir. MPLS L3VPN'de PE router'lar arasi VPN route dagitimi icin kullanilir.

KULLANİM AMACİ

PE-PE arasi VPNv4 route exchange'i saglamak, MPLS VPN'in temel bilesenini olusturmak.

SYNTAX

```
Router(config-router)# address-family vpnv4
Router(config-router-af)# neighbor ip-address activate
Router(config-router-af)#   neighbor   ip-address   send-community
extended
```

ORNEK KULLANİM

```
Router(config)# router bgp 65001
Router(config-router)# address-family vpnv4
Router(config-router-af)# neighbor 2.2.2.2 activate
Router(config-router-af)# neighbor 2.2.2.2 send-community extended
```

SİK YAPİLAN HATA

'send-community extended' unutulursa route-target'ler tasinmaz ve VPN route'lari PE'ler arasi eslesmez. Bu cok yaygin bir hatadir.

CİKTİ ORNEGİ

```
Router# show ip bgp vpnv4 all summary
BGP router identifier 1.1.1.1, local AS number 65001
Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ
Up/Down  State/PfxRcd
2.2.2.2         4        65001    5234    5232       25    0    0
1d02h    5
```

180 ping vrf

ACİKLAMA Belirli bir VRF icerisinden ping atar. Global routing tablosu yerine VRF routing tablosu kullanilir.

KULLANİM AMACİ VRF icindeki komsuluklari ve erisilebilirligi test etmek.

SYNTAX

```
Router# ping vrf vrf-name ip-address [source source-ip] [repeat count] [timeout seconds]
```

ORNEK KULLANİM

```
Router# ping vrf CUSTOMER-A 10.1.1.2 source 10.1.1.1 repeat 100
```

SİK YAPİLAN HATA 'vrf' belirtilmezse ping global routing tablosundan gider ve VRF icindeki hedeflere ulasamaz.

CİKTİ ORNEGİ

```
Router# ping vrf CUSTOMER-A 10.1.1.2 source 10.1.1.1 repeat 5
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.1.1.2, timeout is 2 seconds:
Packet sent with a source address of 10.1.1.1
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/2/4 ms
```

19. Monitoring ve Debug

> Cihaz izleme, sorun giderme, paket analizi ve debug komutlaridir. Dikkatli kullanilmalidir, uretim sistemlerinde etkisi buyuk olabilir.

181 show running-config

ACİKLAMA Cihazin gecerli (RAM'deki) yapilandirmasini goruntuler. Tum aktif komutlari icerir.

KULLANİM AMACİ Mevcut yapilandirmayi incelemek, backup almak, degisiklikleri dogrulamak.

SYNTAX

```
Router# show running-config [interface type number | section section-name | begin line | include line]
```

ORNEK KULLANİM

```
Router# show running-config interface GigabitEthernet0/0/1
Router# show running-config | section router ospf
Router# show running-config | include hostname
```

SİK YAPİLAN HATA 'show running-config' cok uzun olabilir. '| include', '| section', '| begin' pipe filtreleri ile istenen kisma ulasin.

CİKTİ ORNEGİ

```
Router# show running-config | section router ospf
router ospf 1
 router-id 1.1.1.1
 passive-interface default
 no passive-interface GigabitEthernet0/0/1
 network 10.1.1.0 0.0.0.255 area 0
 default-information originate always
```

182 show startup-config

ACİKLAMA Cihazin NVRAM'deki (kalici) yapilandirmasini goruntuler. Cihaz yeniden baslatildiginda yuklenecek yapilandirmadir.

KULLANİM AMACİ Kalici yapilandirmayi incelemek, running-config ile karsilastirmak.

SYNTAX

```
Router# show startup-config
```

ORNEK KULLANİM

```
Router# show startup-config
```

SİK YAPİLAN HATA

'write memory' yapilmadikca running-config'deki degisiklikler startup-config'e yansimaz.

CİKTİ ORNEGİ

```
Router# show startup-config
Using 4521 out of 524288 bytes
!
version 15.9
service timestamps debug datetime msec
service timestamps log datetime msec
!
hostname Router
!
boot-start-marker
```

183 show version

ACİKLAMA Cihaz hakkinda temel bilgileri goruntuler: IOS versiyonu, hardware model, uptime, bellek, lisans.

KULLANİM AMACİ Cihaz bilgilerini toplamak, yazilim versiyonunu kontrol etmek, TAC case acarken gerekli bilgileri almak.

SYNTAX

```
Router# show version
```

ORNEK KULLANİM

```
Router# show version
```

SİK YAPİLAN HATA

'show version' cok uzun bir ciktidir. '| include' ile versiyon, model, uptime gibi bilgilere hizli erisilebilir.

CİKTİ ORNEGİ

```
Router# show version | include Version|uptime|bytes of memory
Cisco IOS Software, ISR Software (X86_64_LINUX_IOSD-UNIVERSALK9-
M), Version 15.9(3)M2
Router uptime is 45 days, 12 hours, 30 minutes
cisco ISR4331/K9 (1RU) processor with 1791999K/6147K bytes of
memory.
Configuration register is 0x2102
```

184 show processes cpu

ACİKLAMA

Cihazin CPU kullanimini ve sureclerin CPU tuketimini gosterir. 5 saniyelik, 1 dakikalik ve 5 dakikalik ortalamalar sunar.

KULLANİM AMACİ

Yuksek CPU kullaniminin nedenini tespit etmek, performans sorunlarini gidermek.

SYNTAX

```
Router# show processes cpu [sorted] [history]
```

ORNEK KULLANİM

```
Router# show processes cpu sorted
Router# show processes cpu history
```

SİK YAPİLAN HATA

'history' ASCII grafik olarak gosterir. Kisa sureli spike'lari gormek icin faydalidir. Surekli yuksek CPU sorununu 'sorted' ile surec bazli tespit edin.

CİKTİ ORNEGİ

```
Router# show processes cpu
CPU utilization for five seconds: 12%/4%; one minute: 10%; five
minutes: 9%
 PID Runtime(ms)     Invoked      uSecs   5Sec   1Min   5Min TTY
Process
   1           0           1          0  0.00%  0.00%  0.00%   0
Chunk Manager
   2           4       12345          0  0.00%  0.00%  0.00%   0
Load Meter
  10         256        4567         56  0.24%  0.18%  0.15%   0
```

185 show memory statistics

ACİKLAMA Cihazin bellek (RAM) kullanim istatistiklerini goruntuler. Bos, kullanilan ve en buyuk bloklari gosterir.

KULLANİM AMACİ Bellek sizintilarini tespit etmek, bellek yetersizligi sorunlarini gidermek.

SYNTAX

```
Router# show memory statistics
```

ORNEK KULLANİM

```
Router# show memory statistics
```

SİK YAPİLAN HATA 'show memory' ile 'show processes memory' farklidir. Birincisi sistem bellegini, ikincisi surec bazli bellek kullanimini gosterir.

CİKTİ ORNEGİ

```
Router# show memory statistics
                Head    Total(b)    Used(b)    Free(b)    Lowest(b)
Largest(b)
Processor     7F0000    1791999168  452341760  1339657408  1234567890
876543210
      I/O     3F0000     268435456   67342345   201093111   187654321
156789012
```

186 debug

ACİKLAMA

Cihaz uzerinde detayli debug (hata ayiklama) mesajlarini etkinlestirir. Cok dikkatli kullanilmalidir.

KULLANİM AMACİ

Protokol calismasini adim adim izlemek, komsuluk kurma sorunlarini gidermek, paket akisini analiz etmek.

SYNTAX

```
Router# debug {ip {rip | ospf | bgp | nat | dhcp | packet} | {cdp
| arp | frame-relay | ppp} | all}
Router# no debug all / undebug all
```

ORNEK KULLANİM

```
Router# debug ip ospf adj
Router# debug ip bgp updates
Router# undebug all
```

SİK YAPİLAN HATA

'debug all' veya 'debug ip packet' uretim cihazlarinda CPU'yu %100 yapabilir ve cihazi kullanilmaz hale getirebilir. KESINLIKLE kullanmayin!

CİKTİ ORNEGİ

```
Router# debug ip ospf adj
OSPF adjacency events debugging is on

Router#
OSPF-1 ADJ    Gi0/0/1: 2.2.2.2 address 10.1.1.2 is dead
OSPF-1  ADJ      Send  DBD  to  2.2.2.2  on  GigabitEthernet0/0/1  seq
0x1A2B opt 0x52 flag 0x7 len 32
OSPF-1  ADJ      Rcv  DBD  from  2.2.2.2  on  GigabitEthernet0/0/1  seq
0x1A2B opt 0x52 flag 0x2 len 132  mtu 1500 state EXSTART
OSPF-1 ADJ    Nbr 2.2.2.2: 2-Way Communication restored
```

187 terminal monitor

ACİKLAMA

VTY (SSH/Telnet) oturumunda log ve debug mesajlarinin gorunmesini saglar. Varsayilan olarak VTY'de log gorunmez.

KULLANİM AMACİ

Uzaktan bagli iken log mesajlarini ve debug ciktilarini gormek.

SYNTAX

```
Router# terminal monitor
Router# terminal no monitor
```

ORNEK KULLANİM

```
Router# terminal monitor
```

SİK YAPİLAN HATA

Konsol (line console 0)'da log'lar varsayilan olarak gorunur. VTY'de 'terminal monitor' gerekir. Oturum kapandiginda etkisi kaybolur.

CİKTİ ORNEGİ

```
Router# terminal monitor
Router#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/1, changed state to
up
%LINEPROTO-5-UPDOWN: Line protocol on Interface
GigabitEthernet0/0/1, changed state to up
! Log mesajlari artik SSH/Telnet oturumunda da gorunuyor
```

188 show users

ACİKLAMA

Cihaza aktif olarak bagli olan tum kullanicilari (konsol, vty, aux) ve durumlarini listeler.

KULLANİM AMACİ

Aktif oturumlari kontrol etmek, yetkisiz erisimi tespit etmek, baska kullanici olup olmadigini gormek.

SYNTAX

```
Router# show users [all]
```

ORNEK KULLANİM

```
Router# show users
```

SİK YAPİLAN HATA

'show users' ile 'show sessions' farklidir. 'show users' cihaza bagli kullanicilari, 'show sessions' sizin suspend ettiginiz oturumlari gosterir.

CİKTİ ORNEGİ

```
Router# show users
    Line       User       Host(s)              Idle       Location
*   0 con 0                idle                 00:00:00
*    2 vty 0       admin        idle                          00:05:23
192.168.1.100
     3 vty 1       operator     idle                          00:15:45
192.168.1.101

  Interface     User                 Mode          Idle      Peer
Address
```

189 show cdp neighbors

ACİKLAMA

CDP (Cisco Discovery Protocol) ile komsu Cisco cihazlarini listeler. Komsu cihazin modeli, portu ve kendi bagli portunuzu gosterir.

KULLANİM AMACİ

Fiziksel topoliyi kesfetmek, komsu cihazlari tanimlamak, baglanti hatalarini tespit etmek.

SYNTAX

```
Router# show cdp neighbors [detail]
```

ORNEK KULLANİM

```
Router# show cdp neighbors
Router# show cdp neighbors detail
```

SİK YAPİLAN HATA

CDP global olarak veya arayuzde kapatilmis olabilir. 'show cdp' ile durumu kontrol edin. Guvenlik icin untrusted portlarda CDP kapatilmalidir.

CİKTİ ORNEGİ

```
Router# show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route
Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P
- Phone,
                  D - Remote, C - CVTA, M - Two-port Mac Relay

Device ID        Local Intrfce     Holdtme    Capability  Platform
Port ID
CORE-SW01        Gig 0/0/1         123             R S I  WS-C3850
```

190 traceroute

ACİKLAMA Hedefe giden yoldaki (hop) router'lari sirasiyla listeler. Her hop'ta TTL artirilarak yanit alinir.

KULLANİM AMACİ Yonlendirme yolunu izlemek, nerede paket kaybinin oldugunu tespit etmek, latency olcumü yapmak.

SYNTAX

```
Router# traceroute {ip-address | hostname} [source source-ip] [probe count] [timeout seconds] [numeric]
```

ORNEK KULLANİM

```
Router# traceroute 8.8.8.8 source 10.1.1.1
```

SİK YAPİLAN HATA '* * *' gorunumu timeout anlamina gelir. Firewall ICMP engelliyor olabilir veya router yanit vermiyor olabilir. Bu yolun olmadigi anlamina gelmez.

CİKTİ ORNEGİ

```
Router# traceroute 8.8.8.8 source 10.1.1.1 numeric
Type escape sequence to abort.
Tracing the route to 8.8.8.8
VRF info: (vrf in name/id, vrf out name/id)
  1 10.1.1.2 1 msec 1 msec 1 msec
  2 192.168.1.254 2 msec 2 msec 2 msec
  3 203.0.113.1 5 msec 5 msec 5 msec
  4 142.250.160.1 8 msec 8 msec 8 msec
  5 142.250.230.45 10 msec 10 msec 10 msec
  6 8.8.8.8 12 msec 12 msec 12 msec
```

191 show flash:

ACİKLAMA

Flash bellek icerigini goruntuler. IOS imaji, yapilandirma dosyalari ve bos alan bilgisi verir.

KULLANİM AMACİ

Flash'taki dosyalari listelemek, bos alani kontrol etmek, IOS yukseltmesi oncesi hazirlik yapmak.

SYNTAX

```
Router# show flash: [all | chips | filesys]
```

ORNEK KULLANİM

```
Router# show flash:
```

SİK YAPİLAN HATA

Bazi platformlarda flash: yerine bootflash:, disk0:, usb0: gibi farkli dosya sistemleri olabilir.

CİKTİ ORNEGİ

```
Router# show flash:
-# - --length-- -----date/time------ path
1           524288000  Jul  10  2025  08:30:00  +00:00    isr4300-
universalk9.16.09.03.SPA.bin
2          45210 Jul 10 2025 09:15:32 +00:00  running-config.bak
3        1048576 Jul 10 2025 09:16:45 +00:00  vlan.dat

524288000 bytes available (314572800 bytes used)
```

192 show arp

ACİKLAMA

ARP (Address Resolution Protocol) onbellegini goruntuler. IP adresleri ile MAC adresleri arasindaki eslesmeleri gosterir.

KULLANİM AMACİ

L2-L3 eslestirmelerini kontrol etmek, ARP spoofing tespiti yapmak, MAC adreslerini ogrenmek.

SYNTAX

```
Router# show arp [ip-address | vrf vrf-name | interface type number]
```

ORNEK KULLANİM

```
Router# show arp
Router# show arp 192.168.1.10
```

SİK YAPİLAN HATA

Switch'lerde ARP tablosu SVI'ler icin tutulur. L2 switch'lerde 'show mac address-table' kullanilmalidir.

CİKTİ ORNEGİ

```
Router# show arp
Protocol  Address          Age (min)  Hardware Addr   Type
Interface
Internet  10.1.1.1               -   aabb.ccdd.ee01  ARPA
GigabitEthernet0/0/1
Internet  10.1.1.2             123   aabb.ccdd.ee02  ARPA
GigabitEthernet0/0/1
Internet  192.168.1.10          45   aabb.ccdd.ee10  ARPA
GigabitEthernet0/0/0
```

193 show mac address-table

ACİKLAMA

Switch'in CAM tablosunu (MAC adres tablosu) goruntuler. Hangi MAC adresinin hangi port/VLAN'da ogrenildigini gosterir.

KULLANİM AMACİ

Cihazin hangi porta bagli oldugunu bulmak, MAC flooding tespiti yapmak, L2 yolunu izlemek.

SYNTAX

```
Switch# show mac address-table [address mac-address | vlan vlan-id
| interface type number | dynamic | static]
```

ORNEK KULLANİM

```
Switch# show mac address-table
Switch# show mac address-table address aabb.ccdd.ee01
Switch# show mac address-table vlan 10
```

SİK YAPİLAN HATA

Eski IOS surumlerinde 'show mac-address-table' tirelidir. Yeni surumlerde 'show mac address-table' boslukludur. Her ikisi de calisabilir.

CİKTİ ORNEGİ

```
Switch# show mac address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----
   1    aabb.ccdd.ee01    DYNAMIC     Gi0/24
  10    aabb.ccdd.ee10    DYNAMIC     Gi0/1
  10    aabb.ccdd.ee11    DYNAMIC     Gi0/2
  20    aabb.ccdd.ee20    DYNAMIC     Gi0/3
```

194 show environment

ACİKLAMA

Cihazin donanim cevre durumunu (sicaklik, voltaj, fan, guc kaynagi) goruntuler.

KULLANİM AMACİ

Donanim sagligini kontrol etmek, sicaklik ve guc sorunlarini erken tespit etmek.

SYNTAX

```
Router# show environment [all | temperature | power | fan]
```

ORNEK KULLANİM

```
Router# show environment
```

SİK YAPİLAN HATA

'show environment' tum platformlarda desteklenmeyebilir. Router'lar genellikle destekler ancak bazi L2 switch'lerde farkli komutlar kullanilir.

CİKTİ ORNEGİ

```
Router# show environment
TEMPERATURE is OK
Temperature Value: 37 Degree Celsius

FAN Status is OK
Fan 1 OK, Speed (rpm) = 5623
Fan 2 OK, Speed (rpm) = 5610

POWER is OK
R0/PS0: Pwr Ok, Vin=230V, Vout=12.1V, Capacity=250W
```

195 reload

ACİKLAMA Cihazi yeniden baslatir. Belirli bir sure sonra veya hemen yapilabilir.

KULLANİM AMACİ Degisiklikleri etkinlestirmek, IOS yukseltmesi sonrasi yeniden baslatmak, sorun giderme amaciyla restart yapmak.

SYNTAX

```
Router# reload [in minutes | at hh:mm [month day | day month]]
[cancel]
```

ORNEK KULLANİM

```
Router# reload in 5
Router# reload at 02:00
Router# reload cancel
```

SİK YAPİLAN HATA

Ayarlanan reload iptal edilmezse cihaz otomatik restart olur. 'reload in' kullandiktan sonra 'reload cancel' ile iptal edilebilir.

CİKTİ ORNEGİ

```
Router# reload in 5
System will reload in 5 minutes. Proceed with reload? [confirm]y

Router# reload cancel

! Reload iptal edildi:
%SYS-5-SCHEDULED_RELOAD_CANCELLED: Scheduled reload cancelled at
13:30:45
```

196 archive log config

ACİKLAMA Yapilandirma degisikliklerini arsivlemeyi ve loglamayi saglar. Kim ne zaman hangi komutu girmis gorulebilir.

KULLANİM AMACİ Yapilandirma degisikliklerini izlemek, TACACS+ olmadan audit trail olusturmak.

SYNTAX

```
Router(config)# archive
Router(config-archive)# log config
Router(config-archive-log-cfg)# logging enable
Router(config-archive-log-cfg)#    notify    syslog    contenttype
plaintext
Router(config-archive-log-cfg)# hidekeys
```

ORNEK KULLANİM

```
Router(config)# archive
Router(config-archive)# log config
Router(config-archive-log-cfg)# logging enable
Router(config-archive-log-cfg)# notify syslog
```

SİK YAPİLAN HATA 'hidekeys' olmadan sifreler log'da clear-text gorunur. Guvenlik icin hidekeys mutlaka eklenmelidir.

CİKTİ ORNEGİ

```
Router# show archive log config all
  1       10     Jul  10  2025  13:30:45    admin       192.168.1.100
|interface gi0/0/1
  2      11    Jul  10  2025  13:30:52   admin      192.168.1.100   |ip
address 10.1.1.1 255.255.255.0
  3      12    Jul  10  2025  13:31:10   admin      192.168.1.100   |no
shutdown
```

197 ip sla / ip sla schedule

ACİKLAMA

IP SLA (Service Level Agreement) ile ag performansi olcumleri yapar. ICMP echo, jitter, HTTP gibi testler destekler.

KULLANİM AMACİ

Ag performansini surekli izlemek, SLA ihlallerini tespit etmek, EEM/tracking ile otomatik aksiyon almak.

SYNTAX

```
Router(config)# ip sla operation-number
Router(config-ip-sla)# icmp-echo {destination-ip | destination-host} [source-ip {ip-address | hostname} | source-interface interface-name]
Router(config-ip-sla-echo)# frequency seconds
Router(config)# ip sla schedule operation-number [life {forever | seconds}] [start-time {hh:mm[:ss] [month day | day month] | pending | now | after hh:mm:ss}] [ageout seconds]
```

ORNEK KULLANİM

```
Router(config)# ip sla 1
Router(config-ip-sla)# icmp-echo 8.8.8.8 source-ip 10.1.1.1
Router(config-ip-sla-echo)# frequency 30
Router(config)# ip sla schedule 1 life forever start-time now
```

SİK YAPİLAN HATA

'ip sla schedule' yapilmadan olcum baslamaz. Hem sla tanimi hem schedule gereklidir.

CİKTİ ORNEGİ

```
Router# show ip sla statistics
IPSLAs Latest Operation Statistics

IPSLA operation id: 1
        Latest RTT: 12 milliseconds
Latest operation start time: 13:45:32 IST Fri Jul 10 2025
Latest operation return code: OK
Number of successes: 234
Number of failures: 2
Operation time to live: Forever
```

198 show archive

ACİKLAMA

Yapilandirma arsivini (configuration archive) goruntuler. Snapshot alinarak yapilandirmanin onceki versiyonlari saklanabilir.

KULLANİM AMACİ

Yapilandirma degisikliklerini geri almak, onceki yapilandirmalari karsilastirmak.

SYNTAX

```
Router# show archive
Router# configure replace archive-config
```

ORNEK KULLANİM

```
Router# show archive
Router# configure replace flash:archive-Jul-10-13-30-45.123-0
```

SİK YAPİLAN HATA

'archive' global yapilandirmada tanimlanmadan 'configure replace' calismaz. Oncelikle archive yolu tanimlanmalidir.

CİKTİ ORNEGİ

```
Router# show archive
The maximum archive configurations allowed is 14.
Archive #  Name
   1       flash:archive-Jul-9-08-30-12.234-1
   2       flash:archive-Jul-9-12-15-45.567-2
   3       flash:archive-Jul-10-13-30-45.123-3
*  4       flash:archive-Jul-10-13-45-00.890-4

! * = Aktif yapilandirma
```

199 copy tftp: running-config

ACİKLAMA

TFTP sunucusundan yapilandirma dosyasini cihaza yukler. Mevcut running-config uzerine merge eder.

KULLANİM AMACİ

Yapilandirmayi TFTP'den geri yuklemek, toplu yapilandirma degisikligi yapmak.

SYNTAX

```
Router# copy tftp: running-config
Address or name of remote host []? tftp-server-ip
Source filename []? config-file-name
Destination filename [running-config]?
```

ORNEK KULLANİM

```
Router# copy tftp: running-config
Address or name of remote host []? 192.168.1.10
Source filename []? router-config.txt
Destination filename [running-config]?
```

SİK YAPİLAN HATA

'copy tftp: running-config' merge yapar, replace yapmaz. Tamamen yeni yapilandirma icin once 'write erase' yapilip 'copy tftp: startup-config' sonrasi reload gerekir.

CİKTİ ORNEGİ

```
Router# copy tftp: running-config
Address or name of remote host []? 192.168.1.10
Source filename []? router-config.txt
Destination filename [running-config]?
Accessing tftp://192.168.1.10/router-config.txt...
Loading     router-config.txt     from     192.168.1.10     (via
GigabitEthernet0/0/0): !
[OK - 4521 bytes]

[OK]
```

200 write erase / erase startup-config

ACİKLAMA NVRAM'deki startup-config dosyasini siler. Cihaz bir sonraki restart'ta bos yapilandirma ile baslar.

KULLANİM AMACİ Cihazi fabrika ayarlarina dondurmek, yeni yapilandirma yuklemek, hazirlik sirasinda temiz baslangic yapmak.

SYNTAX

```
Router# write erase
Router# erase startup-config
```

ORNEK KULLANİM

```
Router# write erase
Erasing the nvram filesystem will remove all configuration files!
Continue? [confirm]y
[OK]
Erase of nvram: complete
```

SİK YAPİLAN HATA 'write erase' sadece NVRAM'i siler. Running-config hala aktiftir. Kalici silme icin 'write erase' + 'reload' yapilmalidir.

CİKTİ ORNEGİ

```
Router# write erase
Erasing the nvram filesystem will remove all configuration files!
Continue? [confirm]y
[OK]
Erase of nvram: complete
%SYS-7-NV_BLOCK_INIT: Initialized the geometry of nvram

Router# show startup-config
startup-config is not present
! NVRAM bos, reload sonrasi fabrika ayarlariyla baslayacak
```